Software Design Description (SDD)

UTeM Micro-Credential Management System (MCMS)

Document Control

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Disclaimer:

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Revision History

Version	Date	Change
1.0	27 Oct 2026	First issue: architecture, technology, module, data, interface, process, security, quality and deployment design for Release 1, traced to the SRS v1.0 and the approved UI/UX Design Specification v1.1.
1.1	27 Oct 2026	Aligned with Glossary v0.6, SRS v1.1, PRD v1.1 and UI/UX Design Specification v1.2 after the cross-document review of 27 October 2026: Laravel 13 replaces Laravel 11 (AD-02); completion window setting added; cross-cutting requirements and the PRD build increments traced (11.2); Release 1 prototype and design prototype distinguished (D-31); assumptions DA-18 and DA-19 added; static public pages assigned to a module. No requirement, module, event or job ID changed.

Glossary

All terms of the Glossary and Decision Log v0.6 apply with exactly the same meaning. The table below adds only the technical terms used in this SDD.

Term	Definition
Action	A single-purpose class that performs one business operation of a module (for example VerifyResults), inside one database transaction.
Adapter	An implementation of an integration interface for one external system. Each interface has a Stub, Manual and, later, Live adapter (AD-08).
AD-XX / DA-XX	Architecture decision / design assumption of this SDD (Sections 1.6 and 11.3).
Domain event (EV-XX)	A record that something happened in a module (for example CredentialIssued), published so other modules can react without reading its tables.
Eloquent	Laravel's object-relational mapper.
Horizon	Laravel's dashboard and supervisor for Redis queues.
Job (JB-XX)	A unit of background work run by a queue worker, for example issuing a batch of credentials.
Livewire	A Laravel library for interactive server-rendered components.
MD-XX	A module of the modular monolith (Section 2.3).
Policy	A Laravel class that decides whether a user may perform an action on a record (role, scope, separation of duties).
ULID	Universally Unique Lexicographically Sortable Identifier, used as the internal primary key.

1. Introduction

1.1 Purpose

This Software Design Description (SDD) describes how the UTeM Micro-Credential Management System (MCMS) is built: its architecture, technology, modules, data, interfaces, processes, security, quality attributes and deployment. It is the last document of the set before the Release 1 build. Developers and AI development agents use it, together with the SRS and the UI/UX Design Specification, to build Release 1 without having to guess.

The SRS states *what* the system shall do, the PRD *what to build first and why*, and the UI/UX Design Specification *how it looks and behaves*. This SDD states *how it is made*. Where this SDD and the SRS differ on a requirement, the SRS prevails. Where this SDD and the UI/UX Design Specification differ on a screen's look or behaviour, the UI/UX Design Specification prevails. Where they differ on a term or decision, the Glossary and Decision Log prevails.

1.2 Scope

The design covers the complete Release 1 system: every requirement of the SRS marked SHALL (Section 11.2 gives the module for each area and the cross-cutting table gives the rest), every Release 1 feature of the PRD, and every screen and document of the approved UI/UX Design Specification v1.2. It shows where Release 2 and 3 capabilities (SHOULD and MAY) will attach, without designing them in detail, so that they can be added without rework (D-24).

In this SDD "the prototype" means the Release 1 prototype, the application designed here. The clickable HTML prototype of the UI/UX Design Specification is called the design prototype (D-31).

MCMS is a new standalone web application with its own database, which reaches other systems only through adapters (D-06). Learning content stays on UCreds. The national ABC platform and the QVC/MQA procedures are outside the system; MCMS prepares and records, and UTeM staff submit (D-14).

1.3 Document Conventions

Convention	Meaning
AD-XX	Architecture decision of this SDD (Section 1.6). Every design element cites the decisions it follows.
DA-XX	Design assumption of this SDD, made where the owner has not yet decided (Section 11.3).
MD-XX	Module of the modular monolith (Section 2.3). MD-00 is the shared platform kernel.
EV-XX / JB-XX	Domain event / background job (Section 4.19).
FR-XX, BR-XX, IR-XX, DR-XX, NFR-XX, SR-XX	Requirement IDs of the SRS v1.1, exactly as published there.
D-XX, A-XX, OI-XX	Decisions, assumptions and open items of the Glossary and Decision Log v0.6.
Screen IDs (DD-12, ST-03 …)	Screens of the UI/UX Design Specification v1.2, Section 6.6.
`code`	Class, table, column, route or configuration names.

1.4 Intended Audience

Reader	Use of this document	Read first
Supervisor and CODL	Check that the design is sound, buildable by a small team and safe for UTeM's records.	Sections 1, 2, 8, 11
Developers and AI development agents	Build the modules, data, interfaces and jobs exactly as designed.	Sections 2 to 7
Testers	Derive integration and end-to-end tests from the processes, events and traceability.	Sections 7, 11
System administrators (CODL)	Install, configure, operate, back up and monitor the system.	Sections 9, 10
Security reviewers	Check authentication, authorisation, key handling, audit integrity and privacy.	Section 8

1.5 References

1. Glossary and Decision Log for the UTeM MCMS, version 0.6, 27 October 2026 (MCMS_FND_V0.6_Oct_2026).
2. Software Requirements Specification, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_SRS_V1.1_Oct_2026).
3. Product Requirements Document, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_PRD_V1.1_Oct_2026).
4. UI/UX Design Specification, UTeM MCMS, version 1.1, 27 October 2026 (MCMS_UIUX_V1.2_Oct_2026), and its approved clickable prototype (`UIUX/v2/prototype`).

5. Garis Panduan Micro-Credential UTeM, Versi 1.0, 18 September 2026 (the Guide).
6. IEEE 1016-2009, Standard for Information Technology: Systems Design, Software Design Descriptions (structure guidance).
7. 1EdTech Open Badges 3.0; W3C Verifiable Credentials Data Model 2.0; W3C Data Integrity EdDSA Cryptosuites.
8. OWASP Application Security Verification Standard 4.0; WCAG 2.1; Personal Data Protection Act 2010 (Act 709).
9. Laravel 13, Livewire 3, PostgreSQL 16 and Microsoft Entra ID (OpenID Connect) documentation.

1.6 Architecture Decisions

Table 1 lists the decisions that shape the whole design. Each later section cites them. Decisions marked Assumed rest on an open item of the Glossary and Decision Log; if the open item closes differently, only the parts that cite that decision change.

Table 1: *Architecture decisions*

ID	Decision	Basis	Status
AD-01	Modular monolith: one Laravel application with code split into modules under `app/Modules/<Module>`. A module owns its tables; other modules use its public service interface or its domain events, never its tables.	D-07	Proposed
AD-02	Laravel 13 on PHP 8.3 (Laravel 11 left security support on 12 March 2026; Laravel 12 security fixes end on 24 February 2027).	D-08	Decided
AD-03	PostgreSQL 16 as the database. Queries go through Eloquent and the query builder, so MySQL 8 stays a fallback.	D-09, OI-02	Assumed
AD-04	Server-rendered user interface: Blade templates with Livewire 3 and Alpine.js. DESIGN.md tokens are CSS custom properties, built with Vite. Shared components mirror the approved prototype.	UI/UX spec 3–4	Proposed
AD-05	Background work as queued jobs on Redis, supervised by Laravel Horizon; timed alerts from the Laravel scheduler.	NFR queue	Proposed
AD-06	Sign-in with Microsoft Entra ID through OpenID Connect. Staff MFA is enforced by Entra Conditional Access and checked through the `amr` claim. A stub identity provider exists only outside production.	D-10, OI-03	Assumed
AD-07	Authorisation by role assignment with scope, checked per record by policies for role, scope and separation of duties. No super-user bypass for academic actions; CODL roles never pass an academic policy.	D-11, D-12, D-28	Decided basis
AD-08	Each external system is an interface with Stub, Manual and Live adapters, chosen by configuration. An integration hub logs and retries calls.	D-06, D-14	Proposed
AD-09	Credentials as Open Badges 3.0 / W3C Verifiable Credentials, signed with Ed25519 (Data Integrity eddsa-rdfc-2022). Keys are held in a key vault owned by CODL. Public verification is by random code, link or QR.	D-19, OI-08	Assumed
AD-10	Printable documents are Blade A4 templates rendered to PDF by headless Chromium in a queue worker; templates are bilingual and versioned.	UI/UX spec 8	Proposed
AD-11	Append-only audit table with a hash chain, written in the same transaction as the change; the application role cannot update or delete it. Audit writes are serialised through a chain-head row lock, which works on PostgreSQL and MySQL at the cost of write throughput (Section 9).	D-20	Proposed
AD-12	Settings versioned with effective dates; each decision stores the setting version it used.	D-16	Proposed
AD-13	Interface strings in Laravel language files (ms, en); user-facing content stored in both languages; locale per user.	D-02	Proposed
AD-14	Files in S3-compatible object storage (MinIO on premises), scanned by ClamAV on upload, served through short-lived signed URLs.	SRS uploads	Assumed
AD-15	Email through an SMTP relay; every notification is stored first and sent by a queued job.	SRS SMTP	Assumed
AD-16	Staff search with PostgreSQL full-text search, always within the user's scope.	NFR search	Proposed
AD-17	Pest feature tests named after the SRS requirement they prove, plus Playwright end-to-end tests reusing the prototype's flows.	D-01	Proposed
AD-18	Hosting on Linux virtual machines in the UTeM data centre: Nginx with PHP-FPM, PostgreSQL, Redis, MinIO and queue workers; environments dev, test, staging and production. Production never runs the stub identity provider or the demo clock; the payment stub and the Manual adapters (UCreds, ABC, QVC/MQA) are the Release 1 production modes.	A-09, D-21, D-14	Assumed
AD-19	Three database roles: `mcms_app` for the application, without UPDATE or DELETE on append-only tables; `mcms_migrator` for schema changes; and `mcms_disposal`, used only by the approved disposal procedure.	D-20, OI-07	Proposed

1.7 Design Principles

The design applies the principles of the UI/UX Design Specification (Section 2.1) to the code:

- **One stage, one record.** Each recognition stage is its own table, its own action and its own event (D-15). No action changes two stages. Verifying results and confirming completion are separate actions; the system issues credentials only from a completion confirmation (D-29).
- **Authority in the policy layer.** Who may decide is enforced in policies on the server for every request, not only by hiding buttons (D-11, D-12).
- **Generated recognition text.** Credit, MQF level and QV wording is computed from verified records and templates, never stored as free text (D-17).
- **Nothing is erased.** Corrections add revisions and events; the audit trail is append-only (D-20).
- **Rules are settings.** Thresholds and approving roles are versioned settings, and decisions record the version in force (D-16).
- **Adapters at the edge.** Every external system can run as a stub, manual exchange or live connection without changing the modules (D-06, D-14).

2. Architecture Overview

This section shows how MCMS is divided: what surrounds it, what runs where, how the code is cut into modules, how modules talk to each other and to external systems, and how one request travels through the layers. It applies decisions AD-01 to AD-18 of Section 1. Module internals are in Section 4, data in Section 5, interfaces in Section 6.

2.1 System Context

MCMS is a new standalone system (SRS Section 2.1). People reach it through a browser. Every external system is reached through an adapter (AD-08, IR-14). Figure 1 shows the systems MCMS exchanges data with in Release 1 and how each one is handled.

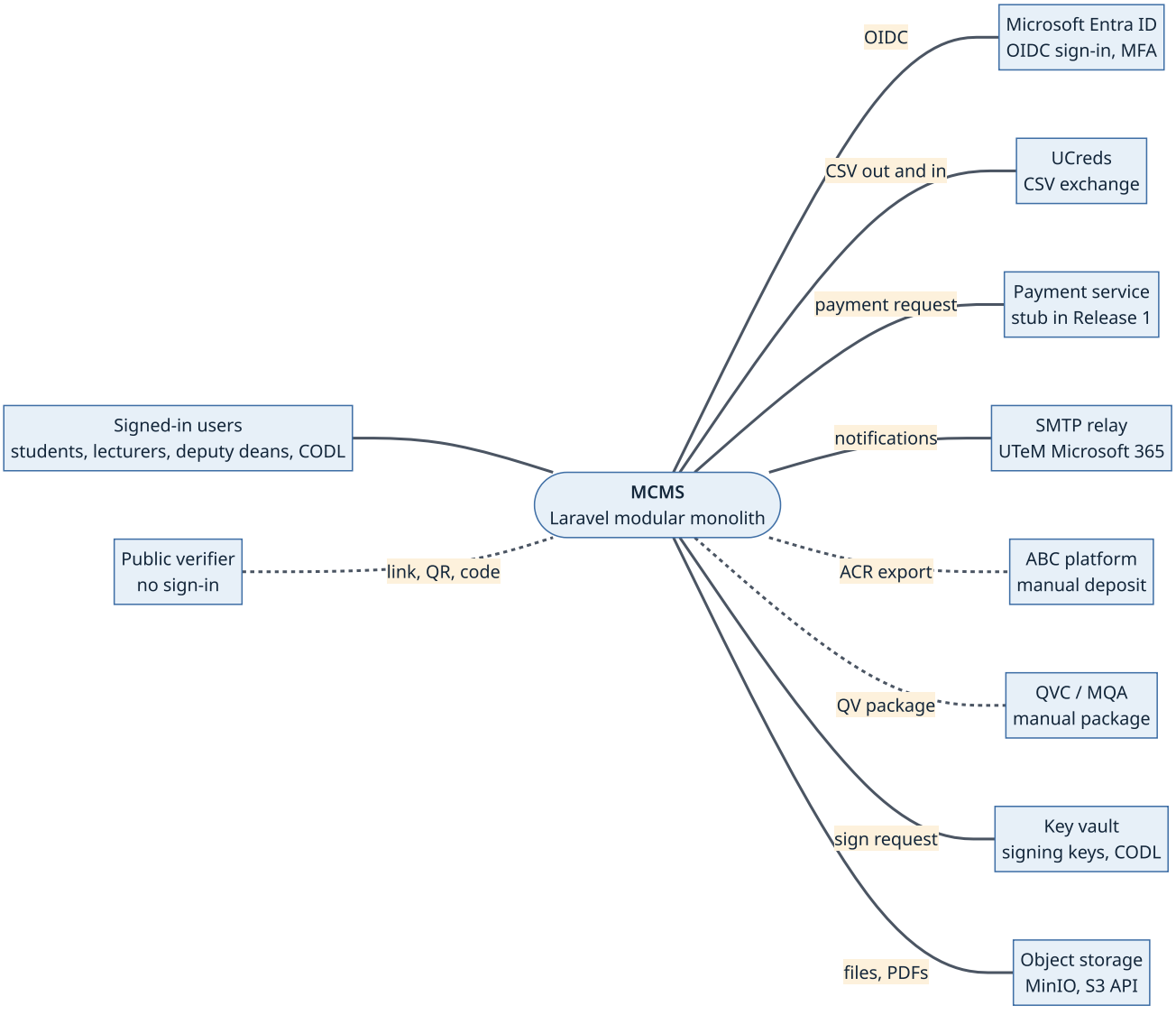


Figure 1: *System context. Solid lines are automated exchanges, dotted lines are manual or file-based exchanges in Release 1.*

Table 2: *External systems and their Release 1 handling*

External system	Adapter interface (MD-16)	Release 1 mode	Basis
Microsoft Entra ID (UTeM SSO)	`IdentityProvider`	Live OIDC; Stub only outside production	AD-06, FR-01, IR-18
UCreds learning platform	`LearningPlatformGateway`	Manual (CSV export and import)	FR-110, IR-19
Payment service	`PaymentGateway`	Stub (dummy payment)	FR-101, A-03
SMTP relay	`MailTransport`	Live SMTP; Stub captures mail outside production	AD-15, IR-23
ABC platform	`AbcGateway`	Manual (export file, deputy dean records the deposit)	FR-188, A-02

External system	Adapter interface (MD-16)	Release 1 mode	Basis
QVC / MQA	`QvGateway`	Manual (package export, result recorded by hand)	FR-169, A-02
Key vault	`SigningKeyStore`	Live vault owned by CODL; Stub key only outside production	AD-09, SR-08, IR-13
Object storage	Laravel filesystem disk `s3`	Live MinIO; local disk in dev	AD-14, IR-26

> **Note:** The student information system is not connected in Release 1 (SRS Section 2.1). The adapter pattern of Section 2.7 leaves room for it.

2.2 Runtime Containers

MCMS is one code base deployed as several processes (AD-05, AD-18). The web process and the queue workers run the same release of the application. No process keeps session state on its local disk (NFR-10): sessions and cache live in Redis.

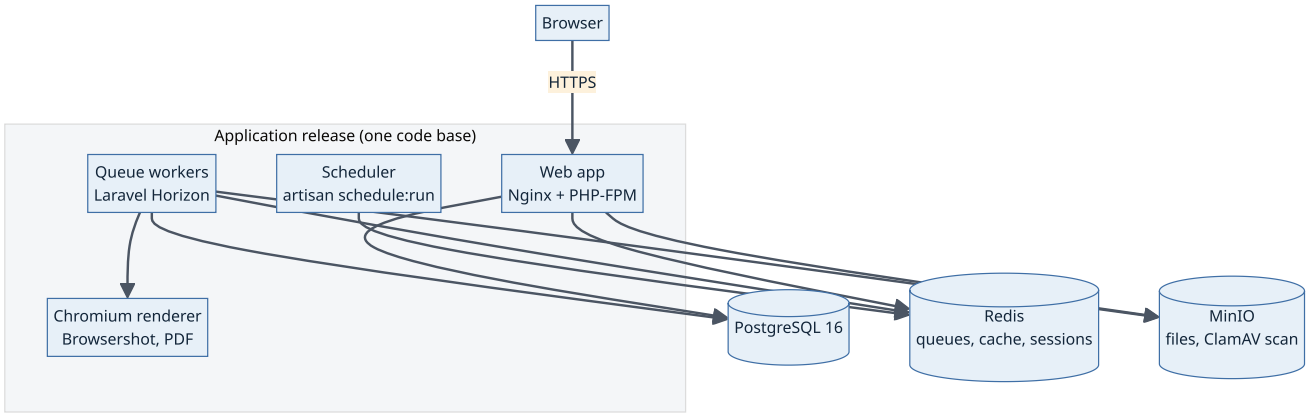


Figure 2: *Runtime containers. The scheduler only dispatches jobs onto Redis; the workers do the work.*

Table 3: *Runtime containers and their duties*

Container	Duty	Scales by	Basis
Web app (Nginx + PHP-FPM)	Serves Blade and Livewire pages, the public verification page and the JSON API; runs policies and actions; writes the database	More web nodes behind a load balancer	AD-04, NFR-01
Queue workers (Horizon)	Run jobs in named queues: `default`, `notifications`, `imports`, `credentials`, `reports`, `integration`	More worker processes per queue	AD-05, NFR-11
Scheduler	One process runs `schedule:run` every minute; each timed job logs its result	One instance only (single-run lock)	NFR-43
PostgreSQL 16	System of record; full-text search; audit table	Vertical, then read replica if needed	AD-03, AD-16
Redis	Queues, cache, sessions, rate limiter, locks	Vertical	AD-05, NFR-10
MinIO	Uploaded files, generated PDFs, export files; ClamAV scans uploads before they are stored for use	Disk	AD-14
Chromium renderer	Headless Chromium started by a worker to print A4 Blade templates to PDF; never started from a web request	Worker count	AD-10

2.3 Module Map

The code is split into the modules of Table 4 (AD-01, NFR-27). The SRS modules M1 to M16 are regrouped as the brief allows: M2 and M3 form one Curriculum module and M13 is split into Transfer and Pathways. MD-00 Platform is the shared kernel. It holds no business rule of its own.

Table 4: *Module map and dependencies*

ID	Module	Responsibility	SRS	Depends on
MD-00	Platform	Settings with effective dates, i18n, files, PDF rendering, UI components, clock, base classes	all	none
MD-01	Identity	Sign-in, users, roles and scopes, delegation, profile	M1	MD-00
MD-02	Curriculum	Category determination, proposal and design workspace, UMC link to parent course	M2, M3	MD-00, MD-01

ID	Module	Responsibility	SRS	Depends on
MD-03	Approval	Review, endorsement, Senate record, versions, material change	M4	MD-02, MD-01
MD-04	Offering	Offering, readiness checks, public catalogue	M5	MD-02, MD-03, MD-10
MD-05	Enrolment	Registration, eligibility, payment, cohorts, waiting list	M6	MD-04, MD-01
MD-06	Learning platform	Cohort export, results import and staging, badge flow with UCreds	M7	MD-05
MD-07	Assessment	Assessment plan, result entry, verification of achievement, appeals	M8	MD-05, MD-06
MD-08	Achievement record	Achievement Record, module status, completion	M9	MD-07, MD-05
MD-09	Credentials	Badge, Certificate, Digital Credential, public verification, correct, replace, revoke	M10	MD-08, MD-02, MD-10
MD-10	Quality verification	QV record for SAMC, evidence, package, result, lapse	M11	MD-02
MD-11	Credit bank	ACR, consent, deposit, credit ledger, validity, redemption	M12	MD-08, MD-09, MD-10
MD-12	Transfer	Credit transfer application, equivalence, decision	M13 (part)	MD-11, MD-02
MD-13	Pathways	Stackable mapping, plans, 30 percent home rule, residency	M13 (part)	MD-02, MD-11, MD-12
MD-14	Quality and change	Indicators, CQI, periodic review, suspension and termination	M14	MD-03, MD-07, MD-10
MD-15	Records	Audit trail, traceability, reports, export, retention	M15	MD-00
MD-16	Integration hub	Notifications, adapters, integration log, job registry	M16	MD-00

Two modules are called by every other module and so are not drawn as arrows: MD-15 offers the `AuditRecorder` contract and MD-16 offers the `Notifier` contract. Both depend only on MD-00 and receive other modules' facts through arguments or events, so they never depend back on a business module. Figure 3 and Figure 4 show the remaining dependencies in two parts; an arrow reads "uses the public contract of".

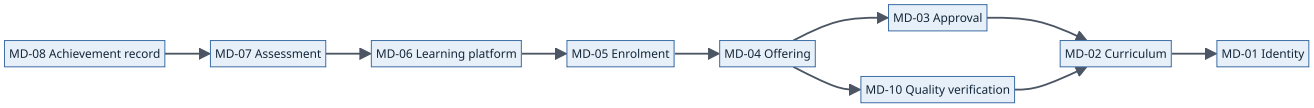


Figure 3: *Dependencies, part 1: from curriculum to the Achievement Record.*

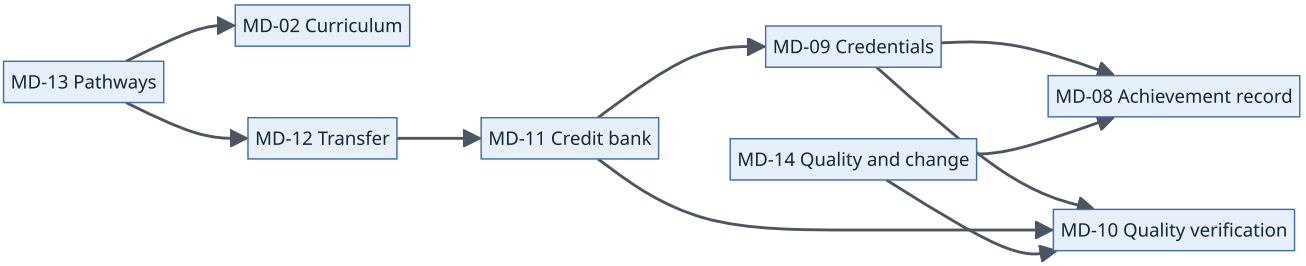


Figure 4: *Dependencies, part 2: from the Achievement Record to credit and monitoring.*

The dependency graph has no cycle. A build check (Section 3, `deptrac`) fails if a module uses a class outside the contracts of the modules it may depend on.

2.4 Layering Inside a Module

Every module has the same layers. A layer may call only the layer below it.

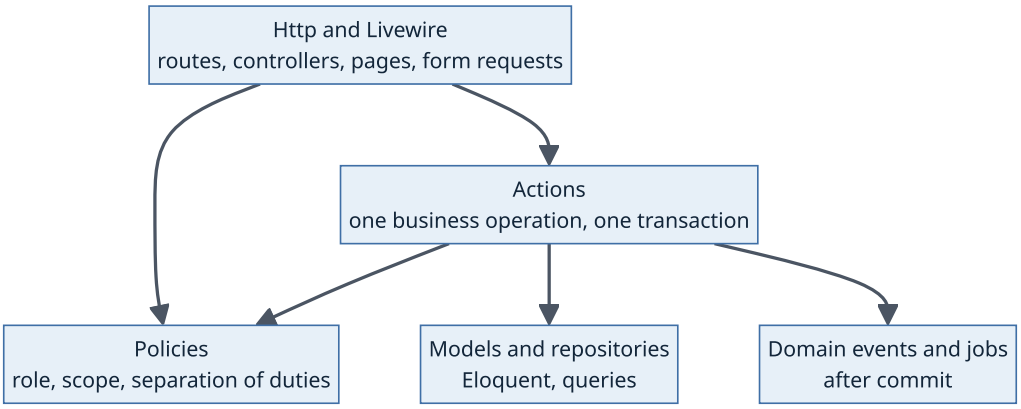


Figure 5: *Layers inside a module.*

Table 5: *Layer rules*

Layer	Holds	Must not
Http and Livewire	Routes, controllers, Livewire components, form requests (input validation), Blade views	Contain a business rule or write the database directly
Policies	One Laravel Policy per model; checks role, scope and separation of duties on the record (AD-07, SR-03)	Query other modules' tables; contain a super-user bypass for academic actions
Actions	Invokable classes such as `VerifyResults`; each runs in one database transaction, checks the policy, applies the rule, writes the audit row (AD-11), then raises events (NFR-20)	Call another module's models; be called from a model
Models and repositories	Eloquent models of the tables the module owns, query scopes, enums	Reach into another module's models
Domain events and jobs	Past-tense events (for example `ResultsVerified (EV-29)`), listeners and queued jobs, dispatched after commit	Carry a whole model; they carry identifiers and the facts needed

The folder example below is the pattern for every module. The full repository tree is in Section 3.

```
app/Modules/Assessment/
  Http/Controllers/  Http/Requests/  Livewire/
  Actions/VerifyResults.php  Actions/VerifyResultsData.php (DTO)
  Models/Result.php  Models/AssessmentPlan.php
  Enums/ResultStatus.php
  Policies/ResultPolicy.php
  Events/ResultsVerified (EV-29).php
  Listeners/  Jobs/
  Contracts/AssessmentService.php   (public interface)
  Database/Migrations/  Database/Factories/
  Resources/views/  Resources/lang/{ms,en}/
  AssessmentServiceProvider.php
  routes.php
```

2.5 Communication Between Modules

Modules talk in two ways only (AD-01, NFR-27).

1. **Public service contract.** A module that needs an answer from another module calls an interface in that module's `Contracts` folder, for example `AchievementRecordService::isComplete()`. The interface returns DTOs or scalars, never Eloquent models.
2. **Domain event.** A module that needs to tell others what happened raises an event. Listeners in other modules react. Events are named in the past tense and listed in Sections 4 and 6.

Table 6: *Rules of inter-module communication*

Rule	Meaning	Enforced by
R1	A module reads and writes only the tables it owns. No cross-module join, no foreign key to another module's table except by identifier column checked in the contract.	Review, `deptrac`, Pest architecture tests (NFR-27)
R2	Reads across modules go through a contract. Reports that need many modules use a read-only query contract that each module offers to MD-15.	`deptrac`

Rule	Meaning	Enforced by
R3	Events are raised only after the transaction commits (`ShouldDispatchAfterCommit`). A listener that changes data runs as a queued job in its own transaction.	Base event class
R4	An action changes one module's data. A change in another module follows from an event, not from a direct write.	Review, Pest architecture test
R5	Anything that must reach the user or an outside system is stored first in the same transaction: a `notifications` row (AD-15) or an `integration_calls` row, then sent by a queued job. A scheduler sweep re-dispatches rows that are still pending, so a lost queue message does not lose a notice. This is the outbox.	`Notifier` and `IntegrationHub` contracts
R6	Recognition stages stay separate records and separate actions (D-15). No event of one stage triggers a decision of the next; it only creates a task or notice for the person who decides.	Review, Pest feature tests
R7	The audit row is written in the same transaction as the change (AD-11, FR-254).	`AuditRecorder` inside every action

> **Note:** Credential issuing after completion is the one place where an event starts automatic work. `CompletionConfirmed (EV-31)` starts the issue job because the deputy dean's confirmation is itself the decision (BR-20). Nothing else is issued without a human decision.

2.6 Transactions and Events

Each action runs in one database transaction so that a multi-record change succeeds or fails as a whole (NFR-20). Inside the transaction an action locks the row it changes, checks the rule, saves, and writes the audit row. After the commit the events fire and jobs are queued. A job is idempotent: it can run twice without a second result, using the record's identifier as the idempotency key (IR-17). Registration takes a row lock on the offering to keep capacity exact under concurrent submissions (NFR-06).

2.7 Adapter Pattern

Each external system has a PHP interface in MD-16 and three implementations (AD-08, IR-14, IR-15). The configuration for the environment chooses which one is bound in the service container, so a business module never knows which one it is using (IR-16, NFR-31).

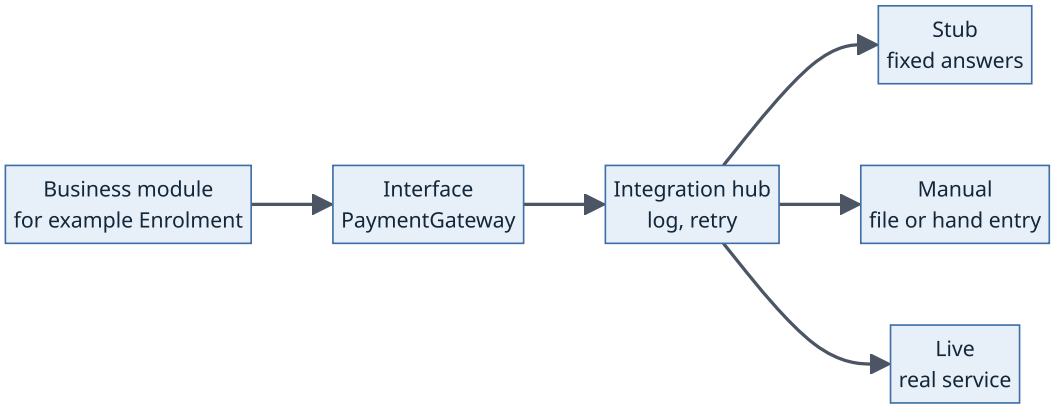


Figure 6: *Adapter pattern. Every call passes through the integration hub, which logs it and retries failures.*

Table 7: *Adapter rules*

Topic	Rule
Selection	`config/integrations.php` maps each interface to `stub`, `manual` or `live` from an environment value such as `INTEGRATION_PAYMENT=stub`. A settings screen shows the current mode for each adapter to administrators and changes it only where the environment permits (FR-25).
Stub	Returns fixed, documented answers, including failure cases, so tests and demonstrations are repeatable. The stub for sign-in and the demo clock exist only when `APP_ENV` is not `production` (AD-06).
Manual	Writes an export file and waits for a person to bring the result back, for example the ABC export and the QV package (FR-188, FR-169).
Live	Calls the real service. Added in a later release without a change to business modules.
Integration log	Every call writes an `integration_calls` row: adapter, operation, correlation identifier, status, duration, attempt number and a redacted summary (FR-270, FR-116). The log never holds identity numbers, tokens or secrets (NFR-41).
Retries	A failed call is retried by its job with back-off (1, 5, 15, 60 minutes, then marked failed and shown to administrators). Business functions that do not depend on the system continue (NFR-18, FR-271).

Topic	Rule
Idempotency	Each outbound request carries a key built from the record identifier and operation, so a repeat is safe (IR-17).
Contract tests	One shared Pest test suite runs against Stub, and against Live when it exists, to prove both follow the interface.

2.8 Request Lifecycle: Deputy Dean Verifies Results

Figure 7 follows one request through the layers. The deputy dean opens a cohort in Assessment and presses Verify results (screen and flow as in the UI/UX specification). The request shows how route, middleware, policy, action, audit, event, job and notification fit together (FR-123, BR-20).

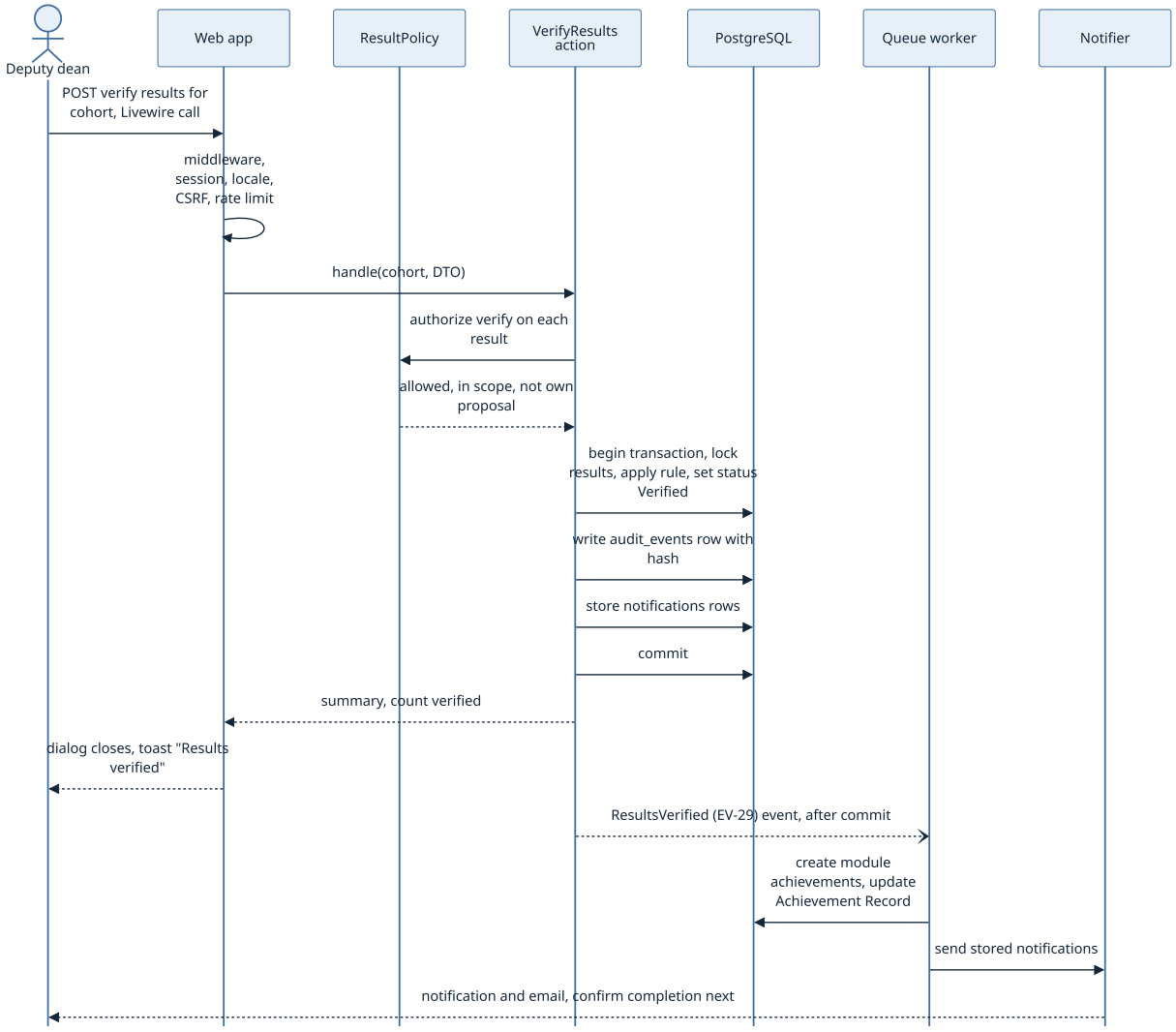


Figure 7: *Lifecycle of a Verify results request. Verifying results creates module achievements only. Completion is confirmed in a separate action.*

Table 8: *Steps of the request lifecycle*

Step	What happens	Where in code
1	Route `POST` with the Livewire update endpoint; sign-in, idle timeout, locale and rate-limit middleware run	MD-00, MD-01 middleware
2	The Livewire component validates the input with a form request and builds a DTO	`Http/Livewire`
3	The component calls the action; the action asks the policy for each record; a refusal returns an access error and a security log entry (SR-03, SR-18)	`Policies`
4	The action opens a transaction, locks the records, checks the blocking conditions (for example a held integrity case, FR-122), and sets the status	`Actions`
5	The audit row and the notification rows are written in the same transaction (FR-254, FR-265)	`AuditRecorder`, `Notifier`
6	After commit the event `ResultsVerified (EV-29)` is dispatched; a queued listener in MD-08 creates the module achievements	`Events`, `Listeners`
7	A notification job sends the stored notices by in-app message and email	MD-16 job
8	The user sees a toast; the cohort page shows the new statuses and the next step	`x-ui.toast`

3. Technology and Code Structure

This section fixes the technology, the repository layout and the coding rules, so that developers and coding agents produce code of one shape. The code follows Laravel conventions and the PHP-FIG standards so that CODL PHP developers can maintain it (NFR-28).

3.1 Technology Stack

Versions are the lines to install. At the start of the build every package line is checked against the supported Laravel release (AD-02, DA-19) and then fixed in `composer.lock`. The exact patch version is fixed in `composer.lock` and `package-lock.json` at the start of the build, and only security patches are taken within a release.

Table 9: *Technology stack*

Component	Choice	Version	Purpose	Ref
Language	PHP	8.3	Application language; `sodium`, `intl`, `pdo_pgsql`, `redis` extensions	AD-02
Framework	Laravel	13	Routing, Eloquent, queues, scheduler, policies, validation	AD-02
UI server side	Blade with Livewire	Livewire 3	Server-rendered pages and interactive tables, forms and dialogs	AD-04
UI browser side	Alpine.js	3	Small behaviours: dialogs, menus, focus handling, toasts	AD-04
Asset build	Vite with plain CSS custom properties	Vite 5, Node 20 LTS (build only)	Builds the DESIGN.md tokens and components into CSS and JS	AD-04
Database	PostgreSQL	16	System of record, full-text search, audit table	AD-03, AD-16
Cache, queue, session	Redis	7	Queues, cache, sessions, locks, rate limits	AD-05
Queue supervisor	Laravel Horizon	5	Runs and monitors queue workers	AD-05
Sign-in	Laravel Socialite with the Microsoft provider	Socialite 5	OIDC sign-in with Microsoft Entra ID	AD-06
PDF	Spatie Browsershot with headless Chromium	Browsershot 4	Prints bilingual A4 Blade templates to PDF in a worker	AD-10
Credential signing	PHP sodium (Ed25519) with an RDFC-1.0 canonicalisation library	PHP 8.3	Open Badges 3.0 Data Integrity proof (`eddsa-rdfc-2022`)	AD-09
Object storage	MinIO (S3 API) through Laravel filesystem	current stable	Files, PDFs, exports	AD-14
Malware scan	ClamAV daemon	current stable	Scans each upload	AD-14, SR-09
Web server	Nginx with PHP-FPM	current stable	HTTPS termination and PHP	AD-18
Health checks	spatie/laravel-health	current for Laravel 13	Health endpoint for monitoring	NFR-40
Unit and feature tests	Pest	3	Feature tests named after SRS requirements	AD-17
End-to-end tests	Playwright	current stable	Browser tests of the prototype's flows	AD-17
Code style	Laravel Pint	current	PSR-12 formatting	NFR-28
Static analysis	Larastan (PHPStan)	level 8	Type checking	NFR-28
Architecture check	Deptrac	current	Fails the build on a forbidden module dependency	AD-01, NFR-27
Mail capture (non-production)	Mailpit	current	Catches test email	AD-15
Containers	Docker Compose for dev and test	current	Repeatable environments, PostgreSQL, Redis, MinIO, ClamAV, Mailpit	NFR-33

> **Note:** No front-end framework other than Livewire and Alpine is used, and no CSS framework. All dependencies must have licences compatible with UTeM ownership and on-premises use (NFR-38); licences are listed in the release notes.

3.2 Repository Layout

One Git repository holds the application. Business code lives only under `app/Modules` (AD-01). Folders outside `app/Modules` follow the Laravel default.

```
mcms/
  app/
    Modules/
      Platform/ Identity/ Curriculum/ Approval/ Offering/ Enrolment/
      LearningPlatform/ Assessment/ AchievementRecord/ Credentials/
      QualityVerification/ CreditBank/ Transfer/ Pathways/
      Monitoring/ Records/ Integration/
        (each module: Http, Livewire, Actions, Models, Enums, Policies,
         Events, Listeners, Jobs, Contracts, Database, Resources, routes.php)
    Providers/            registers module providers
  bootstrap/  config/
    mcms.php              release number, feature flags, limits
    integrations.php      adapter mode per interface
    settings-defaults.php first values of versioned settings
  database/
    migrations/           only platform-wide migrations; modules keep their own
    seeders/              ReferenceSeeder, SampleWorldSeeder
  resources/
    css/tokens.css        DESIGN.md tokens as custom properties
    js/                   Alpine setup
  routes/                 web.php, api.php, public.php
  lang/                   ms/ and en/ shared strings (module strings sit in the module)
  tests/
    Feature/<Module>/    Pest tests named by SRS key
    Architecture/         layer and dependency tests
    Contract/             adapter contract tests
    E2E/                  Playwright specs
  docs/                   SDD, SRS, UI/UX specification (read-only copies)
  deploy/                 Dockerfiles, Nginx and Horizon config, scripts
  .github or .gitlab-ci   pipeline definition
  deptrac.yaml  phpstan.neon  pint.json  composer.json  package.json
```

Each module registers itself through its own service provider, loaded from `config/app.php`, which loads its routes, migrations, views, language files and policies. A module can be read, tested and reasoned about on its own.

3.3 Naming and Coding Conventions

Table 10: *Conventions*

Topic	Rule
Style	PSR-12 enforced by Pint. Every PHP file starts with `declare(strict_types=1);`. Static analysis at level 8 must pass.
Classes	PascalCase. Models are singular (`Offering`). Tables are plural snake_case (`offerings`) and carry no module prefix, except that a table name that would clash gets the module name first.
Actions	One class per business operation, a verb phrase, with a single public `handle()` method: `VerifyResults`, `ConfirmCompletion`, `RecordSenateDecision`. An action runs one transaction, calls the policy, writes the audit row and raises events.
DTOs	Readonly classes ending in `Data` carry input to actions and output of contracts (`VerifyResultsData`). Controllers and Livewire components pass DTOs, not request objects. Contracts return DTOs, not models.
Enums	Every status and every closed list is a backed PHP enum in the owning module, with the exact words of the UI/UX specification (Section 4.8) in the label method, in both languages. Statuses are never stored as free text. Enum names are the stored values, in snake_case.
Events	Past tense, PascalCase, one fact: `ResultsVerified (EV-29)`, `CompletionConfirmed (EV-31)`, `CredentialIssued (EV-32)`. Jobs are verb phrases ending in `Job` only where a class of the same name would clash.
Policies	One per model, named `<Model>Policy`, with a method per action (`verify`, `revoke`). Methods check role, scope and separation of duties and return a reason string on refusal.
Identifiers	Internal primary keys are ULIDs; public codes and human numbers are separate columns (Section 5).

Topic	Rule
Migrations	One migration per change, in the owning module, named `YYYY_MM_DD_HHMMSS_create_<table>_table` or `..._add_<column>_to_<table>_table`. Never edit a migration that has run in staging; add a new one (NFR-32). Vendor-specific SQL only inside migrations, with a note on the MySQL fallback (NFR-30).
Routes	Staff routes under `/staff`, participant routes under `/me`, public routes at the root, API under `/api/v1`. Route names are `module.resource.action`.
Strings	No user-visible text in PHP or Blade; use `__('module::key')` (Section 3.5).
Business rules	Thresholds, lists and periods are versioned settings read through `Settings::get()`, never constants (C4, AD-12).
Tests	A Pest test is named after the requirement it proves: `it('FR-123 refuses to verify results of a held case')` and grouped with `->group('FR-123')`. Business rules carry the BR key the same way (NFR-29). Test files mirror the module folder.
Commits and branches	One topic per branch; the commit message starts with the module ID, for example `MD-07: verify results action`.

3.4 UI Component Library

The shared components live in MD-00 under `Platform/Resources/views/components` and `Platform/Livewire`, with anonymous Blade components used wherever no server logic is needed. They mirror the component catalogue of UI/UX specification Section 3.5 and the interaction rules of Section 4, and take their colours, type and spacing from `tokens.css` (DESIGN.md). Business modules use these components and write no component CSS of their own. The prototype's `core.js` shows the behaviour that each component must reproduce.

Table 11: *UI components and their Blade or Livewire names*

Component (UI spec)	Component name	Key props and behaviour
Shell (top bar, side navigation, content)	`<x-ui.shell>`	`nav` (menu from the user's roles), `title`, `breadcrumb`, `status`; slot `actions`. Top bar holds search (staff), language switch, help, bell and profile menu. Reflows below 760 px.
Navy side navigation	`<x-ui.side-nav>`	`groups[]` of items with `label`, `route`, `icon`, `count` (from live task data); highlights the current route. Colour #1e2a6e as in DESIGN.md.
Public site shell	`<x-ui.public-shell>`	Top navigation, sign-in link, footer; used by the catalogue and verification pages.
Maintenance banner	`<x-ui.maintenance-banner>`	Reads the announced maintenance setting; shown 48 hours before (NFR-15).
Buttons	`<x-ui.button>`	`variant` (primary, secondary, tertiary, ghost, danger), `disabled` with `reason` text shown on the button; one primary per screen.
Form fields	`<x-ui.field>`	`label`, `name`, `required`, `hint`, `error` (plain-language message under the field); wraps input, select, textarea, date.
Bilingual field	`<x-ui.bilingual-field>`	`name`; two inputs (ms, en) stored in two columns; marks a missing language (NFR-49).
Choice cards	`<x-ui.choice-card>`	`type` (radio, checkbox), `title`, `consequence` (one line).
Status tag	`<x-ui.status-tag>`	`:status` (a status enum); renders pale tint, the status word in the user's language and a short explanation on request (NFR-22); never colour alone (NFR-21).
Category mark	`<x-ui.category-mark>`	`category` (UMC, SAMC, MCE).
Data table	`<livewire:ui.data-table>`	`query` (scoped builder class), `columns[]` (label, key, sortable, align), `filters[]`, `searchable`, `rowRoute`, `perPage` (5, 10, 25, 50; default 10), `bulk` action. Search uses full-text search inside scope (AD-16); counts follow the filter; numbers right-aligned.
Tabs	`<x-ui.tabs>`	`tabs[]` with `label`, `count`; keyboard arrow navigation; selected tab kept in the query string.
Dialog	`<x-ui.dialog>`	`kicker`, `title` (question with count), slot body, `fields`, `confirmLabel` (a verb), `destructive` (danger style, red top edge, reason required, focus on Cancel). Traps focus, closes on Escape, returns focus to the opener (IR-06).
Drawer	`<x-ui.drawer>`	`side`, `title`; same focus rules as the dialog.
Toast	`<x-ui.toast>`	Raised by the `toast` browser event with `title`, `message`, `tone`; stays 5 seconds at the top right; a polite live region.
Menu	`<x-ui.menu>`	`items[]`; arrow keys, Escape; used for profile and row actions.
KPI strip	`<x-ui.kpi-strip>`	`items[]` of label and value; one bordered container with hairlines.
Stage track	`<x-ui.stage-track>`	`stages[]` and `current`; done stages in ink, current in navy, later grey; each stage has a text state for screen readers.

Component (UI spec)	Component name	Key props and behaviour
Notices	`<x-ui.notice>`	`tone` (info, warning, error, success); inline bar with a coloured left edge.
Checks list	`<x-ui.checks>`	`items[]` of state, statement and evidence; used for readiness, pre-issuance and eligibility checks.
Long-job progress	`<x-ui.job-progress>`	`job` identifier; polls the job record and shows queued, running with a count, done.
Empty and error blocks	`<x-ui.empty>`, `<x-ui.error-block>`	Empty: one sentence and first action. Error: message, reference code, Retry (NFR-19).
Printable document layout	`<x-doc.layout>`	`title`, `reference`, `lang`; A4 page, header, footer with page numbers and QR slot; used by the certificate, ACR, proposal and report templates and printed by Chromium (AD-10).

Every component is covered by a Blade component test and by an accessibility check in the end-to-end suite. A component that cannot be used with the keyboard alone is not accepted (NFR-21).

3.5 Internationalisation

Interface text is bilingual (AD-13, FR-08, IR-02).

- **Lang files.** Each module keeps `Resources/lang/ms/<module>.php` and `en/<module>.php`, loaded under the namespace of the module, for example `__('assessment::results.verify_title')`. Shared strings are in `lang/`. A test fails if a key exists in one language and not the other. Bahasa Melayu terms follow the Guide (NFR-46).
- **Locale per user.** The locale is stored in the user's profile and held in the session. The language switch posts to the same page and keeps entered data (NFR-25). The public site takes the locale from a query value or cookie, with Bahasa Melayu as the default.
- **Content columns.** Text that users see and that belongs to a record is stored in two columns, `title_ms` and `title_en`, read through a `Bilingual` cast that returns the user's language and falls back to the other (DR-10, NFR-49). Credentials, notices and documents are produced in the language chosen for them (FR-145, FR-266).
- **Formats.** Helpers in MD-00: `Fmt::date()` returns DD/MM/YYYY, `Fmt::time()` returns 24-hour Malaysia time, `Fmt::money()` returns "RM 450.00" (IR-25, IR-24, NFR-50). Dates are stored in UTC and displayed in `Asia/Kuala_Lumpur` (DR-08). Names are never shortened or reformatted (NFR-45).
- **Statuses.** Status labels come from the enum label method in both languages, so the same word is used on every screen.

3.6 Configuration and Feature Flags

Environment values (addresses, keys, adapter modes) are configuration, not code (NFR-31). Secrets come from the environment and are never committed. Business thresholds and lists are versioned settings in the database, not configuration (AD-12).

Table 12: *Configuration files and what they hold*

File or key	Holds	Changed by
`.env`	Database, Redis, MinIO, SMTP, Entra ID tenant and client, vault address; per environment	Operations
`config/integrations.php`	Mode (`stub`, `manual`, `live`) for each adapter, from `INTEGRATION_*` values	Operations
`config/mcms.php` `release`	Active release number: 1 now	Release process
`config/mcms.php` `features`	Switches: `live_ucreds`, `live_abc`, `live_qv`, `live_payment`, `sis_link`; all off in Release 1	Operations at release
`config/mcms.php` `demo_clock`	Allowed only when `APP_ENV` is not `production`	Operations
Settings (database)	Thresholds, lists, approving bodies, retention periods, idle timeout, notification texts, with effective dates (FR-20, FR-21)	Administrator in the interface

Rules for flags and non-production switches:

- A flag guards a whole capability (a route group, a menu item, an adapter binding). Code inside a flagged capability is written and tested in Release 1 where practicable, so that turning it on is a configuration change.
- A **Release 2 or 3 switch** is off by default. A route behind an off switch returns 404 and its menu item is not drawn.
- **Demo clock.** A `Clock` service in MD-00 returns the current time. In non-production a setting can move it so that expiry alerts and overdue reminders can be shown, as the prototype does. The service provider refuses to boot when `APP_ENV=production` and the demo clock or the stub identity provider is on (AD-06, AD-18).
- Code never calls `now()` directly for a business date; it uses `Clock::now()`, so tests and the demo clock work.

4. Module Design

This section designs the seventeen modules of MCMS (AD-01). Each module has the same sub-structure: a fact sheet, a table of actions, and, where the module owns a record with a life cycle, a state machine. Section 4.19 collects the domain events (EV-xx) and the jobs (JB-xx) in two catalogues. Section 6 lists the events again and Section 7 shows the main flows as sequences.

4.1 Conventions for Module Descriptions

The following conventions apply to every module in this section.

- **Fact sheet.** Purpose, the SRS requirements delivered, the tables owned (snake_case plural, columns defined in Section 5), the screens served by their UI/UX specification ID, the events published and consumed, and the jobs run.
- **Tables owned.** Only the owning module reads or writes its tables. Other modules call its public service interface or react to its domain events (AD-01).
- **Actions.** An action is one public service operation, called by a Livewire component or a controller. Every action runs under a Laravel Policy that checks role, scope and separation of duties on the record (AD-07). The policy column names the rule that is not obvious from the role alone.
- **Confirmation.** "Yes" means the interface shows a confirmation dialog before the action runs (IR-06). An action with "Yes" is irreversible, submits, approves, issues, revokes or changes a setting. The server does not rely on the dialog: it checks the same rules again.
- **Roles.** Student (P8), Lecturer (P1), Deputy dean of the PTj (P2), colleague reviewer (X1), industry expert (X2), public verifier (X3), CODL system administrator (S1) and CODL technical operator (P5). CODL roles never pass an academic policy (D-12). The Senate has no login: the deputy dean records each Senate decision with its meeting reference (D-11).
- **Audit.** Every action that changes a record writes an `audit_events` row in the same database transaction (AD-11). The tables in this section do not repeat this.
- **No deletion.** A correction creates a new revision linked to the old record. No module offers a delete action on a business record (D-20, DR-04).
- **Events.** A service publishes an event after its transaction commits (`afterCommit`). Listeners are queued on the `default` queue and are idempotent: each listener stores the event ID it has handled. Events carry record IDs and version numbers, not personal data. Event names are `PastTense` class names.
- **Jobs.** Every job has a queue, an idempotency key and a retry rule (Section 4.19). A job that exhausts its retries writes an `integration_jobs` row in the failed state and raises `AdapterCallFailed` or a module-specific failure event.
- **State machines.** Status words are those of UI/UX Specification v1.1 section 4.8. Each transition names the action that causes it. A state change is made only by the owning module's service.
- **Derived wording.** Credit, MQF level and QV wording is read from verified records at the time of use and is never typed or stored as free text (D-17, DR-05).

Table 13: *Summary of modules*

ID	Module	Namespace	SRS section	Key records	Key status vocabulary
MD-00	Platform	`Platform`	cross-cutting	settings, setting_versions, templates, stored_files	Setting version effective or superseded
MD-01	Identity	`Identity`	3.1	users, role_assignments	Account active or deactivated
MD-02	Curriculum	`Curriculum`	3.2	micro_credentials, mc_versions	Proposal or version (see MD-03)
MD-03	Approval	`Approval`	3.3	reviews, approval_decisions	Draft to Terminated
MD-04	Offering	`Offering`	3.4	offerings, readiness_items	Not ready to Cancelled
MD-05	Enrolment	`Enrolment`	3.5	enrolments, payments	Registered to Completed
MD-06	Learning platform	`LearningPlatform`	3.6	import_files, export_files	Import file accepted or rejected
MD-07	Assessment	`Assessment`	3.7	assessment_results	Provisional, Held, Verified
MD-08	Achievement record	`AchievementRecord`	3.8	achievement_records	Not started, In progress, Complete
MD-09	credentials	`Credentials`	3.9	credentials, credential_events	Issued to Revoked
MD-10	Quality verification	`QualityVerification`	3.10	qv_records, qv_evidence	Not applied to Withdrawn
MD-11	Credit bank	`CreditBank`	3.11, 3.13	acrs, consents, abc_deposits, credit_ledger_entries	ACR prepared to Deposited, Available to Expired
MD-12	Transfer	`Transfer`	3.12	transfer_applications, transfer_decisions	Submitted to Not approved
MD-13	Pathways	`Pathways`	3.14	pathways, stackable_plans	Suitable, Needs review, Not yet suitable
MD-14	Quality and change	`Monitoring`	3.15	cqi_records, cqi_actions, suspensions	Open to Closed

ID	Module	Namespace	SRS section	Key records	Key status vocabulary
MD-15	Records	`Records`	3.16	audit_events, corrections	Discrepancy open or resolved
MD-16	Integration hub	`Integration`	3.17	notifications, integration_jobs	Delivery sent or failed

4.2 MD-00 Platform

Table 14: *MD-00 fact sheet*

Item	Content
Purpose	The shared kernel. It holds versioned settings, bilingual templates, file storage, PDF rendering, global search, the UI component library and the demo clock. It contains no business rules of other modules.
SRS requirements	FR-20, FR-21, FR-22, FR-08, IR-07, IR-26, SR-09, NFR-02, NFR-31, NFR-48, NFR-49, DR-10
Tables owned	settings, setting_versions, templates, stored_files
Screens served	CMN-01 Personas, CMN-08 Help, CMN-09 Search, CMN-10 Something went wrong, CODL-04 Settings, CODL-05 Templates, CODL-06 Template, DOC-17 Documents and PDFs. The task queue DD-01 is assembled by the shell from the task providers that each module registers.
Events published	EV-05 SettingChanged
Events consumed	None
Jobs	JB-04 RenderPdfJob, JB-17 ScanUploadJob

Table 15: *MD-00 actions*

Action	Actor	Policy rule	Confirm
Change a setting	S1	Reason and effective date required. Writes a new `setting_versions` row and never updates the old one (FR-21). Decisions store the version they used (AD-12).	Yes
Publish a template version	S1	BM and EN bodies both required. A new `templates` row with the next version number is created, and old versions stay (FR-22).	Yes
Store a file	Any signed-in user in scope	Type and size checked, then JB-17 scans it. The file is unavailable until the scan passes (SR-09).	No
Get a signed file URL	Any user allowed to see the owning record	URL lives a few minutes and is bound to the user (AD-14).	No
Render a document	System	Only records in a final state are rendered. The template version and record revision are stored with the PDF (AD-10).	No
Search	Any signed-in user	Full-text query always joined to the caller's scope (AD-16, NFR-02).	No
Set demo clock	Tester	Available only when `APP_ENV` is not `production`. The application refuses to start in production with the clock on (AD-18).	No

State machine. The key record is a setting. It has no status field: the version in force at a date is the row with the latest effective date not after that date. A decision reads the version in force on the decision date.

4.3 MD-01 Identity

Table 16: *MD-01 fact sheet*

Item	Content
Purpose	Signs users in through Microsoft Entra ID, holds accounts, role assignments with scope, delegation and the reference data (PTj, programmes, parent courses). It is the source for every authorisation decision.
SRS requirements	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-09, FR-10, FR-11, FR-12, FR-13, FR-14, FR-15, FR-16, FR-17, FR-18, FR-19, FR-23, FR-24, SR-01, SR-02, SR-03, SR-04
Tables owned	users, role_assignments, delegations, conflict_declarations, ptjs, programmes, parent_courses
Screens served	CMN-02 Sign in, CMN-03 Verify it is you, CMN-04 Account not recognised, CMN-05 Test sign-in, CMN-06 My profile, CODL-01 Users and roles, CODL-02 User, CODL-03 Reference data
Events published	EV-01 RoleAssigned, EV-02 RoleRevoked, EV-03 ConflictDeclared, EV-04 ReferenceDataChanged
Events consumed	None
Jobs	None (reference data CSV import runs inside the request for files within the limit)

Table 17: *MD-01 actions*

Action	Actor	Policy rule	Confirm
Sign in (OIDC callback)	All users	The account is created on first sign-in with the Participant role only. There is no public sign-up. Staff roles need the MFA claim `amr` (AD-06).	No
Sign in through stub identity provider	Tester	Route exists only when `APP_ENV` is not `production` (FR-05).	No
Edit own profile	Any user	Language, notification preferences and phone only. Name and identifiers are read-only.	No
Assign or revoke a role	S1	Role, scope, reason and effective date required. S1 cannot assign itself an academic role, and S1 holds no approve, verify, issue, ACR or transfer action (FR-15).	Yes
Map decision type to approver role	S1	Stores `approval.route.*` through MD-00 settings. The mapped role must be an academic role, never an S1 or P5 role (D-12).	Yes
Deactivate or reactivate account	S1	Linked records stay. No delete action exists.	Yes
Merge two accounts (Release 2)	S1	Moves linked records to one account and writes an audit entry.	Yes
Declare conflict of interest	X1	Must be done before the review opens. The reviewer is removed from the task.	Yes
Delegate approver role for a period	Approver	Only users already eligible for the role, with start and end dates (FR-19).	Yes
Maintain reference data	S1	Form or CSV. An import validates all rows first, then applies.	Yes

State machine. A user account is Active or Deactivated, set only by the deactivate and reactivate actions. A role assignment is Assigned or Revoked. Neither record is deleted, so no diagram is drawn.

4.4 MD-02 Curriculum

Table 18: *MD-02 fact sheet*

Item	Content
Purpose	Determines the category (UMC, MCE, SAMC), holds the proposal workspace and the versions of a micro-credential, and checks the category rules before submission. It owns the version record and its status; MD-03 changes the status through the Curriculum service interface.
SRS requirements	FR-26, FR-27, FR-28, FR-29, FR-30, FR-32, FR-33, FR-34, FR-35, FR-36, FR-37, FR-38, FR-39, FR-40, FR-41, FR-42, FR-43, FR-44, FR-45, FR-46, FR-47, FR-48, FR-49, FR-50, FR-51, FR-52, FR-53, FR-54, FR-55, FR-56, FR-57, FR-58, FR-74, FR-75, FR-76, FR-77, FR-79, BR-01, BR-07, BR-08, BR-18
Tables owned	micro_credentials, mc_versions, mc_modules, learning_outcomes, assessment_components, proposal_collaborators, change_requests
Screens served	LEC-01 My micro-credentials, LEC-02 Micro-credential, LEC-03 Proposals and versions, LEC-04 New proposal, LEC-05 Proposal workspace, LEC-06 Request a change, DOC-07 Proposal print
Events published	EV-06 ProposalSubmitted, EV-07 MaterialChangeRecorded, EV-08 MinorChangeLogged, EV-15 VersionSuperseded
Events consumed	EV-04 ReferenceDataChanged (flags linked UMC versions when a parent course changes), EV-11 ProposalReturned, EV-13 ProposalNotSupported, EV-14 SenateDecisionRecorded, EV-59 SuspensionDecided (version status updates through the service interface)
Jobs	None

Table 19: *MD-02 actions*

Action	Actor	Policy rule	Confirm
Determine category	P1	Rules read from settings `slt.min_hours_samc`. UMC origin allows UMC only. Own offering below the threshold allows MCE only. The result is stored as proposed until P2 verifies it in MD-03.	No
Create proposal (New, Revision, Re-offer)	P1	Owning PTj must be inside the actor's role scope. A Revision or Re-offer starts from an Approved version.	No
Edit proposal sections	P1, co-developer of the same PTj	Allowed only while the version is Draft or Returned for amendment. Edits are attributed to the user. Autosave runs every edit.	No
Declare IP and copyright	P1	Required before submission.	Yes
Enter credit and MQF	P1	Stored as proposed credit. It is not shown in the catalogue or on credentials until a verified record confirms it (FR-42, D-17).	No
Submit proposal	P1	The validation service runs the category rules. Blocking errors stop submission. Warnings need a written response. UMC checks: accredited parent programme, module SLT sum equals parent SLT, minimum SLT, one model only. The content is locked on submit.	Yes

Action	Actor	Policy rule	Confirm
Reclassify MCE to SAMC	P1	Allowed only from MCE. Creates a new version in Draft (FR-33).	Yes
Set separately recognisable credit unit	P2 or academic authority role	Default No. Only a recorded case decision sets Yes. S1 cannot (FR-58, D-18).	Yes
Classify a change	P1	Listed triggers force material. Material creates a new version in Draft. Minor needs a description and impact note.	Yes
Print proposal	P1, P2	Queues JB-04.	No

State machine. The version status machine is shown in Section 4.5 (Figure 8), because the transitions are caused by approval actions. MD-02 itself changes status only for Draft to Submitted and for the fixed set of events it consumes.

4.5 MD-03 Approval

Table 20: *MD-03 fact sheet*

Item	Content
Purpose	Runs review, endorsement, category verification, the recording of Senate decisions, change decisions and version transition rules. Approval always binds to one exact version and is a separate record from Quality Verification, ACR or transfer (D-15).
SRS requirements	FR-60, FR-61, FR-62, FR-63, FR-64, FR-65, FR-66, FR-67, FR-68, FR-69, FR-70, FR-71, FR-72, FR-73, FR-31, FR-78, FR-80, FR-81, FR-82, FR-17, FR-18, FR-14, BR-16, BR-39, BR-42
Tables owned	reviews, review_assignments, review_comments, approval_decisions, version_transition_rules
Screens served	LEC-07 Reviews for colleagues, LEC-08 Review a proposal, EXT-01 My review invitations, EXT-02 Comment on a proposal, DD-02 Proposals and approval, DD-03 Proposal review, DD-04 Record Senate decision, DD-05 Decision pack, DOC-08 Senate decision pack
Events published	EV-09 ReviewerAssigned, EV-10 ReviewRecorded, EV-11 ProposalReturned, EV-12 ProposalEndorsed, EV-13 ProposalNotSupported, EV-14 SenateDecisionRecorded, EV-16 TransitionRuleRecorded
Events consumed	EV-03 ConflictDeclared (removes the reviewer and reopens the task), EV-06 ProposalSubmitted (creates the endorsement task for P2), EV-07 MaterialChangeRecorded (creates a new approval cycle), EV-08 MinorChangeLogged (creates an acknowledgement task for P2), EV-59 SuspensionDecided (records the decision)
Jobs	JB-10 DecisionOverdueRemindersJob

Table 21: *MD-03 actions*

Action	Actor	Policy rule	Confirm
Assign reviewers	P2	Reviewer must not be a co-developer (no self-review) and must have no declared conflict. Due date and scope required.	No
Submit review	X1, X2 (comment only)	X2 sees no decision controls. Rating per item is Satisfactory or Needs amendment, and the recommendation is Support, Amend or Not support.	Yes
Respond to a comment	P1	Comment status is open, answered or resolved. A blocking comment that is not resolved stops the decision.	No
Request amendment	P2	Moves the version to Returned for amendment and unlocks it.	Yes
Endorse and verify category	P2	Stores the verified category in `approval_decisions`, separate from the proposed category. Not possible while a blocking comment is open.	Yes
Record not supported	P2	Ends the cycle. Reason required.	Yes
Send to Senate	P2	Allowed only from Endorsed. Builds the decision pack.	No
Record Senate decision	P2	Decision, date, Senate meeting reference and comments required. The record names the P2 who entered it, not the Senate (D-11). The approver role comes from `approval.route.*`. S1 and P5 are refused (D-12).	Yes
Batch decision (Release 2)	P2	One meeting reference for many items. Each item is checked on its own.	Yes
Record transition rule	P2	Finish old version, move to new version, or equivalence. Required when a new version takes effect (FR-80).	Yes
Acknowledge minor change	P2	The change is logged to CQI through EV-08.	Yes
Record suspension or termination decision	P2 of the configured authority	Requires a transition plan before Terminated (FR-251). Performed in MD-14 and recorded here.	Yes

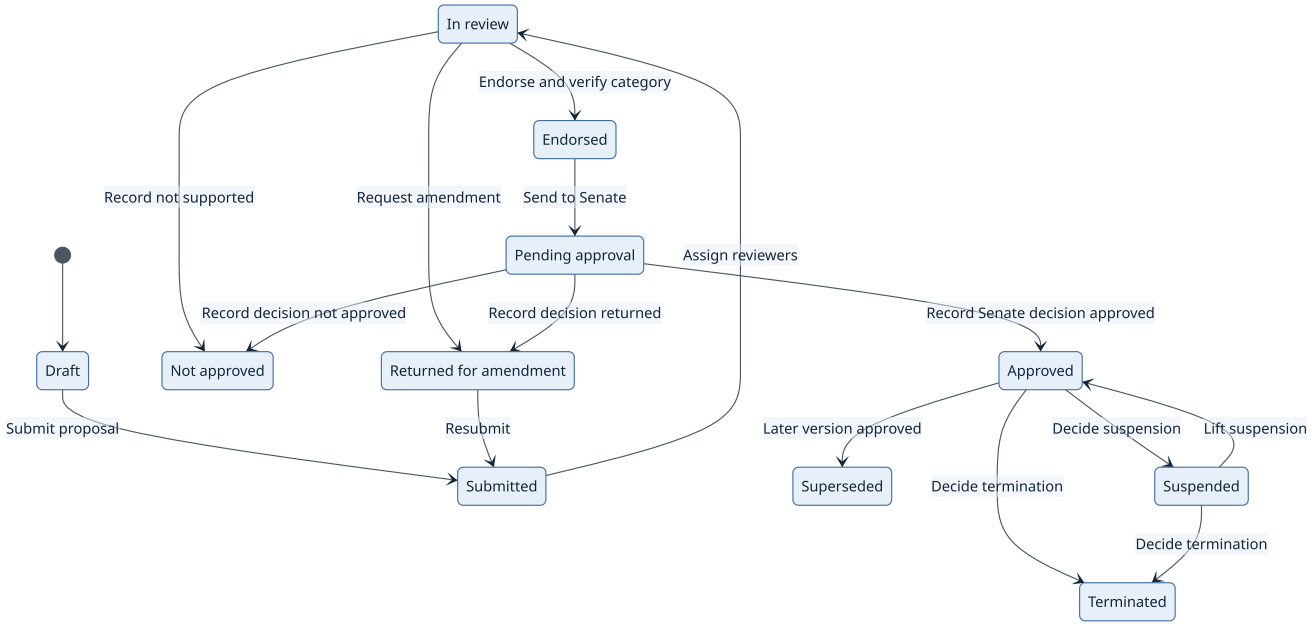


Figure 8: *Proposal or version status. Each arrow is caused by the action named on it.*

> **Note:** The SRS also names the end state "Not supported" for a PTj refusal. UI/UX Specification v1.1 section 4.8 has only "Not approved", so both refusals use that word and the stage (endorsement or Senate) is stored in `approval_decisions`.

4.6 MD-04 Offering

Table 22: *MD-04 fact sheet*

Item	Content
Purpose	Creates cohorts (offerings) of Approved versions, runs the readiness checklist and its confirmation, opens registration, and serves the public catalogue.
SRS requirements	FR-83, FR-84, FR-85, FR-86, FR-87, FR-88, FR-89, FR-90, FR-91, FR-92, FR-93, FR-94, FR-95, FR-96, FR-150, FR-151, BR-17, BR-22
Tables owned	offerings, readiness_items, readiness_confirmations
Screens served	LEC-11 Create an offering, LEC-12 Readiness checklist, DD-06 Offerings and readiness, DD-07 Confirm readiness, PUB-02 Catalogue, PUB-03 Offering, ST-02 Find a micro-credential
Events published	EV-17 ReadinessConfirmed, EV-18 OfferingOpened, EV-19 OfferingHeld
Events consumed	EV-14 SenateDecisionRecorded (makes the version selectable), EV-07 MaterialChangeRecorded (holds open offerings when the change needs approval), EV-39 QvVerified and EV-42 QvLapsed (recompute which QV, credit and MQF wording the catalogue may show), EV-59 SuspensionDecided (holds or cancels offerings)
Jobs	JB-21 OpenCloseRegistrationJob

Table 23: *MD-04 actions*

Action	Actor	Policy rule	Confirm
Create offering	P1, P2	Only an Approved version can be chosen. The caller's scope must include the owning PTj.	No
Answer readiness item	P1	Yes, Not yet or Not applicable with an evidence note for each of B2 to B7.	No
Confirm readiness	P2	Ready, Ready with conditions or Not ready. Conditions must be recorded. The first confirmation by the same user who answered the items is allowed, but the confirmation is stored as its own record (D-15).	Yes
Open registration	System	Runs when readiness is Ready, or Ready with conditions recorded, and the registration date is reached.	No
Recheck for a new cohort	P1, P2	Items are pre-filled. A material change puts the offering on hold until approved.	Yes
Cancel offering	P2	Enrolled participants are notified through EV-19.	Yes

Action	Actor	Policy rule	Confirm
Publish catalogue entry	System	Recognition statements come from templates. Free text is scanned for forbidden claims, and a hit blocks publication (FR-151). QV, credit and MQF show only while the QV record is valid to the cohort end date (D-17).	No

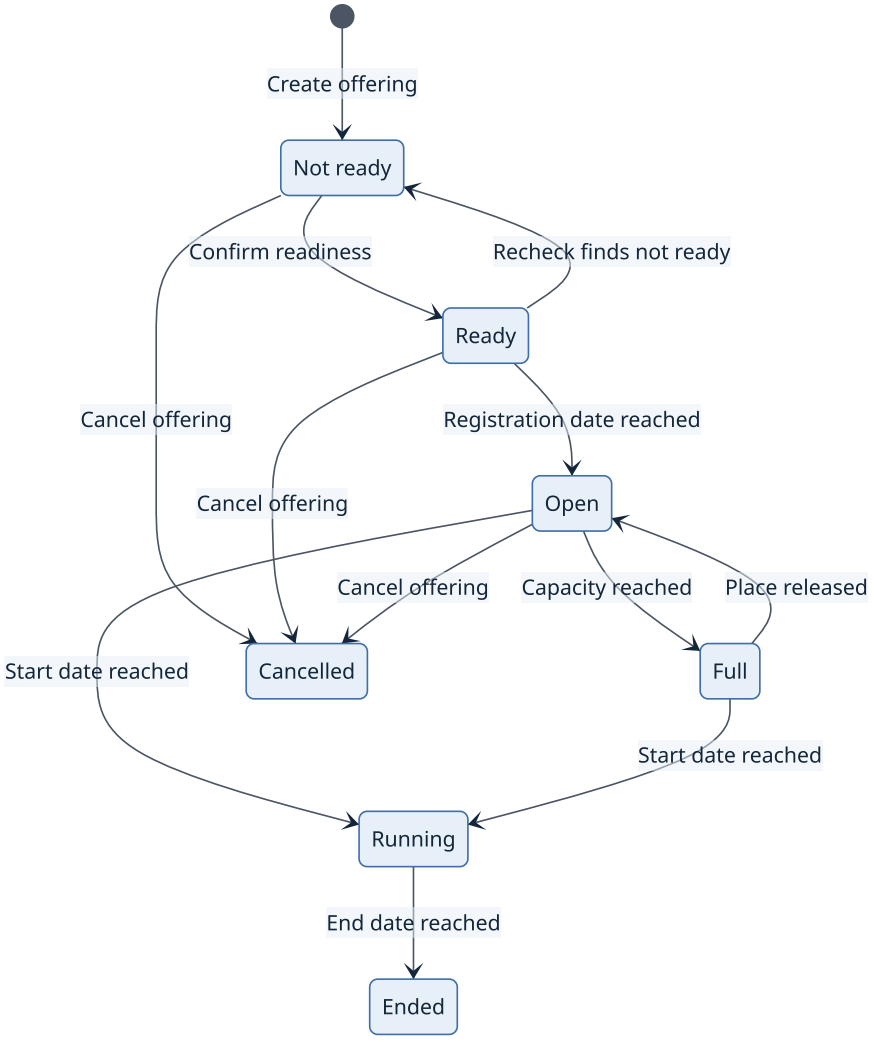


Figure 9: *Offering status. "Ready with conditions" is a value of the readiness confirmation, not a status.*

4.7 MD-05 Enrolment

Table 24: *MD-05 fact sheet*

Item	Content
Purpose	Registers participants, checks eligibility, takes payment through the payment adapter, tracks module enrolment, withdrawal and the cohort list.
SRS requirements	FR-97, FR-98, FR-99, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-109, FR-80, IR-17, NFR-20
Tables owned	enrolments, module_enrolments, payments, pdpa_acknowledgements
Screens served	ST-01 My learning, ST-03 Register, ST-04 Registrations and payments, ST-05 Registration, LEC-09 Cohorts and results, DOC-01 Payment receipt, DOC-02 Enrolment confirmation
Events published	EV-20 StudentRegistered, EV-21 PaymentConfirmed, EV-22 EnrolmentConfirmed, EV-23 EnrolmentWithdrawn
Events consumed	EV-16 TransitionRuleRecorded (rebinds enrolments to the new version and notifies), EV-31 CompletionConfirmed (sets Completed), EV-59 SuspensionDecided
Jobs	JB-20 CancelUnpaidEnrolmentsJob, JB-03 ExportUcredsEnrolmentJob (requested here, run by MD-06)

Table 25: *MD-05 actions*

Action	Actor	Policy rule	Confirm
Check eligibility	System	Group and prerequisite rules run before registration. Each failed rule gives its own reason.	No
Register	P8	Window open and cohort not full. The participant must acknowledge the recognition statements, and the text version is stored with the enrolment. PDPA notice is acknowledged on first registration.	Yes
Pay	P8, payment adapter	Payment reference is the idempotency key. Failure keeps the enrolment in Pending payment with a message and reference.	Yes
Waive fee or record sponsorship (Release 2)	P2	Reason required. Registered moves to Enrolled.	Yes
Withdraw	P8	Allowed only before the offering deadline and only from Enrolled. Refunds happen outside MCMS.	Yes
Cancel unpaid enrolment	System	Pending payment moves to Cancelled when the payment window ends.	No
Set Not completed	System	Runs when the cohort closes for an Enrolled participant without confirmed completion.	No
Register a group by CSV (Release 2)	P5	Per-row checks equal to single registration.	Yes
Export cohort list	P1, P2	Within scope. File type CSV or Excel.	No

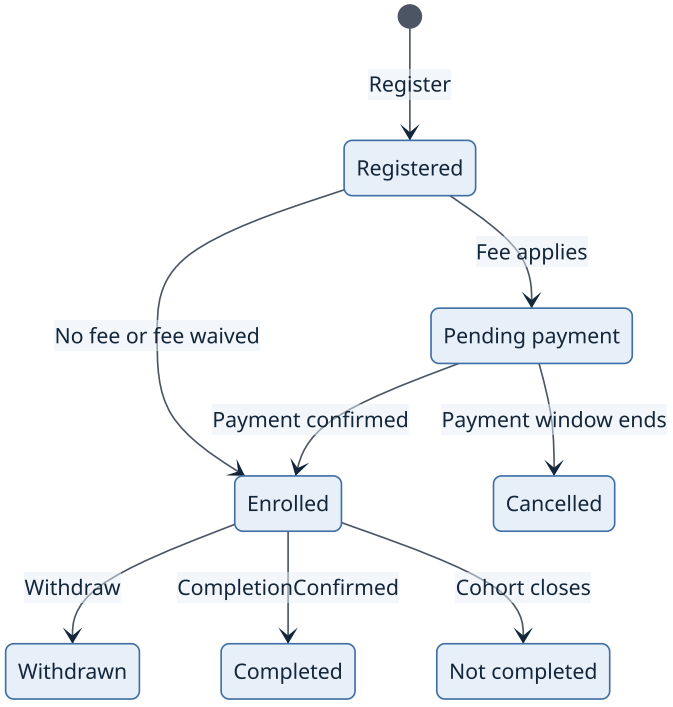


Figure 10: *Enrolment status. A transfer to a new version changes the version binding, not the status.*

4.8 MD-06 Learning Platform

Table 26: *MD-06 fact sheet*

Item	Content
Purpose	Exchanges enrolments and results with UCreds through the adapter interface (AD-08). In Release 1 the adapter is manual: MCMS exports an enrolment CSV and imports result and badge CSV files.
SRS requirements	FR-110, FR-111, FR-112, FR-113, FR-114, FR-115, FR-116, FR-117, IR-19, IR-14, NFR-07
Tables owned	import_files, import_row_errors, export_files
Screens served	CODL-07 UCreds exchange, DOC-14 Import validation report, DOC-15 UCreds enrolment export
Events published	EV-24 ImportExportLogged, EV-25 ResultsStaged, EV-26 ImportRejected, EV-27 BadgeImported
Events consumed	None
Jobs	JB-02 ImportResultsJob, JB-03 ExportUcredsEnrolmentJob

Table 27: *MD-06 actions*

Action	Actor	Policy rule	Confirm
Export enrolment CSV	P5	Uses the UCreds template version in settings. The file name, hash and count are logged.	No
Upload results, progress or badge file	P5	Template version is checked. The file is stored and hashed, then JB-02 validates it. A file already imported with the same hash is refused.	Yes
Validate and stage	System	Format, row match, score range and duplicates are checked. A file-level error rejects the whole file. Valid rows are handed to MD-07 as Provisional results. No achievement and no badge exists until P2 verifies.	No
Match imported badges	System	A badge is linked to a verified achievement. Unmatched badges are flagged for P5 and P2.	No
Download row error report	P5	PDF or CSV of rejected rows.	No

State machine. An import file is Received, then Accepted, Partly accepted or Rejected. These words are local to the integration log and do not appear in the participant interface, so no diagram is drawn. Results taken from the file follow the result states in Section 4.9.

4.9 MD-07 Assessment

Table 28: *MD-07 fact sheet*

Item	Content
Purpose	Holds the results of each participant and component, integrity cases, and the verification of results by the deputy dean. Verifying results is one action. Confirming completion is a different action in MD-08 (D-15).
SRS requirements	FR-119, FR-120, FR-121, FR-122, FR-123, FR-124, FR-125, FR-126, FR-127, FR-128, FR-114, BR-19, BR-20, BR-38
Tables owned	assessment_results, assessment_result_revisions, integrity_cases, result_change_requests, result_appeals
Screens served	LEC-10 Cohort results, DD-08 Results and completion, DD-09 Verify results
Events published	EV-28 IntegrityCaseOpened, EV-29 ResultsVerified, EV-30 ResultChangeApproved
Events consumed	EV-25 ResultsStaged (creates Provisional results and the verification task for P2), EV-22 EnrolmentConfirmed (creates empty result rows)
Jobs	None

Table 29: *MD-07 actions*

Action	Actor	Policy rule	Confirm
Enter results	P1	Per participant and component of the Approved version. Totals and pass or fail are computed against the pass standard of that version. Attendance alone cannot give a pass (FR-121).	No
Record integrity mechanism and flag a case	P1	A flagged result is Held and cannot be verified.	Yes
Resolve integrity case	P2	Reason required. Held returns to Provisional.	Yes
Verify results (single or bulk)	P2	The verifier is not the person who entered the result (separation of duties, FR-14). An open case on any selected result blocks it. Verifier, date and identity evidence link are stored. Verifying does not complete the micro-credential.	Yes
Request result change	P1	Reason required. Old value is kept in `assessment_result_revisions`.	Yes
Approve result change	P2	The approver is not the requester. If a credential was issued, EV-30 tells MD-09 to open a correction task.	Yes
Raise a query or appeal (Release 2)	P8	Opens a case linked to the result.	Yes
View own verified results	P8	Verified results only and released feedback only.	No

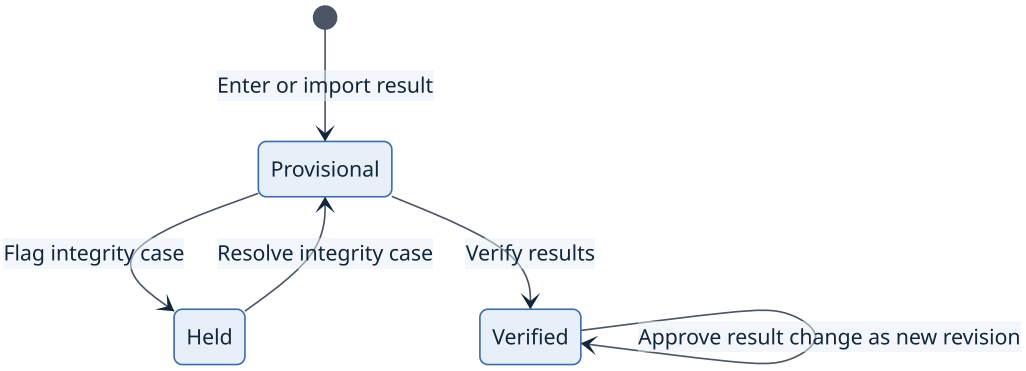


Figure 11: *Result status. "Achieved" is a module status in MD-08, derived from verified results.*

4.10 MD-08 Achievement Record

Table 30: *MD-08 fact sheet*

Item	Content
Purpose	Keeps one internal Achievement Record per participant per micro-credential. It derives module status from verified results and records completion only when the deputy dean confirms it. The record is labelled "Internal record, not ACR" on every screen and export.
SRS requirements	FR-129, FR-130, FR-131, FR-132, FR-133, FR-134, FR-135, FR-136, FR-126, BR-21, BR-25, BR-20
Tables owned	achievement_records, module_achievements
Screens served	ST-06 Achievement record, DD-10 Confirm completion, DOC-05 Achievement record
Events published	EV-31 CompletionConfirmed
Events consumed	EV-22 EnrolmentConfirmed (creates the record), EV-23 EnrolmentWithdrawn, EV-29 ResultsVerified (updates module status), EV-30 ResultChangeApproved (recomputes), EV-32 CredentialIssued (adds the credential to the record)
Jobs	JB-11 CompletionWindowRemindersJob

Table 31: *MD-08 actions*

Action	Actor	Policy rule	Confirm
Derive module achievement	System	A module is Achieved when all its outcomes and assessments are met and verified. Badges or modules of different UMCs are never combined (FR-132).	No
Show completion summary	System	Automatic check of every requirement, with the missing items listed. Complete is blocked until none is missing.	No
Confirm completion	P2	A separate action from verifying results (D-15). Only P2 of the owning PTj. Blocked if any requirement is missing. Stores the confirmer and date. It publishes EV-31, which is the only trigger for issuing a certificate or Digital Credential (D-29).	Yes
View own record	P8	Own record only, including progress across periods.	No
View records in offerings	P1, P2	Within scope.	No

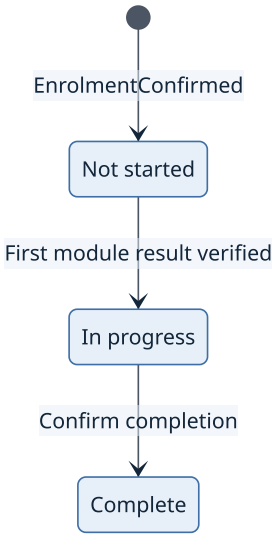


Figure 12: *Achievement record status. Module status Achieved is derived from verified results and does not complete the record.*

4.11 MD-09 Credentials

Table 32: *MD-09 fact sheet*

Item	Content
Purpose	Issues Digital Badges, certificates and Digital Credentials automatically after the pre-issuance checks pass, signs them, serves the wallet and the public verification page, and handles correction, replacement and revocation. The system issues; no person presses an issue button (D-29).
SRS requirements	FR-137, FR-138, FR-139, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, FR-148, FR-150, FR-151, FR-152, FR-153, FR-154, FR-155, FR-156, FR-157, FR-158, FR-159, FR-160, FR-161, FR-162, IR-20, IR-21, IR-13, SR-08, SR-13, SR-17, NFR-03, NFR-08
Tables owned	credentials, credential_events, digital_badges, verification_logs
Screens served	ST-07 Credentials, ST-08 Credential, DD-11 Credentials, DD-12 Credential, PUB-07 Verify a credential, PUB-08 Verification result, DOC-03 Certificate, DOC-04 Digital credential
Events published	EV-32 CredentialIssued, EV-33 BadgeIssued, EV-34 IssuanceFailed, EV-35 CredentialCorrected, EV-36 CredentialReplaced, EV-37 CredentialRevoked
Events consumed	EV-29 ResultsVerified (badge for a verified module, if the design includes badges), EV-31 CompletionConfirmed (certificate and Digital Credential), EV-27 BadgeImported (matches a UCreds badge), EV-30 ResultChangeApproved (opens a correction task), EV-42 QvLapsed (new credentials carry no QV claim)
Jobs	JB-01 IssueCredentialsBatchJob, JB-22 IssueCredentialJob, JB-04 RenderPdfJob

Table 33: *MD-09 actions*

Action	Actor	Policy rule	Confirm
Run pre-issuance checks	System	Checks (App E8) include: completion confirmed (for certificate and Digital Credential), results verified, template exists, no blocked claim, QV valid when QV is stated. Any failing check blocks and lists the check.	No
Issue credentials	System (JB-01)	Triggered only by EV-29 for badges or EV-31 for certificate and Digital Credential. Credit, MQF and QV wording is read from verified records and cannot be edited (D-17). Signs with Ed25519 through the key vault (AD-09). Template change after issue does not alter an issued payload.	No
Retry failed issue	P2	Re-queues the failed items listed to P2. The same idempotency key prevents a duplicate.	No
Correct a credential	P2 of the owning PTj	Confirmed error only, reason required. Creates a new revision. The original is kept (FR-158). S1 and P5 are refused (D-12).	Yes
Replace a credential	P2 of the owning PTj	New credential ID linked to the old one. The old one shows the new ID.	Yes
Revoke a credential	P2 of the owning PTj, or the authorised body in settings	Recorded decision and stated reason required. Revoked is final.	Yes

Action	Actor	Policy rule	Confirm
View and share own credential	P8	Own credentials only. JSON and PDF download, and a share link.	No
Verify publicly	X3	Code, link or QR. Only fields in `verification.public_fields` are shown. An unknown code gives a neutral message that does not reveal similar codes. Requests are rate-limited and logged by time, code, outcome and IP only.	No

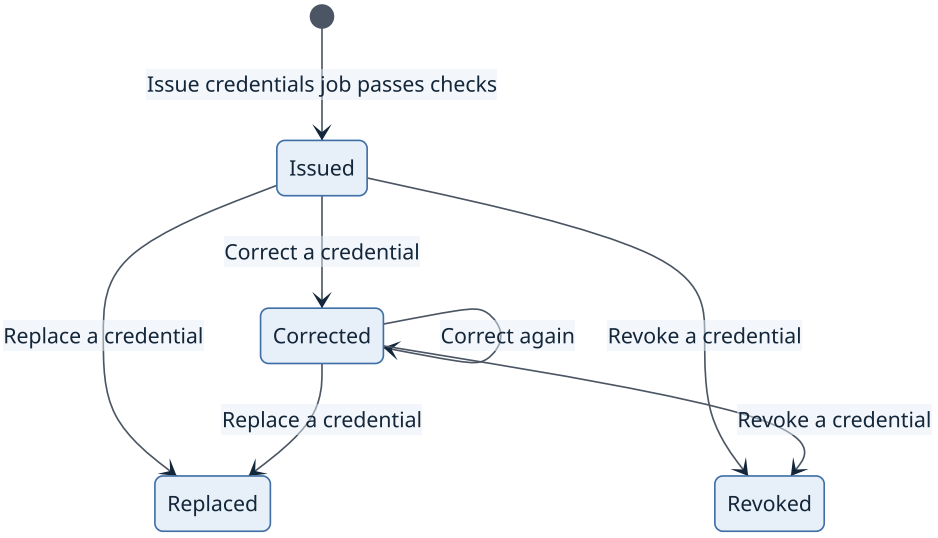


Figure 13: *Credential status. Replaced and Revoked are final. Nothing is deleted.*

4.12 MD-10 Quality Verification

Table 34: *MD-10 fact sheet*

Item	Content
Purpose	Manages the Quality Verification (QV) of a SAMC by QVC/MQA: eligibility, evidence checklist, internal readiness, submission package, recorded result, expiry and lapse. The QV record is the single source for credit, MQF and QV wording.
SRS requirements	FR-163, FR-164, FR-165, FR-166, FR-167, FR-168, FR-169, FR-170, FR-171, FR-172, FR-173, FR-174, FR-175, FR-176, BR-02, BR-03, BR-23, BR-04
Tables owned	qv_records, qv_evidence, qv_submissions
Screens served	LEC-15 QV evidence, LEC-16 QV evidence, DD-13 Quality, QV and review, DD-14 QV record, DOC-11 QV submission package
Events published	EV-38 QvSubmitted, EV-39 QvVerified, EV-40 QvNotVerified, EV-41 QvReverificationRequired, EV-42 QvLapsed
Events consumed	EV-07 MaterialChangeRecorded (starts re-verification for a Quality-Verified SAMC), EV-22 EnrolmentConfirmed and EV-31 CompletionConfirmed (count completed cohorts as evidence)
Jobs	JB-06 QvExpiryAlertsJob, JB-08 LapseExpiredQvJob, JB-14 BuildQvPackageJob

Table 35: *MD-10 actions*

Action	Actor	Policy rule	Confirm
Start preparing	P1	Only for a confirmed SAMC. Other categories do not see QV.	Yes
Set Ready to submit	P1	Blocked if not stand-alone, SLT below `slt.min_hours_samc`, provider not registered, fewer than `qv.min_completed_cohorts` completed cohorts with evidence, or unjustified proposed credit or MQF (D2 conditions).	Yes
Fill evidence checklist	P1	Available, Not yet or Not applicable with file reference. MCMS evidence is linked automatically.	No
Record readiness decision	P1 records, P2 confirms	Ready, Ready with minor corrections or Not ready. Critical missing evidence listed. The confirmer is not the recorder.	Yes
Generate submission package	Staff in scope	Evidence index, summary and compressed evidence.	No
Record submission	P2	Date, QVC/MQA reference and submitter. Requires Ready.	Yes

Action	Actor	Policy rule	Confirm
Record QV result	P2	Verified or not verified, reference, credit, MQF, start, expiry, conditions and document. Expiry after `qv.max_validity_years` from the start is blocked. The record becomes the source for catalogue, credentials and ACR.	Yes
Flag material change	P1	Component review. Sets the re-verification flag.	Yes
Lapse QV	System (JB-08) or P2	On expiry, suspension or withdrawal. Stops QV claims and new ACR eligibility, and creates a task for the PTj.	No

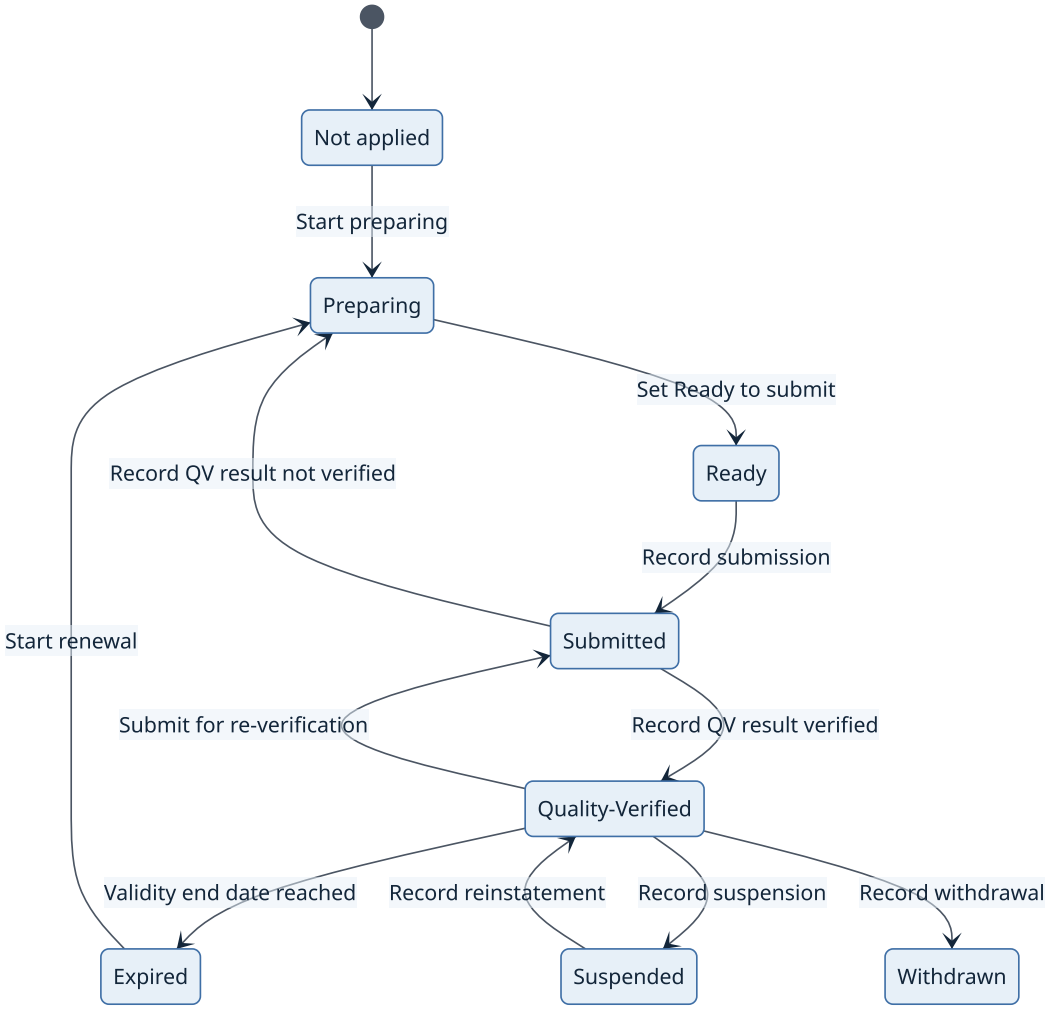


Figure 14: *QV status. Re-verification is a flag on a Quality-Verified record, and Not verified returns the record to Preparing.*

> **Note:** The SRS state diagram names "Not verified" and "Re-verification required" as states. UI/UX Specification v1.1 section 4.8 does not, so MCMS stores them as the recorded outcome and as a flag. This keeps the status words identical to the interface.

4.13 MD-11 Credit Bank

Table 36: *MD-11 fact sheet*

Item	Content
Purpose	Prepares the Academic Credit Record (ACR), asks for consent, records the ABC deposit, keeps the credit ledger and applies single redemption. An ACR exists only for an accredited UMC or a QV-valid SAMC, never for an MCE, badge, module or certificate.
SRS requirements	FR-177, FR-178, FR-179, FR-180, FR-181, FR-182, FR-183, FR-184, FR-185, FR-186, FR-187, FR-188, FR-189, FR-190, FR-191, FR-193, FR-194, FR-195, FR-196, FR-197, FR-213, FR-214, FR-215, FR-216, FR-217, FR-218, FR-219, FR-220, BR-24, BR-26, BR-27, BR-28, BR-29
Tables owned	acrs, acr_checks, consents, abc_deposits, credit_ledger_entries, redemption_events
Screens served	ST-09 Credit ledger and ABC, ST-10 Academic Credit Record, DD-18 ACR and ABC deposits, DD-19 Confirm ACR, DD-20 Record ABC deposit, DOC-06 Academic Credit Record, DOC-10 ABC deposit export

Item	Content
Events published	EV-43 AcrPrepared, EV-44 AcrConfirmed, EV-45 ConsentGiven, EV-46 ConsentDeclined, EV-47 DepositRecorded, EV-48 CreditExpired, EV-49 CreditRedeemed, EV-50 CreditReleased
Events consumed	EV-32 CredentialIssued (prepares an ACR for an eligible UMC or a QV-valid SAMC, and never for an MCE), EV-39 QvVerified and EV-42 QvLapsed (change SAMC eligibility), EV-52 TransferDecided (an approved decision sets the credit Provisionally redeemed), EV-60 CorrectionRecorded
Jobs	JB-07 CreditExpiryAlertsJob, JB-09 ExpireCreditsJob, JB-13 ExportAbcDepositJob

Table 37: *MD-11 actions*

Action	Actor	Policy rule	Confirm
Prepare ACR	System	Runs the F2 checks and records each result. A failing check blocks. Fields come from verified records and cannot be typed (D-17).	No
Confirm ACR	P2	Review of the generated record only. Not editable. S1 and P5 are refused (D-12).	Yes
Give consent	P8	Explicit consent stores date, wording version and channel. Without consent the ACR is not in the deposit queue.	Yes
Decline or postpone	P8	The ACR is kept and consent can be given later.	Yes
Export deposit data	P2	Only consented, confirmed ACR. Queues JB-13. Format open (OI-11).	No
Record deposit	P2	Date, ABC reference, credits, deposit status and recorder required. Sets the credit Available and the expiry to deposit date plus `credit.abc_validity_years` (10 years at the time of writing). The setting version used is stored (AD-12). Notices say that a deposit is storage and not a credit transfer.	Yes
Record programme completion	Authorised user in the receiving PTj	Evidence reference required. Provisionally redeemed moves to Redeemed. Manual in Release 1.	Yes
Record withdrawal or non-completion	Authorised user in the receiving PTj	Evidence reference required. Moves to Available if still valid, otherwise Expired.	Yes
Record whether ABC was updated	Authorised user	Date and reference.	Yes
Record external credit (Release 2)	Staff	Marked external, with provider, evidence and QV or accreditation.	Yes
Select credit for transfer	System	Only credit in Available. Redeemed credit is never selectable, and a second active application on the same credit is blocked (single redemption).	No

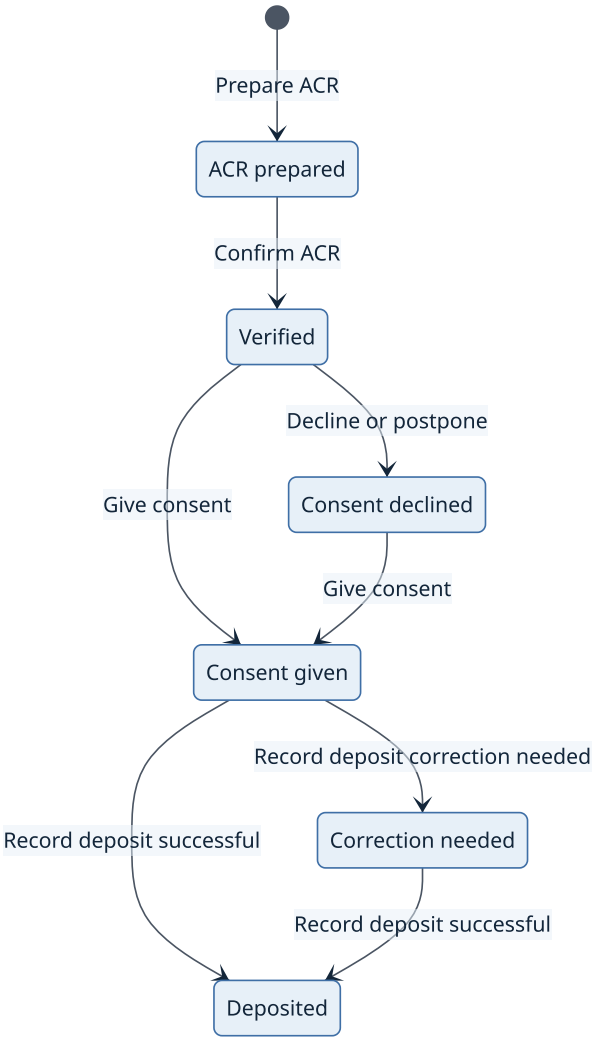


Figure 15: *ACR and deposit status. A deposit does not transfer credit.*

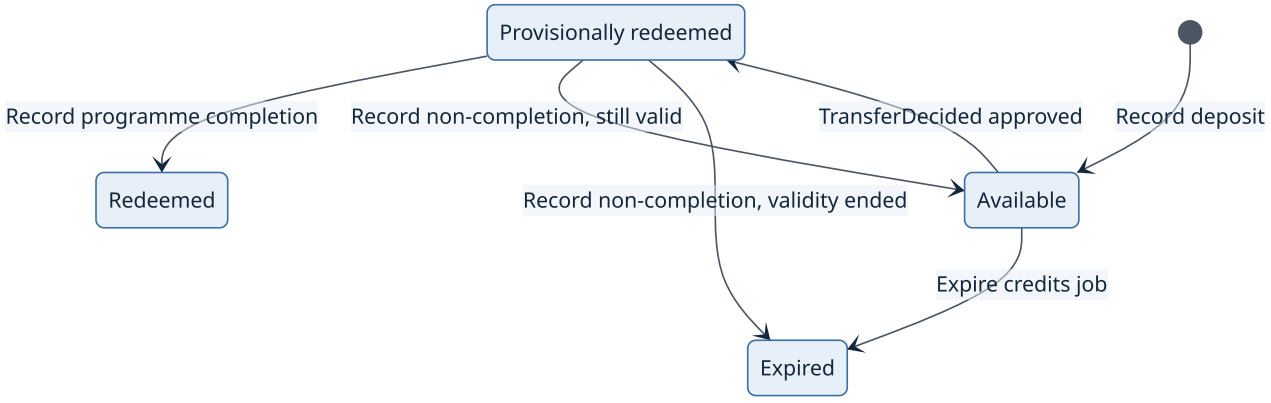


Figure 16: *Credit ledger status for a deposited credit. Redeemed is final.*

4.14 MD-12 Transfer

Table 38: *MD-12 fact sheet*

Item	Content
Purpose	Handles an application to transfer credit into a degree programme: validity check, equivalence evaluation on learning outcomes, limit warnings and the academic decision of the deputy dean of the receiving PTj. It also records programme admission as a separate fact.
SRS requirements	FR-198, FR-199, FR-200, FR-201, FR-202, FR-203, FR-204, FR-205, FR-206, FR-207, FR-208, FR-209, FR-210, FR-211, FR-212, FR-175, BR-30, BR-31, BR-32, BR-33, BR-34

Item	Content
Tables owned	transfer_applications, transfer_lines, equivalence_assessments, transfer_decisions, admissions
Screens served	ST-11 Credit transfer, ST-12 New transfer application, ST-13 Transfer application, DD-21 Credit transfer, DD-22 Transfer decision, DOC-12 Credit transfer decision letter
Events published	EV-51 TransferApplied, EV-52 TransferDecided, EV-53 AdmissionRecorded
Events consumed	EV-48 CreditExpired (flags an affected application for re-evaluation), EV-14 SenateDecisionRecorded (flags an earlier equivalence when a material change is approved)
Jobs	None

Table 39: *MD-12 actions*

Action	Actor	Policy rule	Confirm
Submit application	P8 or staff	Source must be a UTeM or other eligible UMC, or a QV-valid SAMC with verifiable status. An MCE is refused with the reason. Only Available credit is selectable. The conflict check against active applications on the same credit runs first.	Yes
Record validity check	P2 as evaluator	Authenticity, ABC status and QV or accreditation. A failure ends in Not approved.	Yes
Complete equivalence criteria	P2 of the receiving PTj	Each F7 criterion needs a finding and a result. The criterion on learning outcomes is judged on outcomes, not on credit or SLT alone.	Yes
Decide	P2 of the receiving PTj, or the authority in `approval.route.*`	Approved is blocked if any learning-outcome criterion is Not equivalent (FR-203). A partly completed UMC or a badge set is not full credit unless the separately recognisable flag is Yes. Programme limit and non-transferable components show as warnings. Credits, component, date, authority and conditions required.	Yes
Request more evidence	P2	Returns the application to Submitted.	Yes
Record admission	Authorised user of the programme	A separate fact. A transfer approval never implies admission.	Yes
Consult precedent (Release 2)	P2	A precedent does not replace the decision.	No

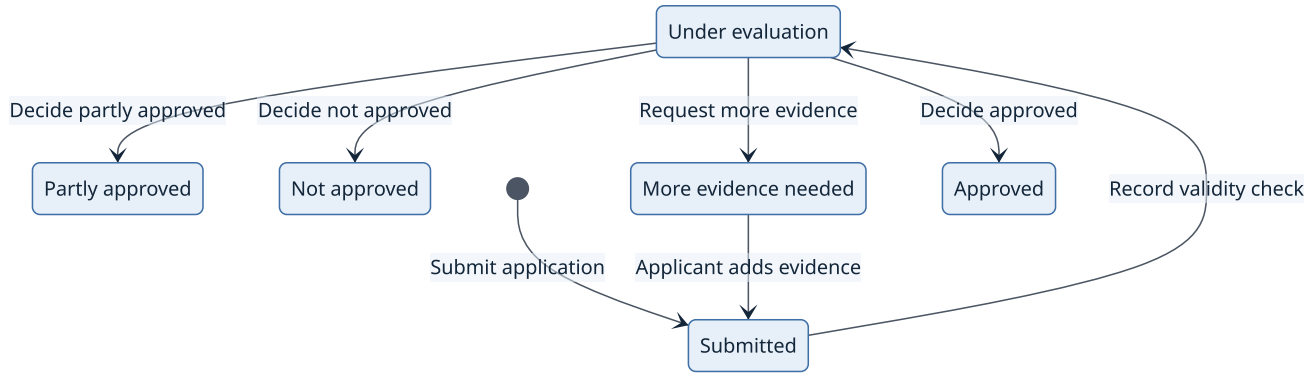


Figure 17: *Transfer application status. Only Approved and Partly approved publish an effect on credit (EV-52).*

4.15 MD-13 Pathways

Table 40: *MD-13 fact sheet*

Item	Content
Purpose	Defines stackable degree pathway mappings between credit-bearing micro-credentials and a degree programme, and keeps an individual stackable plan with the 30 percent UTeM credit and residency checks. A mapping transfers no credit; each credit still goes through MD-12.
SRS requirements	FR-221, FR-222, FR-223, FR-224, FR-225, FR-226, FR-227, FR-228, FR-229, FR-230, FR-231, FR-232, FR-233, FR-234, FR-235, BR-05, BR-06, BR-35, BR-36, BR-37
Tables owned	pathways, pathway_items, stackable_plans, plan_requirements
Screens served	ST-14 Stackable plan, DD-23 Stackable pathways, DD-24 Pathway mapping, DD-25 Individual stackable plan, PUB-04 Pathways to a degree
Events published	EV-54 PathwayPublished, EV-55 PathwayFlaggedForReview, EV-56 StackablePlanDecided
Events consumed	EV-04 ReferenceDataChanged (a material change to a target course flags the mapping), EV-53 AdmissionRecorded (allows Suitable to proceed), EV-14 SenateDecisionRecorded (records the Senate approval of a mapping)
Jobs	None

Table 41: *MD-13 actions*

Action	Actor	Policy rule	Confirm
Define and map pathway	P1, P2	Only credit-bearing credentials can be mapped. An MCE is blocked. Equivalence is High, Partial or None, and the preliminary result is Suitable, Needs review or Not suitable.	No
Approve and publish mapping	P2 records the Senate decision	Approval is required before publication (D-11). The page states that an approved mapping transfers no credit.	Yes
Create individual plan	P1, P2	Computes the share of credit earned at UTeM. A share below `stackable.min_home_credit_percent` marks the plan as not meeting the rule. The interface states that 30 percent does not mean 70 percent is transferable.	No
Record residency	P1, P2	At least `stackable.residency_semesters` for conventional delivery. An ODL scope flag goes to review.	No
Record plan decision	P1, P2	Suitable to proceed needs a recorded admission (FR-233). Conditions and follow-up recorded.	Yes
Record award	Authorised user from programme records	Needs admission and all requirements. MCMS records the award and never gives it.	Yes

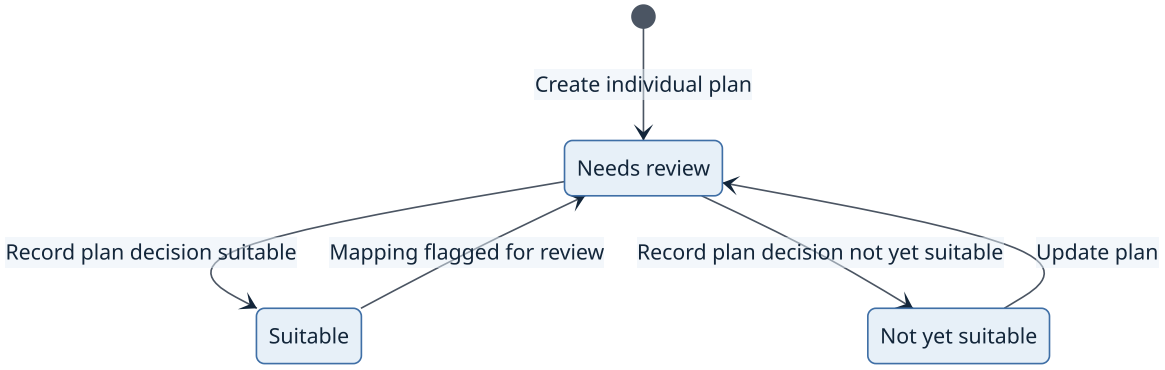


Figure 18: *Stackable plan status. A pathway mapping itself moves Draft, Approved, Published.*

> **Note:** The SRS plan decisions are Suitable to proceed, Suitable with conditions, Not yet suitable and Not suitable. UI/UX Specification v1.1 section 4.8 shows only Suitable, Needs review and Not yet suitable. The design stores the SRS decision in `stackable_plans.decision` and shows the section 4.8 words as status. The mapping is a design choice that needs confirmation (reported).

4.16 MD-14 Quality and Change

Table 42: *MD-14 fact sheet*

Item	Content
Purpose	Calculates cohort indicators, runs continuous quality improvement (CQI) records and actions, schedules periodic reviews, and records suspension or termination with a transition plan. It does not delete any earlier achievement, credential or ACR.
SRS requirements	FR-237, FR-238, FR-239, FR-240, FR-241, FR-242, FR-243, FR-244, FR-245, FR-246, FR-247, FR-248, FR-249, FR-250, FR-251, FR-252, FR-253, BR-41, BR-40
Tables owned	cqi_records, cqi_actions, periodic_reviews, suspensions, feedback_responses, quality_domain_checks
Screens served	LEC-13 Quality review (CQI), LEC-14 Quality review record, DD-13 Quality, QV and review, DD-15 Quality review, DD-16 Periodic review, DD-17 Suspension or termination, DOC-13 Suspension notice
Events published	EV-57 CqiRecordConfirmed, EV-58 PeriodicReviewDue, EV-59 SuspensionDecided
Events consumed	EV-08 MinorChangeLogged (adds the minor change to CQI), EV-31 CompletionConfirmed (updates cohort indicators), EV-42 QvLapsed (adds a QV item to the review list)
Jobs	JB-10 DecisionOverdueRemindersJob, JB-12 PeriodicReviewRemindersJob

Table 43: *MD-14 actions*

Action	Actor	Policy rule	Confirm
Submit feedback questionnaire	P8	One response per enrolment, after completion or cohort close.	Yes
Record quality domains	P1	Satisfactory or Needs action with evidence.	No

Action	Actor	Policy rule	Confirm
Record CQI issue and action	P1	Evidence, root cause, impact and priority (High, Medium, Low). Action has owner, due date and success indicator.	No
Evaluate effectiveness	P1	Effective, Partly effective or Not effective.	Yes
Close CQI record	P2	Needs a prior evaluation. The confirmer is not the evaluator. A CQI action that affects a listed component creates a change request in MD-02.	Yes
Record periodic review outcome	P1, P2	One of five outcomes. Routes to a change request or to a suspension decision.	Yes
Decide suspension or termination	P2 of the configured authority	Lists active participants. A transition plan is required before Terminated. Publishes EV-59, and affected participants are notified (mandatory notice).	Yes

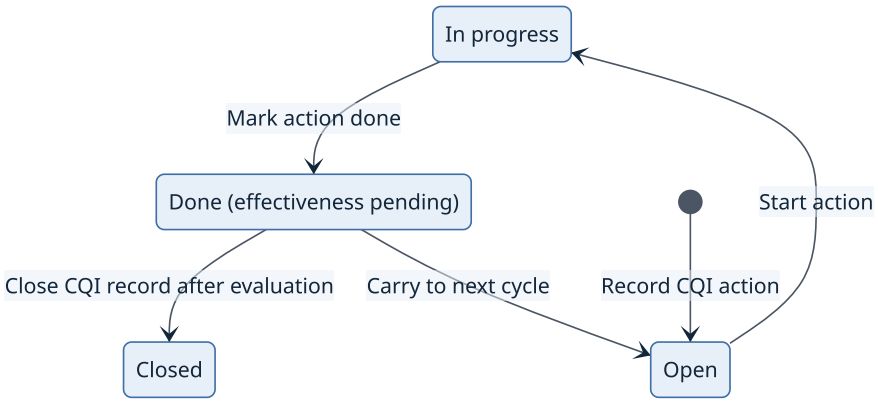


Figure 19: *CQI action status. Suspension and termination use the version statuses in Figure 8.*

4.17 MD-15 Records

Table 44: *MD-15 fact sheet*

Item	Content
Purpose	Provides the append-only audit trail, scoped viewing, corrections of records, discrepancy handling, the traceability chain, reports and exports, and retention with holds. Corrections create revisions and nothing is deleted.
SRS requirements	FR-254, FR-255, FR-256, FR-257, FR-258, FR-259, FR-260, FR-261, FR-262, FR-263, BR-38, SR-10, NFR-04, NFR-37, DR-04, DR-03
Tables owned	audit_events, corrections, discrepancies, retention_holds, disposal_records, report_runs
Screens served	DD-26 Reports, DD-27 Report, DD-28 Traceability, DD-29 Trace a participant, LEC-17 Reports, LEC-18 Report, CODL-10 Audit trail, CODL-11 Reports, CODL-12 Report, CODL-13 Retention, DOC-09 Report, DOC-16 Audit trail export
Events published	EV-60 CorrectionRecorded, EV-61 DisposalApproved
Events consumed	All events EV-01 to EV-63 as audit entries when the owning service did not already write one in its own transaction. EV-62 NotificationFailed. The audit write for a change is always made by the owning service in the same transaction (AD-11).
Jobs	JB-15 GenerateReportJob, JB-18 VerifyAuditChainJob, JB-19 ListDisposalCandidatesJob

Table 45: *MD-15 actions*

Action	Actor	Policy rule	Confirm
Append audit event	System	Hash chain: each row stores the previous hash. The application database role has no UPDATE or DELETE on the table (AD-11).	No
View audit trail	S1, P2, authorised users	Search within scope. CODL sees technical events but never academic content beyond its scope.	No
Record correction	Party with authority over the record (P2 for academic records)	Stores original, reason, action, approver and date. The target module applies the revision through its own service. Participants are notified if a credential, ACR, deposit or credit is affected.	Yes
Log and resolve discrepancy	P2	Open or resolved. Resolution is through a correction.	Yes
Trace a participant	P2, authorised users	Chain from source to award, within scope.	No

Action	Actor	Policy rule	Confirm
Run and export report	Authorised users in scope	CSV, Excel or PDF. Queues JB-15.	No
Set retention	S1	Setting change through MD-00.	Yes
Approve disposal	Approver in settings	Only after the period and when no hold exists. A disposal record is the only way a row leaves the system.	Yes
Place hold (Release 2)	Authorised user	Stops disposal.	Yes

State machine. A discrepancy is Open or Resolved, and a hold is On or Off. The audit event has no status because it is never changed.

4.18 MD-16 Integration Hub

Table 46: *MD-16 fact sheet*

Item	Content
Purpose	Stores and sends notifications, registers each adapter with its mode (Stub, Manual, Live), logs every external call, and retries failures without blocking the business process.
SRS requirements	FR-265, FR-266, FR-267, FR-268, FR-269, FR-270, FR-271, FR-25, FR-81, FR-252, IR-14, IR-15, IR-16, IR-17, IR-23, NFR-18, NFR-11
Tables owned	notifications, notification_preferences, integration_jobs, adapter_configs, job_runs, alert_log
Screens served	CMN-07 Notifications, CODL-08 Adapter status, CODL-09 Test mailbox
Events published	EV-62 NotificationFailed, EV-63 AdapterCallFailed
Events consumed	EV-03, EV-06, EV-09, EV-10, EV-11, EV-12, EV-13, EV-14, EV-16, EV-17, EV-18, EV-19, EV-20, EV-21, EV-22, EV-23, EV-26, EV-29, EV-30, EV-32, EV-34, EV-35, EV-36, EV-37, EV-38, EV-39, EV-40, EV-42, EV-43, EV-44, EV-45, EV-47, EV-48, EV-52, EV-55, EV-56, EV-58, EV-59, EV-60 (each creates a notification from a template, see 4.19). EV-63 (notifies P5).
Jobs	JB-05 SendNotificationJob, JB-16 RetryAdapterCallJob

Table 47: *MD-16 actions*

Action	Actor	Policy rule	Confirm
Create notification	System	Stored first in `notifications`, in the recipient's language, then sent by JB-05 (AD-15). Mandatory notices (version change, suspension, correction, replacement, revocation) cannot be switched off.	No
Set notification preference	Any user	Optional notices only. Mandatory notices are shown locked.	No
Register adapter and set mode	S1, P5	Stub, Manual or Live per environment. Stub and the demo clock are rejected in production (AD-18).	Yes
Call adapter	System	Every call writes an `integration_jobs` row with a stable idempotency key.	No
Resolve failed call	P5	Retry, mark handled or fix the data. P5 cannot change academic records.	No
Email digest (Release 2)	Approvers, lecturers	Daily digest of open tasks.	No

State machine. A notification is Pending, Sent or Failed. An `integration_jobs` row is Queued, Succeeded, Retrying or Failed. These are technical states shown only on CODL-08 and CODL-09.

4.19 Event and Job Catalogues

4.19.1 Domain events

The event IDs below are final for this document. Section 6 and Section 7 reuse them. Cross-module reactions that matter most are:

- EV-29 ResultsVerified lets MD-09 issue a Digital Badge for a verified module. It does not complete the micro-credential.
- EV-31 CompletionConfirmed starts JB-01 in MD-09, which runs the pre-issuance checks and issues the certificate and Digital Credential. No other event issues these (D-29).
- EV-32 CredentialIssued lets MD-11 prepare an ACR for an eligible UMC or a QV-valid SAMC. It never prepares one for an MCE.
- EV-45 ConsentGiven puts a confirmed ACR in the deposit queue. Without it no deposit can be recorded.
- EV-47 DepositRecorded sets the ledger credit to Available and the expiry to the deposit date plus the credit validity setting (10 years).

- EV-52 TransferDecided with an approved outcome sets the credit to Provisionally redeemed, so a second application cannot use it.
- EV-42 QvLapsed stops QV wording on new credentials and stops new ACR eligibility for the SAMC. Earlier credentials keep their status valid at the issue date.

Table 48: *Domain event catalogue*

ID	Event	Published by	Consumed by	Effect
EV-01	RoleAssigned	MD-01	MD-15, MD-16	Audit entry. User is notified of the role and scope.
EV-02	RoleRevoked	MD-01	MD-15, MD-16	Audit entry. Open tasks of the user are listed for reassignment.
EV-03	ConflictDeclared	MD-01	MD-03, MD-16	The reviewer is removed from the review task. The person who assigned the reviewer is notified.
EV-04	ReferenceDataChanged	MD-01	MD-02, MD-13	Linked UMC versions and pathway mappings are flagged for review when a parent or target course changes.
EV-05	SettingChanged	MD-00	MD-15	New setting version stored and audited. Later decisions use the version in force on their date.
EV-06	ProposalSubmitted	MD-02	MD-03, MD-16	Version locked. Endorsement task created for P2, who is notified.
EV-07	MaterialChangeRecorded	MD-02	MD-03, MD-04, MD-10	New version in Draft goes through approval. Open offerings of the old version are held. A Quality-Verified SAMC gets a re-verification task.
EV-08	MinorChangeLogged	MD-02	MD-03, MD-14	P2 gets an acknowledgement task. The change is added to CQI.
EV-09	ReviewerAssigned	MD-03	MD-16	Each reviewer is notified with the due date.
EV-10	ReviewRecorded	MD-03	MD-16	P2 is notified. The review summary is rebuilt.
EV-11	ProposalReturned	MD-03	MD-02, MD-16	Version returns to Returned for amendment and unlocks. P1 is notified.
EV-12	ProposalEndorsed	MD-03	MD-16	Category verified. The version moves to Pending approval when sent to Senate.
EV-13	ProposalNotSupported	MD-03	MD-02, MD-16	Version ends as Not approved. P1 is notified.
EV-14	SenateDecisionRecorded	MD-03	MD-02, MD-04, MD-12, MD-13, MD-16	Version becomes Approved, Not approved or Returned for amendment. An approved version becomes selectable for offerings. An approved material change flags earlier equivalences.
EV-15	VersionSuperseded	MD-02	MD-16	Earlier Approved version becomes Superseded. Enrolled participants are notified with the transition rule.
EV-16	TransitionRuleRecorded	MD-03	MD-05, MD-16	Enrolments are bound to the version by the rule. Participants are notified (mandatory).
EV-17	ReadinessConfirmed	MD-04	MD-16	Offering becomes Ready. Conditions are shown to P1.
EV-18	OfferingOpened	MD-04	MD-16	Offering is listed in the public catalogue and registration opens.
EV-19	OfferingHeld	MD-04	MD-16	Offering is held or cancelled. Registered participants are notified.
EV-20	StudentRegistered	MD-05	MD-16	Participant is notified. Payment is requested when a fee applies.
EV-21	PaymentConfirmed	MD-05	MD-05, MD-16	Pending payment becomes Enrolled. Receipt is rendered.
EV-22	EnrolmentConfirmed	MD-05	MD-07, MD-08, MD-10, MD-16	Achievement record and result rows are created. Confirmation with access instructions is sent.
EV-23	EnrolmentWithdrawn	MD-05	MD-08, MD-16	Record is closed as withdrawn. No credential is issued.
EV-24	ImportExportLogged	MD-06	MD-15	Export file, hash and count are logged.
EV-25	ResultsStaged	MD-06	MD-07, MD-16	Valid rows become Provisional results. A verification task is created for P2.
EV-26	ImportRejected	MD-06	MD-16	P5 is notified with the row error report. Nothing is staged for a file-level error.
EV-27	BadgeImported	MD-06	MD-09	Badge is matched to a verified achievement or flagged.
EV-28	IntegrityCaseOpened	MD-07	MD-16	Affected results become Held. P2 is notified.
EV-29	ResultsVerified	MD-07	MD-08, MD-09, MD-16	Achievements and module status are created. A badge is issued for a verified module if the design includes badges. Participants are notified.
EV-30	ResultChangeApproved	MD-07	MD-08, MD-09, MD-16	A new result revision applies. If a credential exists, a correction task is created for P2.
EV-31	CompletionConfirmed	MD-08	MD-05, MD-09, MD-10, MD-14, MD-16	Enrolment becomes Completed. JB-01 issues the certificate and Digital Credential after the pre-issuance checks.

ID	Event	Published by	Consumed by	Effect
EV-32	CredentialIssued	MD-09	MD-08, MD-11, MD-16	Credential is added to the Achievement Record. MD-11 prepares an ACR for an eligible UMC or QV-valid SAMC, never for an MCE. Participant is notified.
EV-33	BadgeIssued	MD-09	MD-08	Badge is added to the Achievement Record. It carries no credit wording.
EV-34	IssuanceFailed	MD-09	MD-16	P2 is told which participants failed and which check failed.
EV-35	CredentialCorrected	MD-09	MD-16	New revision is current. Participant is notified (mandatory).
EV-36	CredentialReplaced	MD-09	MD-16	Old credential shows the new ID. Participant is notified (mandatory).
EV-37	CredentialRevoked	MD-09	MD-16	Credential shows Revoked on verification. Participant is notified (mandatory).
EV-38	QvSubmitted	MD-10	MD-16	Status becomes Submitted. P1 and P2 are notified.
EV-39	QvVerified	MD-10	MD-04, MD-11, MD-16	QV record becomes the source for QV, credit and MQF wording. SAMC becomes ACR-eligible.
EV-40	QvNotVerified	MD-10	MD-16	Record returns to Preparing. Findings are listed to P1.
EV-41	QvReverificationRequired	MD-10	MD-16	Re-verification flag set after a material change.
EV-42	QvLapsed	MD-10	MD-04, MD-09, MD-11, MD-14, MD-16	Catalogue and new credentials drop QV wording. No new ACR for the SAMC. A task is created for the PTj.
EV-43	AcrPrepared	MD-11	MD-16	P2 is notified to confirm.
EV-44	AcrConfirmed	MD-11	MD-16	Participant gets the consent request.
EV-45	ConsentGiven	MD-11	MD-11, MD-16	ACR enters the deposit queue.
EV-46	ConsentDeclined	MD-11	MD-16	ACR stays unqueued. Participant can consent later.
EV-47	DepositRecorded	MD-11	MD-15, MD-16	Ledger credit becomes Available with expiry at deposit date plus 10 years. Participant is notified that a deposit is not a transfer.
EV-48	CreditExpired	MD-11	MD-12, MD-16	Credit becomes Expired. An affected application is flagged for re-evaluation.
EV-49	CreditRedeemed	MD-11	MD-15, MD-16	Credit becomes Redeemed and is final.
EV-50	CreditReleased	MD-11	MD-16	Credit returns to Available, or becomes Expired if validity ended.
EV-51	TransferApplied	MD-12	MD-16	Application is under evaluation. Evaluator is notified.
EV-52	TransferDecided	MD-12	MD-11, MD-16	An approved outcome sets the credit Provisionally redeemed. Not approved changes nothing in ACR, credentials or credit.
EV-53	AdmissionRecorded	MD-12	MD-13	Admission fact is available to the stackable plan.
EV-54	PathwayPublished	MD-13	MD-16	Mapping is shown on the public pathway page.
EV-55	PathwayFlaggedForReview	MD-13	MD-16	P1 and P2 are notified of a changed target curriculum.
EV-56	StackablePlanDecided	MD-13	MD-16	Plan status is updated. Participant is notified.
EV-57	CqiRecordConfirmed	MD-14	MD-15	CQI record is confirmed and audited.
EV-58	PeriodicReviewDue	MD-14	MD-16	Lecturer and PTj are reminded.
EV-59	SuspensionDecided	MD-14	MD-02, MD-03, MD-04, MD-05, MD-16	Version becomes Suspended or Terminated. Offerings are held. Affected participants are notified (mandatory). Credentials and ACR are untouched.
EV-60	CorrectionRecorded	MD-15	MD-09, MD-11, MD-16	The target module applies the revision. The participant is notified when a credential, ACR, deposit or credit is affected.
EV-61	DisposalApproved	MD-15	MD-15	Disposal record is written and rows are removed only after the approved disposal process.
EV-62	NotificationFailed	MD-16	MD-15	Failure is logged after the last retry.
EV-63	AdapterCallFailed	MD-16	MD-16	P5 is notified. The failed call appears in the failed-call queue.

4.19.2 Jobs

Queues are named in AD-05: `default`, `notifications`, `imports`, `credentials`, `reports`, `integration` and `default`. Scheduled jobs are started by the Laravel scheduler and dispatch to a queue. Long jobs report progress in `job_runs`. Alert jobs record each sent alert in `alert_log` so a re-run sends nothing twice. The idempotency key is stored on the job record, and a second run with the same key does nothing. Retry delays grow at each attempt (backoff).

Table 49: *Job catalogue*

ID	Job	Triggered by	Queue	Idempotency key	Retry
JB-01	IssueCredentialsBatchJob (MD-09)	EV-31 CompletionConfirmed for certificate and Digital Credential, EV-29 ResultsVerified for badges	credentials	enrolment ID, version ID, credential type	3 attempts with backoff 1, 5 and 15 minutes. A failed item raises EV-34 and stays listed to P2. Other items continue.
JB-02	ImportResultsJob (MD-06)	P5 upload	imports	SHA-256 of the file plus template version	No retry on validation errors. 2 retries on storage errors. Whole-file rejection leaves no staged row.
JB-03	ExportUcredsEnrolmentJob (MD-06)	P5 request	imports	cohort ID plus export timestamp rounded to the minute	3 attempts with backoff.
JB-04	RenderPdfJob (MD-00)	Any document request	reports	document type, record ID, revision, locale	3 attempts with backoff. Failure shows a retry action to the user.
JB-05	SendNotificationJob (MD-16)	Notification row created	notifications	notification ID	Back-off 1, 5, 15 and 60 minutes, then Failed and EV-62.
JB-06	QvExpiryAlertsJob (MD-10)	Scheduler, daily 07:00	default	QV record ID and threshold (`alerts.qv_expiry_days`: 180, 90, 30)	Re-run safe. A missed day is picked up next run.
JB-07	CreditExpiryAlertsJob (MD-11)	Scheduler, daily 07:00	default	deposit ID and threshold (`alerts.credit_expiry_days`: 365, 90)	Re-run safe.
JB-08	LapseExpiredQvJob (MD-10)	Scheduler, daily 00:10	default	QV record ID and expiry date	Re-run safe. Raises EV-42 once.
JB-09	ExpireCreditsJob (MD-11)	Scheduler, daily 00:20	default	ledger entry ID and expiry date	Re-run safe. Raises EV-48 once.
JB-10	DecisionOverdueRemindersJob (MD-14)	Scheduler, daily 07:30	default	task ID and reminder date	Re-run safe. Covers decisions older than `alerts.decision_overdue_days`, reviews due and CQI actions due.
JB-11	CompletionWindowRemindersJob (MD-08)	Scheduler, daily 07:40	default	record ID and reminder date	Re-run safe. Reads the offering's completion period, else the setting `umc.completion_window_months`; sends nothing while neither has a value (OI-13).
JB-12	PeriodicReviewRemindersJob (MD-14)	Scheduler, daily 07:50	default	review ID and due date	Re-run safe. Raises EV-58.
JB-13	ExportAbcDepositJob (MD-11)	P2 request	reports	batch ID	3 attempts with backoff.
JB-14	BuildQvPackageJob (MD-10)	Staff request	reports	QV record ID and evidence version	3 attempts with backoff.
JB-15	GenerateReportJob (MD-15)	User request	default	report type, scope, filters hash	2 attempts. The user can start it again.
JB-16	RetryAdapterCallJob (MD-16)	A failed adapter call	integration	integration job ID	Back-off 1, 5, 15 and 60 minutes, then Failed and EV-63. It never blocks the business process.
JB-17	ScanUploadJob (MD-00)	File stored	default	file ID	3 attempts. If the scanner is down the file stays unavailable.
JB-18	VerifyAuditChainJob (MD-15)	Scheduler, daily 02:00	default	check date	Re-run safe. A broken chain raises a security alert to CODL.
JB-19	ListDisposalCandidatesJob (MD-15)	Scheduler, monthly	default	month	Re-run safe. It only lists candidates. Disposal needs EV-61 approval.
JB-20	CancelUnpaidEnrolmentsJob (MD-05)	Scheduler, hourly	default	enrolment ID and window end	Re-run safe. Pending payment becomes Cancelled.
JB-21	OpenCloseRegistrationJob (MD-04)	Scheduler, every 15 minutes	default	offering ID and window date	Re-run safe. Opens a Ready offering on its date and closes it on its deadline.

ID	Job	Triggered by	Queue	Idempotency key	Retry
JB-22	IssueCredentialJob (MD-09)	Child of JB-01, one per credential	credentials	enrolment ID, version ID, credential type	3 attempts with backoff. Failure raises EV-34.

5. Data Design

This section turns the conceptual model of SRS Section 6 into the physical database of Release 1. It follows the data integrity rules DR-01 to DR-10: each recognition stage has its own table, approved records are never updated in place, nothing is physically deleted outside the disposal process, and relationships are enforced by the database (AD-03, AD-01). Every table is owned by one module (MD-xx). Another module reads or changes that table only through the owner's service interface or domain events (AD-01), even though the database holds foreign keys across modules.

5.1 Conventions

Table 50: *Database conventions*

Topic	Convention
Names	Tables are plural snake_case (`mc_versions`). Columns are snake_case. Foreign keys are `{table_singular}_id`. Enumeration columns carry the enum name in Section 5.3.
Primary key	`id`, a ULID stored as `char(26)`. Generated in the application when the model is created, so that sort order follows creation time and no sequence leaks a count.
Timestamps	`timestamptz`, stored in UTC and shown in Asia/Kuala_Lumpur (DR-08, IR-24). Every table has `created_at`. Mutable tables also have `updated_at`. Append-only tables have no `updated_at`.
Dates	Calendar dates without a time (`start_date`, `issued_on`) use `date`. They are exchanged as ISO 8601 (IR-25).
Status	A string column holding the stored value of a PHP backed enum (Section 5.3), with a database CHECK constraint listing the allowed values. A status column is always named `status` unless the table holds two stages (for example `deposit_status` and `credit_status`), as DR-01 requires.
Money	`numeric(10,2)` in Ringgit, cast to a decimal string in PHP. No floating point type is used for money or credit.
Foreign keys	All relationships are declared as foreign keys with `ON DELETE RESTRICT` (DR-09). Cascading deletes are not used. The only soft references (no foreign key) are `subject_id` on `audit_events`, `integration_jobs`, `job_runs` and `alert_log`, because those rows must outlive the records they describe.
Bilingual text	Text that participants see is stored as a pair of columns `x_ms` and `x_en` (Section 5.8).
Soft delete	Not used. A record that is no longer valid gets a status, an end date or a successor row (Section 5.7).
Version reference	A table that belongs to one version of a micro-credential holds `mc_version_id` itself, even when it could be reached through a parent (DR-02).
Derived values	Credit value, MQF level and QV status shown on another record are copied only with the id of the source record (`source_*_id`), so they can be traced (DR-05).
Framework tables	Laravel's own tables (`migrations`, `failed_jobs`, `job_batches`, `cache`) are not listed in the catalogue. Sessions and queues use Redis (AD-05).

5.2 Entity-Relationship Diagrams

The model is drawn in seven areas. Each diagram shows the tables of one area, their relationships and a few key columns. The full column lists are in the catalogues of Section 5.4. A relationship that crosses areas is drawn once, where the child table is owned. Table names follow the shared list used by the module design in Section 4.

5.2.1 Identity and Settings

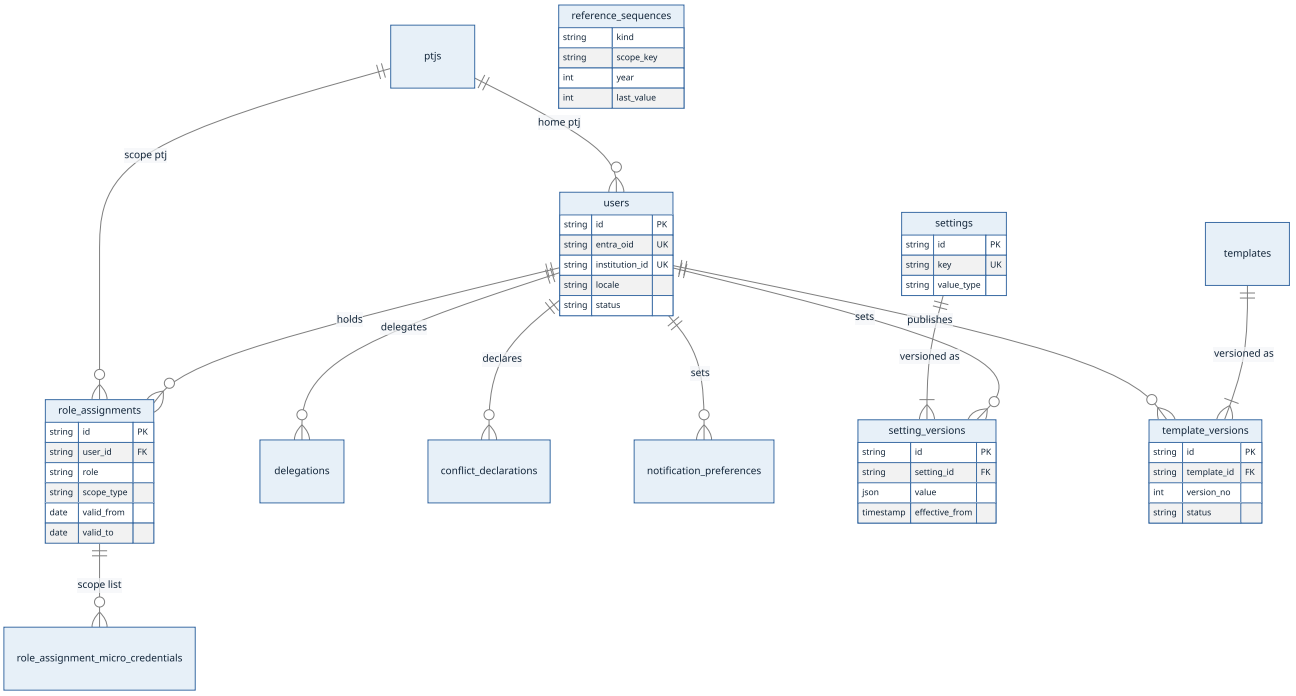


Figure 20: *Identity, role scope, settings and templates.* `reference_sequences` *stands alone: it is read only by the number generator.*

5.2.2 Curriculum and Approval

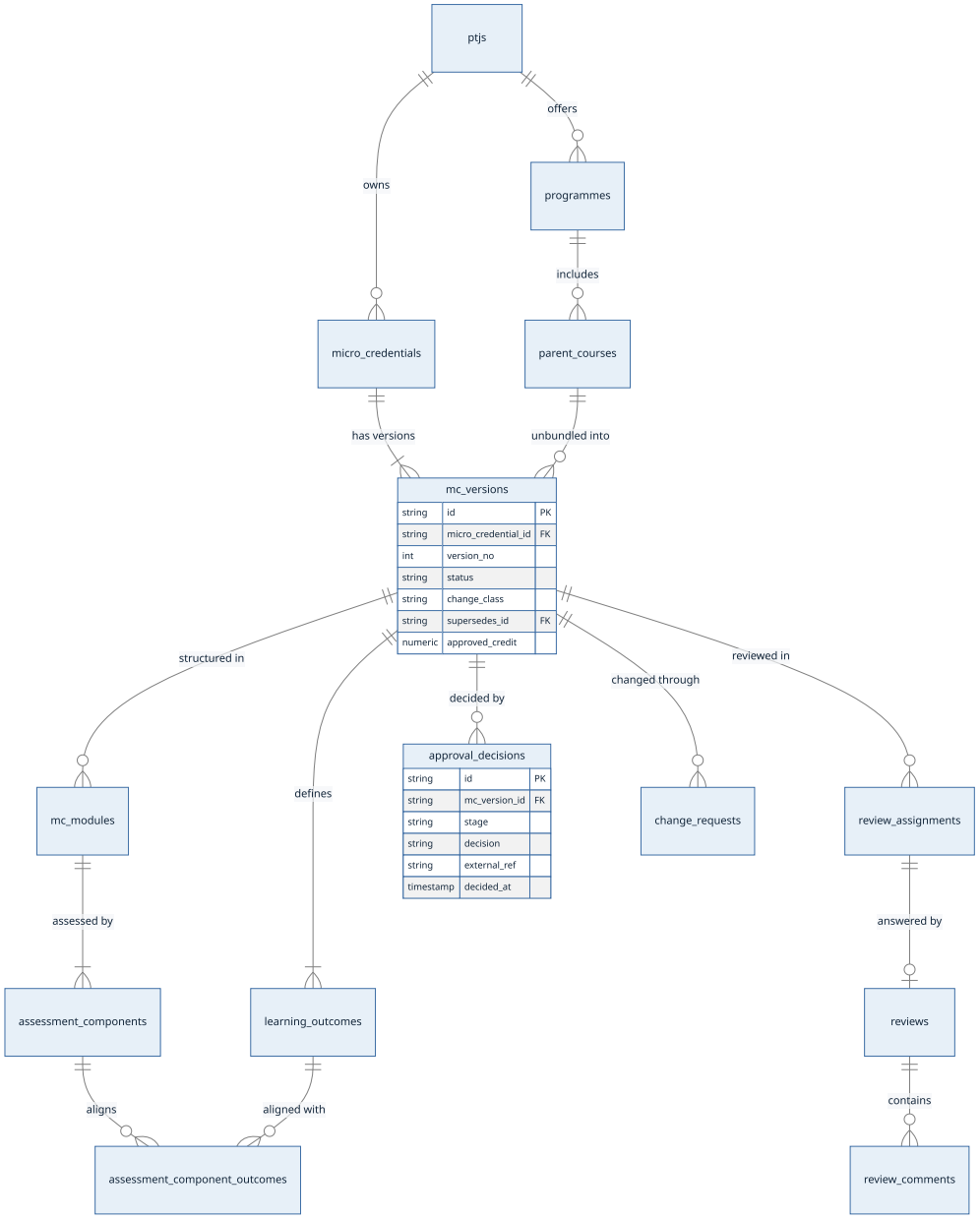


Figure 21: *Micro-credentials, immutable versions, learning design, reviews and approval decisions.* `proposal_collaborators` *and* `version_transition_rules` *also point to* `mc_versions` *and are left out to keep the figure small.*

5.2.3 Offering, Enrolment and Results

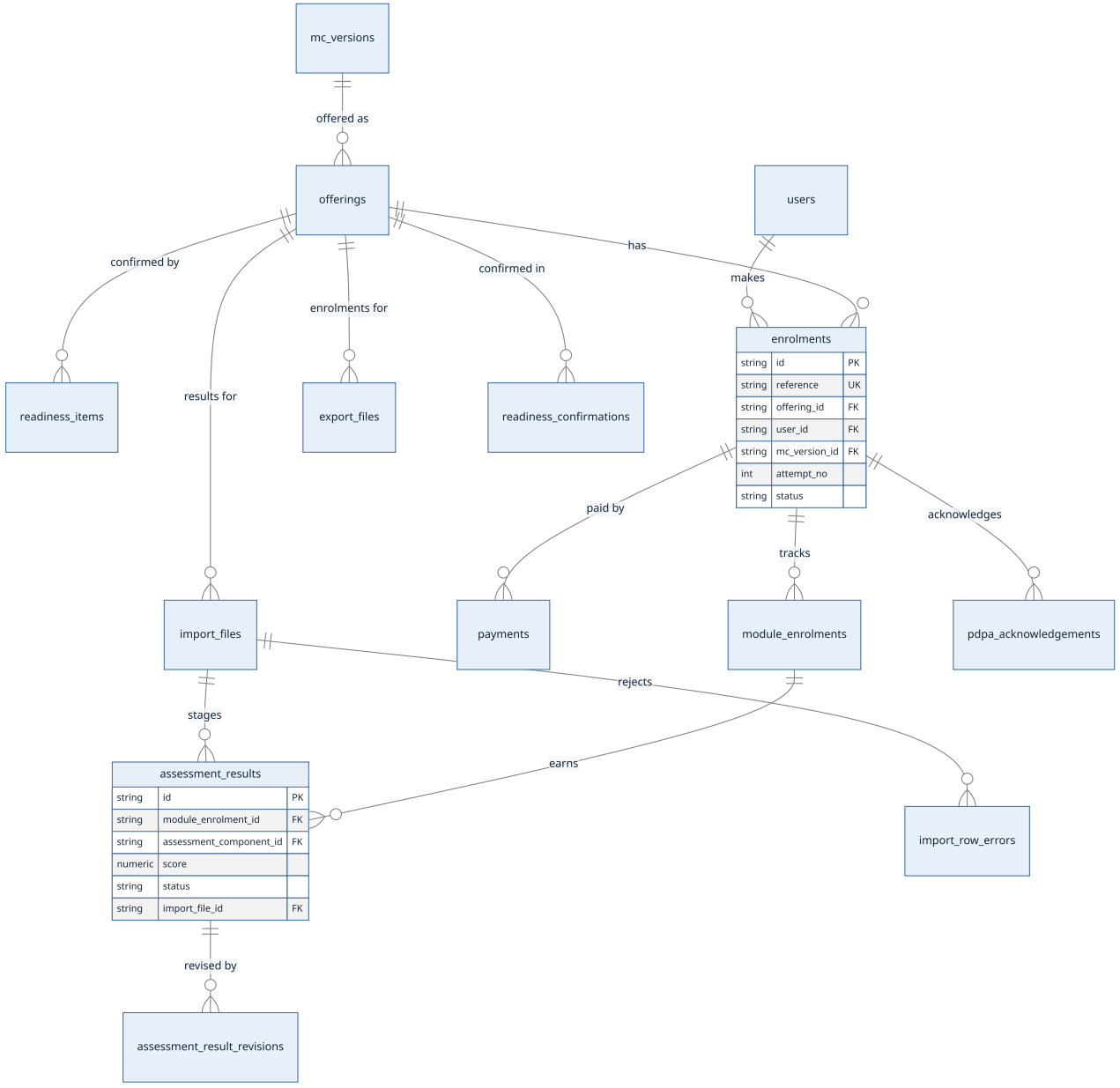


Figure 22: *Offerings, registration, payment, UCreds exchange and results.*

5.2.4 Credentials and Achievement

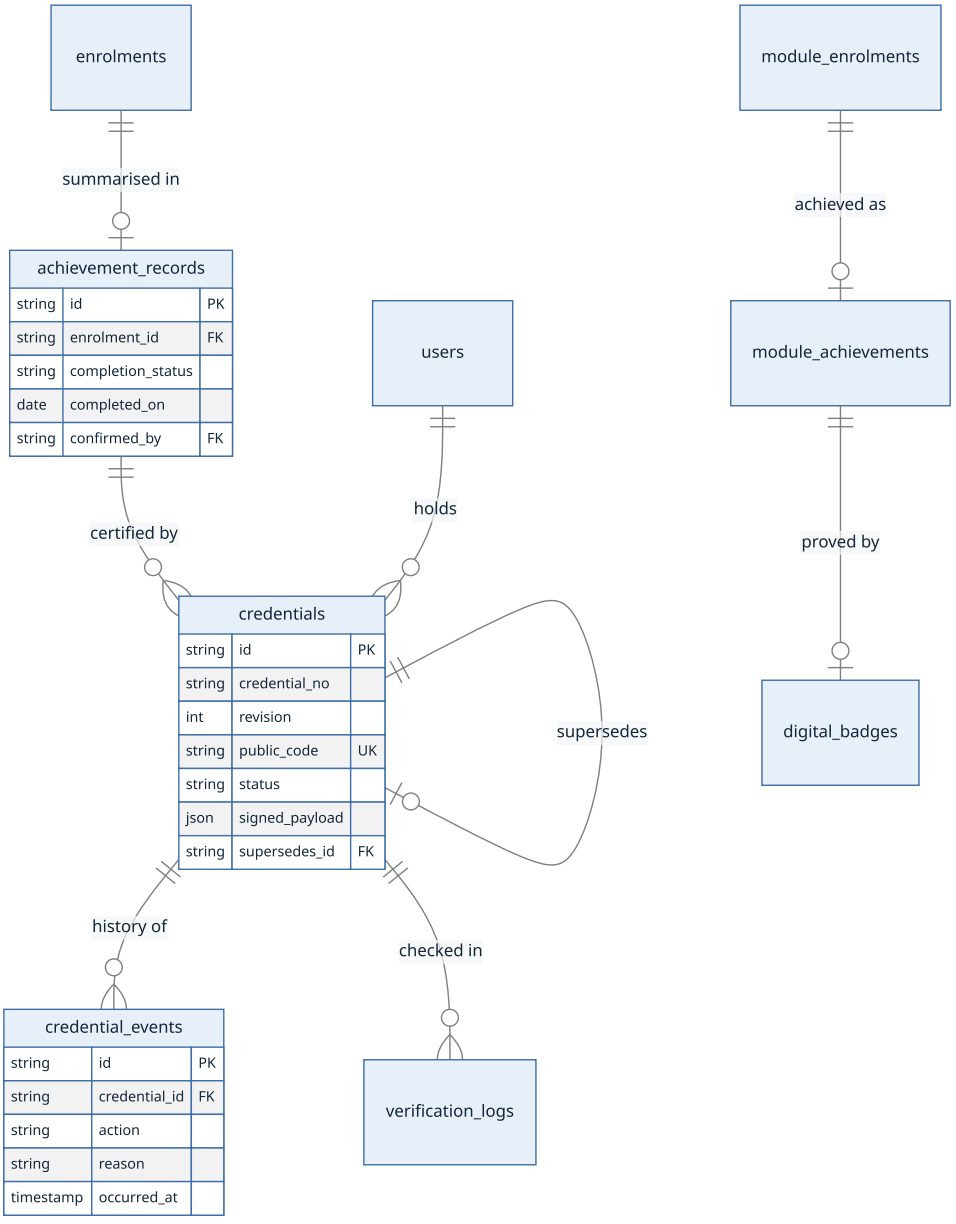


Figure 23: *The Achievement Record, module badges, certificates and credential history. A credential row is never edited after issue.*

5.2.5 Quality Verification and Quality Assurance

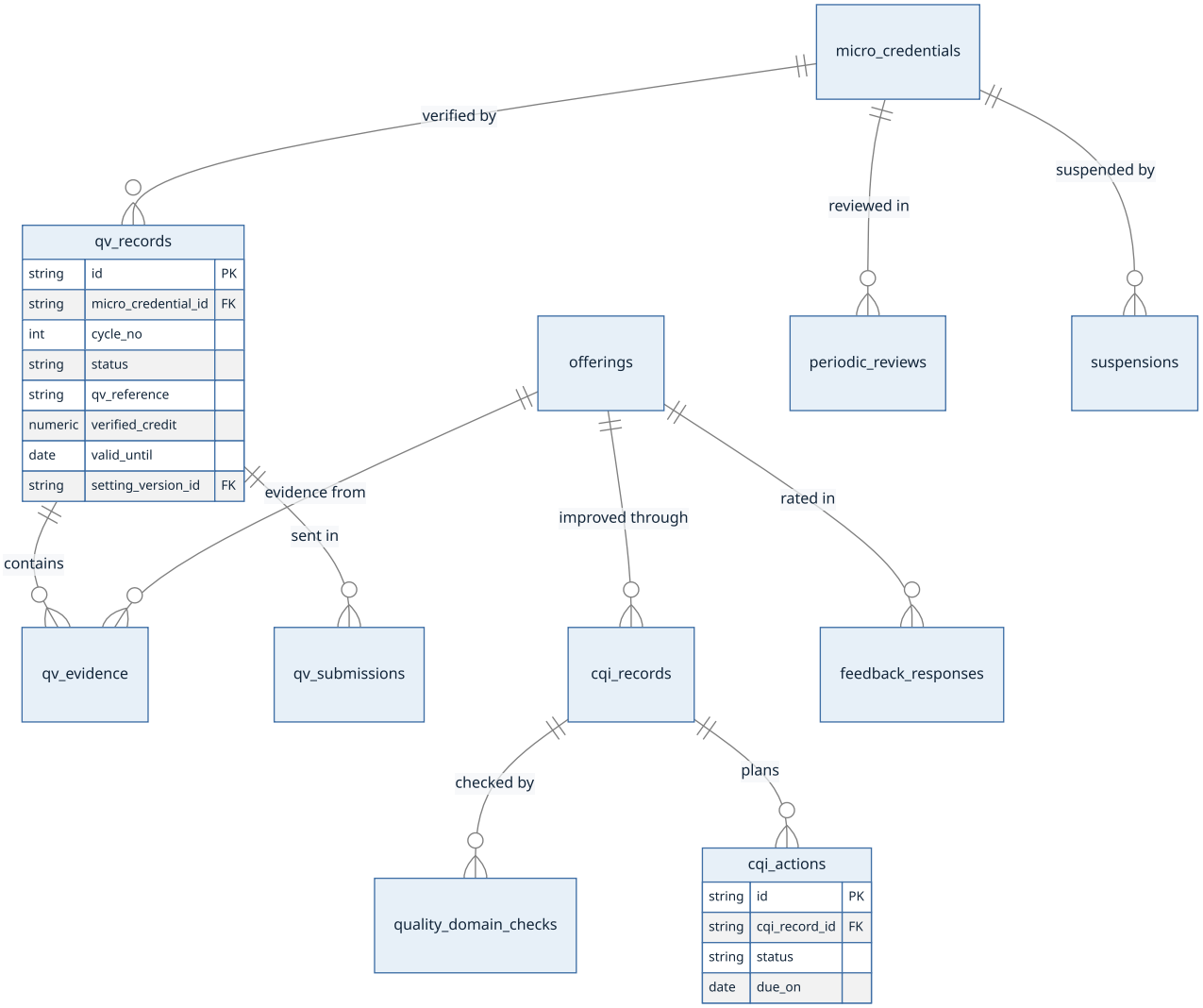


Figure 24: *QV cycles and evidence, CQI, periodic review and suspension.*

5.2.6 Credit Bank, Transfer and Pathways

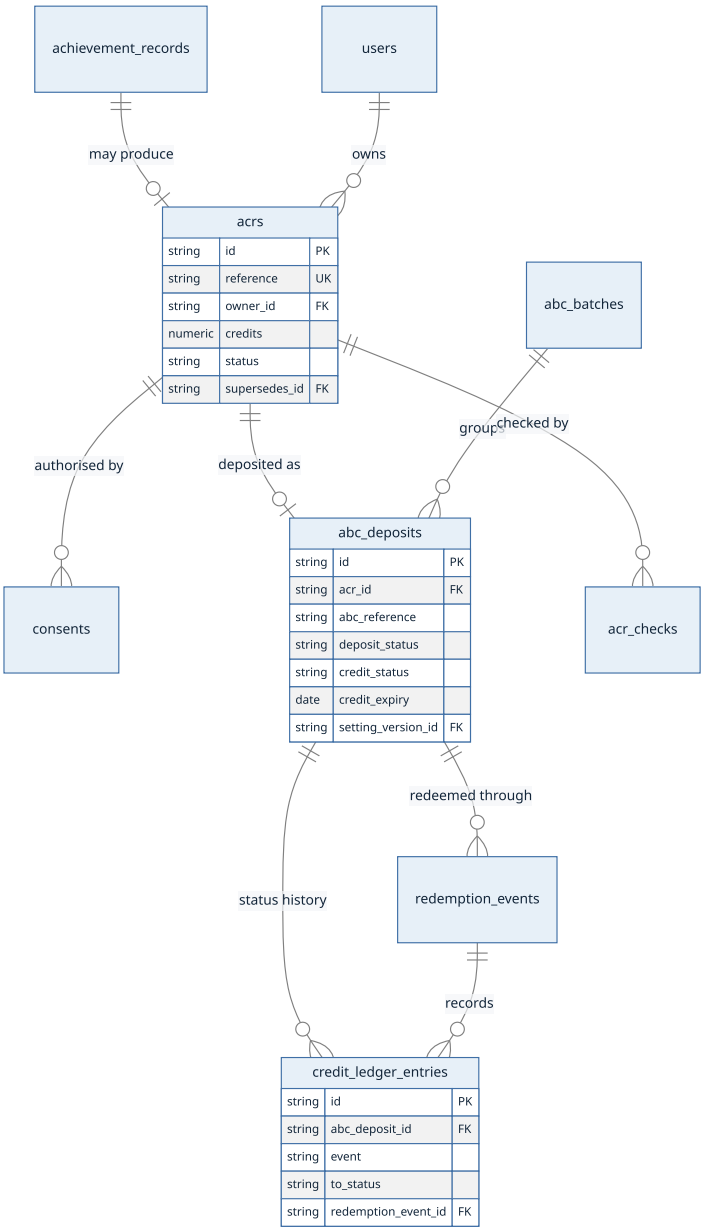


Figure 25: *ACR, consent, ABC deposit and the credit ledger.*

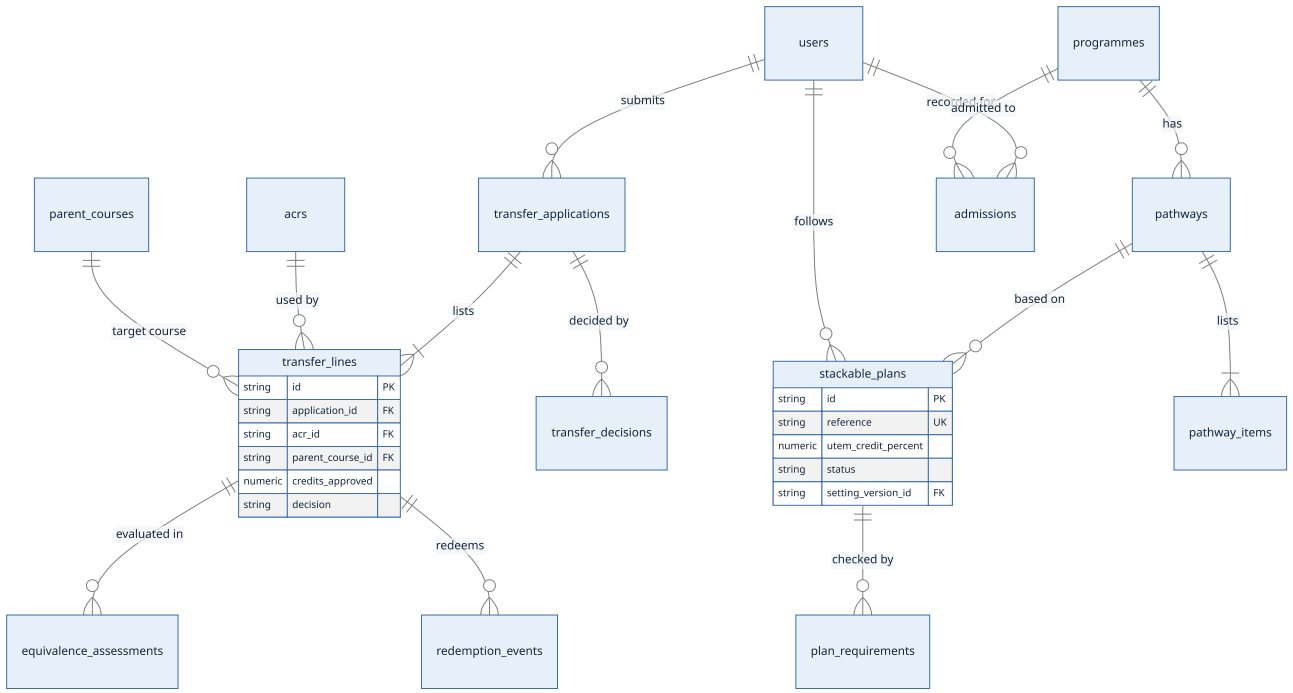


Figure 26: *Credit transfer, stackable pathways and admission records.*

5.2.7 Audit, Notifications and Integration

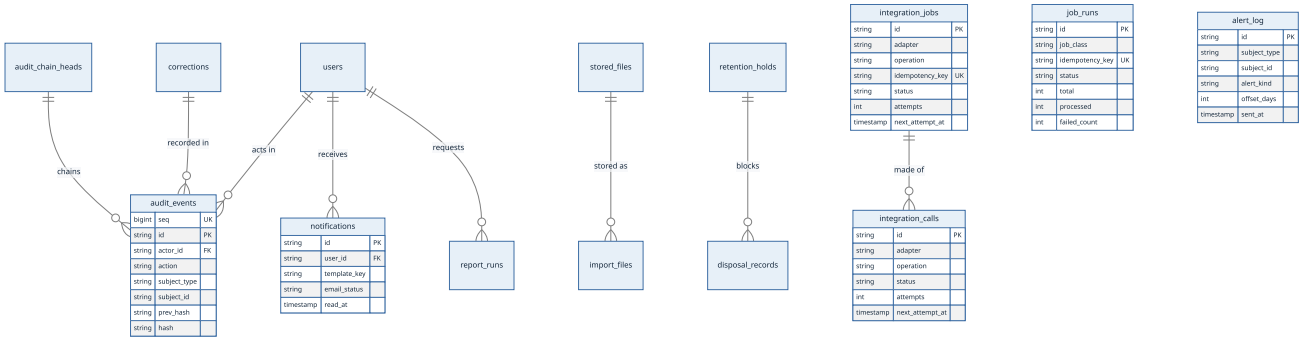


Figure 27: *Audit trail, notifications, integration jobs, files, corrections, reports and retention holds.* `integration_jobs` *refers to its subject by soft reference.*

5.3 Status Enumerations

Statuses use exactly the words of UI/UX Specification v1.1, Section 4.8. In PHP each status is a string-backed enum under `app/Modules/<Module>/Enums`. The case name is the PascalCase of the label, and the stored value is the snake_case of the label, with brackets removed (for example `Held (integrity)` is case `HeldIntegrity`, stored as `held_integrity`). The label shown to the user comes from the lang file key `status.<stored value>` in Bahasa Melayu and English (AD-13). The label is never stored. Negative statuses are tested before positive ones in policies and views, so "Not approved" is never read as an approval.

Table 51: *Status enumerations (UI/UX Specification Section 4.8)*

Record	PHP enum	Column	Cases (labels)
Proposal or version	`VersionStatus`	`mc_versions.status`	Draft; Submitted; In review; Returned for amendment; Endorsed; Pending approval; Approved; Not approved; Superseded; Suspended; Terminated (stored values in Section 5.3.1)
Offering	`OfferingStatus`	`offerings.status`	Not ready; Ready; Open; Full; Running; Ended; Cancelled
Enrolment	`EnrolmentStatus`	`enrolments.status`	Registered; Pending payment; Enrolled; Cancelled; Withdrawn; Completed; Not completed
Module or result	`ResultStatus`	`module_enrolments.status`, `assessment_results.status`	Not started; In progress; Provisional; Verified; Held (integrity); Achieved

Record	PHP enum	Column	Cases (labels)
Completion	`CompletionStatus`	`achievement_records.completion_status`	Complete
Credential	`CredentialStatus`	`credentials.status`	Issued; Corrected; Replaced; Revoked
QV	`QvStatus`	`qv_records.status`	Not applied; Preparing; Ready; Submitted; Quality-Verified; Expired; Suspended; Withdrawn (stored values in Section 5.3.1)
ACR or deposit	`AcrStatus`	`acrs.status`	ACR prepared; Verified; Consent given; Consent declined; Deposited; Correction needed
Credit	`CreditStatus`	`abc_deposits.credit_status`, `credit_ledger_entries.to_status`	Available; Provisionally redeemed; Redeemed; Expired
Transfer	`TransferStatus`	`transfer_applications.status`	Submitted; Under evaluation; Approved; Partly approved; Not approved; More evidence needed
Stackable plan	`PlanSuitability`	`stackable_plans.status`	Suitable; Needs review; Not yet suitable (stored values in Section 5.3.1)
CQI action	`CqiActionStatus`	`cqi_actions.status`	Open; In progress; Done (effectiveness pending); Closed

Notes on the use of these enums:

- **Result and module status.** One enum serves both columns. `assessment_results.status` allows only Provisional, Verified and Held (integrity). `module_enrolments.status` allows Not started, In progress, Provisional, Held (integrity) and Achieved. A module becomes Achieved only when its results are Verified (FR-123, FR-126).
- **Display statuses.** A label such as "Completed, waiting for verification" in the prototype is a view-model string built from `EnrolmentStatus::Completed` and the result status. It is never stored.
- **Deposit outcome.** The deputy dean records a deposit outcome of Successful, In process or Correction needed (FR-189). It is stored in `abc_deposits.deposit_status` (enum `DepositOutcome`). Successful sets `acrs.status` to Deposited, Correction needed sets it to Correction needed, and In process leaves it at Consent given.
- **Public status.** The public verification page uses valid, corrected, replaced and revoked (FR-154). The service maps Issued to valid.

5.3.1 Stored Values That Differ From the Displayed Word

Where the SRS state diagrams hold a state that UI/UX Specification Section 4.8 does not show, the database stores the SRS value and a mapping gives the displayed word. The enum has one case for each SRS state. The mapping is a method on the enum (`label()`), so a view never contains the rule.

Table 52: *SRS states stored, and the word displayed*

Enum	Stored case (SRS state)	Displayed word (Section 4.8)
`VersionStatus`	`EndorsedByPtj` (Endorsed by PTj)	Endorsed
`VersionStatus`	`NotSupported` (Not supported, the PTj does not support)	Not approved
`VersionStatus`	`Rejected` (Rejected by the approving authority)	Not approved
`QvStatus`	`PreparingEvidence`	Preparing
`QvStatus`	`ReadyToSubmit`	Ready
`QvStatus`	`SubmittedToQvc` (Submitted to QVC / MQA)	Submitted
`QvStatus`	`NotVerified` (QV not obtained)	Preparing, with the findings shown
`QvStatus`	`ReverificationRequired`	Preparing, with a "re-verification required" note. QV claims stop, because only Quality-Verified allows them
`PlanSuitability`	`SuitableToProceed`	Suitable
`PlanSuitability`	`SuitableWithConditions`	Needs review
`PlanSuitability`	`NotYetSuitable`	Not yet suitable
`PlanSuitability`	`NotSuitable`	Not yet suitable

The mappings for `NotVerified`, `ReverificationRequired` and `NotSuitable` are the design's proposal where the UI/UX specification has no word; they are listed in Section 11 for confirmation. All other cases keep the same word in both lists. The decision text (for example "Suitable with conditions") is also shown on the plan screen (FR-232).

Enumerations that the UI/UX specification does not list are internal to the design and are shown in Table 53. Their labels are not shown to users as statuses, except where a screen already uses the word.

Table 53: *Supporting enumerations (design only)*

PHP enum	Column	Cases
`MicroCredentialCategory`	`micro_credentials.category`	UMC; SAMC; MCE
`ChangeClass`	`mc_versions.change_class`, `change_requests.class`	Initial; Minor; Material
`ApprovalStage`	`approval_decisions.stage`	Endorsement; Senate
`DecisionOutcome`	`approval_decisions.decision`	Endorsed; Returned; Approved; Not approved
`ReviewStatus`	`review_assignments.status`	Invited; In progress; Submitted; Declined
`CommentState`	`review_comments.state`	Open; Resolved
`UserStatus`	`users.status`	Active; Deactivated
`ScopeType`	`role_assignments.scope_type`	University; PTj; Listed micro-credentials
`PaymentStatus`	`payments.status`	Pending; Paid; Failed; Waived; Refunded
`ImportStatus`	`import_files.status`	Queued; Running; Done; Rejected
`BadgeSource`	`digital_badges.source`	UCreds; MCMS
`CredentialAction`	`credential_events.action`	Issued; Corrected; Replaced; Revoked
`VerifyOutcome`	`verification_logs.outcome`	Found; Not found; Rate limited
`ConsentDecision`	`consents.decision`	Given; Declined; Postponed; Withdrawn
`AbcBatchStatus`	`abc_batches.status`	Prepared; Sent, waiting for ABC reference; Recorded
`MatchLevel`	`equivalence_assessments.match`	Full; Partial; None
`RequestStatus`	`result_change_requests.status`, `result_appeals.status`, `corrections.status`	Requested; Approved; Rejected; Done
`CqiRecordStatus`	`cqi_records.status`	Draft; Submitted; Confirmed
`ReviewOutcome`	`periodic_reviews.outcome`	Continue; Amend; Suspend; Terminate
`JobStatus`	`integration_jobs.status`	Queued; Running; Done; Retrying; Failed
`CallStatus`	`integration_calls.status`	Pending; Sent; Failed; Succeeded
`RunStatus`	`job_runs.status`	Queued; Running; Done; Failed
`EmailStatus`	`notifications.email_status`	Pending; Sent; Failed; Not required
`ScanStatus`	`stored_files.scan_status`	Pending; Clean; Infected
`TemplateStatus`	`template_versions.status`	Draft; Published; Retired

5.4 Table Catalogue

The catalogues list every table by area. **Module** is the owner. **Key columns** lists the columns that carry meaning, the foreign keys (FK) and the status column. Every table also has `id`, `created_at` and, where mutable, `updated_at`. The extra columns of each table are decided in its migration. Tables marked with an asterisk are not owned by any fact sheet of Section 4 and are named in Section 5.4.8.

5.4.1 Identity and Settings

Table 54: *Tables for identity, settings and platform services*

Table	Module	Purpose	Key columns
`users`	MD-01	A person who signs in with the UTeM Microsoft account: participant, staff or invited expert (FR-01, FR-07)	`entra_oid` (unique), `institution_id` (staff or matric number, unique), `upn`, `name`, `name_on_certificate`, `email`, `phone`, `locale`, `ptj_id` FK, `status` (UserStatus), `last_login_at`
`role_assignments`	MD-01	A role with a scope, start and end date, optionally delegated (FR-13, FR-12, FR-19)	`user_id` FK, `role`, `scope_type`, `ptj_id` FK, `valid_from`, `valid_to`, `assigned_by` FK, `reason`
`role_assignment_micro_credentials`*	MD-01	The micro-credentials listed in a role assignment whose scope is a list	`role_assignment_id` FK, `micro_credential_id` FK; unique pair
`delegations`	MD-01	A temporary hand-over of a role from one user to another (FR-19)	`role_assignment_id` FK, `delegator_id` FK, `delegate_id` FK, `valid_from`, `valid_to`, `reason`, `ended_at`
`conflict_declarations`	MD-01	A declared conflict of interest of a reviewer or approver (FR-17)	`user_id` FK, `mc_version_id` FK, `declared_at`, `reason`, `reviewed_by` FK

Table	Module	Purpose	Key columns
`ptjs`	MD-01	Faculties and centres (PTj) as reference data (FR-24)	`code` (unique), `name_ms`, `name_en`, `active`
`settings`	MD-00	One row per setting key (`credit.abc_validity_years`, `alerts.qv_expiry_days` and so on) (FR-20)	`key` (unique), `value_type`, `unit`, `description_ms`, `description_en`, `source_ref`
`setting_versions`	MD-00	Every value a setting has had, with the date it takes effect (Section 5.6) (FR-21)	`setting_id` FK, `value` json, `effective_from`, `set_by` FK, `reason`, `supersedes_id` FK
`templates`	MD-00	A notification, certificate, credential display, recognition statement or document template identified by key (FR-22)	`key` (unique), `kind`, `channel`, `description`
`template_versions` *	MD-00	A versioned bilingual body of a template	`template_id` FK, `version_no`, `subject_ms`, `subject_en`, `body_ms`, `body_en`, `status` (TemplateStatus), `published_by` FK, `published_at`
`stored_files`	MD-00	Metadata of every stored file; the bytes are in object storage (AD-14)	`storage_key`, `original_name`, `mime_type`, `size_bytes`, `sha256`, `category`, `scan_status`, `retain_until`, `uploaded_by` FK
`reference_sequences` *	MD-00	Per-year counters behind the human reference numbers (Section 5.5)	`kind`, `scope_key`, `year`, `last_value`; unique (kind, scope_key, year)

5.4.2 Curriculum and Approval

Table 55: *Tables for curriculum and approval*

Table	Module	Purpose	Key columns
`programmes`	MD-01	Target degree programmes (FR-24)	`code` (unique), `name_ms`, `name_en`, `ptj_id` FK, `total_credits`, `mqf_level`, `mode`, `accredited_until`
`parent_courses`	MD-01	Courses of a programme that a UMC can be unbundled from (FR-50, FR-51)	`code` (unique), `programme_id` FK, `name_ms`, `name_en`, `credits`, `slt_hours`, `clo_count`, `accredited_until`
`micro_credentials`	MD-02	A micro-credential across all its versions (FR-26, FR-31)	`code` (null for SAMC and MCE), `category`, `ptj_id` FK, `owner_id` FK, `category_confirmed_by` FK, `category_confirmed_at`, `search_vector` (full-text)
`mc_versions`	MD-02	One version of a micro-credential. An approved version is frozen (FR-74, FR-68)	`micro_credential_id` FK, `version_no`, `name_ms`, `name_en`, `parent_course_id` FK, `credit_model`, `total_slt_hours`, `approved_credit`, `mqf_level`, `change_class`, `supersedes_id` FK, `effective_date`, `status` (VersionStatus), `content_hash`
`learning_outcomes`	MD-02	The learning outcomes of a version (FR-37)	`mc_version_id` FK, `code`, `text_ms`, `text_en`, `sort_order`
`mc_modules`	MD-02	The modules of a version (FR-52, FR-53)	`mc_version_id` FK, `seq`, `name_ms`, `name_en`, `slt_hours`, `credit`
`assessment_components`	MD-02	An assessed item of a module with its weight and pass mark (FR-39)	`mc_module_id` FK, `mc_version_id` FK, `csv_key` (the mark column in the UCreds template), `name_ms`, `name_en`, `weight_percent`, `max_mark`, `pass_mark`
`assessment_component_outcomes` *	MD-02	Alignment of components with learning outcomes (FR-40)	`assessment_component_id` FK, `learning_outcome_id` FK; unique pair
`proposal_collaborators`	MD-02	Co-authors who may edit a draft proposal (FR-45)	`mc_version_id` FK, `user_id` FK, `access` (edit or view), `invited_by` FK, `invited_at`
`review_assignments`	MD-03	The invitation of a reviewer to review a version, with a due date (FR-60)	`mc_version_id` FK, `reviewer_id` FK, `kind` (colleague or industry expert), `scope`, `due_on`, `invited_by` FK, `coi_declared`, `status` (ReviewStatus)
`reviews`	MD-03	The completed review: form ratings and recommendation (FR-61, FR-62)	`review_assignment_id` FK (unique), `form` json, `recommendation` (Support, Amend, Do not support), `submitted_at`
`review_comments`	MD-03	Comments of a review and their resolution (FR-63)	`review_id` FK, `section`, `body`, `blocking`, `state` (CommentState), `resolved_by` FK, `resolved_at`

Table	Module	Purpose	Key columns
approval_decisions	MD-03	One row per decision at a stage: endorsement or Senate decision. Never updated (FR-67)	mc_version_id FK, stage, decision, decided_by FK, external_ref (Senate meeting reference), decided_at, notes
version_transition_rules	MD-03	The rule for learners already enrolled when a new version is approved: finish on the old version, move to the new one, or by recorded equivalence (FR-80)	mc_version_id (new version), from_version_id, rule, approval_decision_id, created_at
change_requests	MD-02	A request to change an approved version, classified as minor or material (FR-75, FR-76)	micro_credential_id FK, from_version_id FK, to_version_id FK, class, reason, requires_qv, requested_by FK, status (VersionStatus of the new version)

5.4.3 Offering, Enrolment and Results

Table 56: *Tables for offering, enrolment and results*

Table	Module	Purpose	Key columns
offerings	MD-04	A cohort of an approved version (FR-83, FR-87)	reference (for example DS-2026-2, unique), mc_version_id FK, owner_id FK, start_date, end_date, places, fee, language, qv_record_id FK (SAMC), is_migrated, status (OfferingStatus)
readiness_items	MD-04	The readiness checklist of an offering (FR-85)	offering_id FK, item_key, label_ms, label_en, done, done_by FK, done_at, evidence_file_id FK
readiness_confirmations	MD-04	The deputy dean's confirmation, with conditions, that an offering is ready (FR-86, FR-88)	offering_id FK, confirmed_by FK, confirmed_at, conditions, items_snapshot json
enrolments	MD-05	A participant's registration in an offering (FR-103)	reference (for example E-2026-0644, unique), offering_id FK, user_id FK, mc_version_id FK, attempt_no, registered_at, withdrawn_at, waiver_ref, is_migrated, status (EnrolmentStatus); unique (offering_id, user_id, attempt_no)
pdpa_acknowledgements	MD-05	The participant's acknowledgement of the recognition statement and the personal data notice, with the wording version (FR-99, FR-100)	enrolment_id, user_id, statement_version, pdpa_version, acknowledged_at
module_enrolments	MD-05	Progress of an enrolment in each module (FR-104)	enrolment_id FK, mc_module_id FK, started_at, achieved_at, status (ResultStatus)
payments	MD-05	Payment attempts of an enrolment (FR-101, FR-102)	enrolment_id FK, amount, currency, method, provider_ref, idempotency_key (unique), status (PaymentStatus), failure_reason, paid_at
import_files	MD-06	Every import of a UCreds results or badge file (FR-116)	kind (results or badges), template_version, offering_id FK, file_id FK, sha256, row_count, staged_count, rejected_count, duplicate_of_id FK, run_by FK, status (ImportStatus)
import_row_errors	MD-06	A rejected row of an import with the reason (FR-113). Valid rows are not stored here: they become Provisional results (FR-114)	import_file_id FK, row_no, learner_ref, reason; marks are not copied
export_files	MD-06	Every export of an enrolment file (FR-111, FR-116)	offering_id FK, template_version, file_id FK, sha256, row_count, run_by FK
assessment_results	MD-07	One score per module enrolment and component. A verified row is never updated (FR-120, FR-123)	module_enrolment_id FK, assessment_component_id FK, score, source (import or manual), import_file_id FK, entered_by FK, verified_by FK, verified_at, hold_reason, status (ResultStatus)
assessment_result_revisions	MD-07	Each approved change to a verified result; the effective score is the latest revision (FR-125)	assessment_result_id FK, revision_no, old_score, new_score, result_change_request_id FK, approved_by FK, approved_at
integrity_cases	MD-07	An academic-integrity case that holds results (FR-122)	offering_id FK, opened_by FK, reason, outcome, closed_at, closed_by FK
result_change_requests	MD-07	A request to change a result after entry or verification (FR-125)	assessment_result_id FK, requested_by FK, new_score, reason, decided_by FK, status (RequestStatus)

Table	Module	Purpose	Key columns
result_appeals	MD-07	A participant's appeal against a result (FR-127)	assessment_result_id FK, participant_id FK, reason, decided_by FK, outcome, status (RequestStatus)

5.4.4 Credentials and Achievement

Table 57: *Tables for the Achievement Record and credentials*

Table	Module	Purpose	Key columns
achievement_records	MD-08	The internal record of modules, badges and completion of one enrolment. It is not an ACR (FR-129, FR-130)	enrolment_id FK (unique), mc_version_id FK, completed_on, final_result json, confirmed_by FK, confirmed_at, completion_status (CompletionStatus, null until confirmed)
module_achievements	MD-08	A module achieved by verified results (FR-126, FR-135)	achievement_record_id FK, module_enrolment_id FK (unique), mc_module_id FK, achieved_on, verified_by FK
credentials	MD-09	A certificate with its Digital Credential, one signed object with two renderings. The signed payload is never updated (FR-140, FR-141)	credential_no (MCMS-C-...), revision, public_code (unique), holder_id FK, holder_name, achievement_record_id FK, mc_version_id FK, issued_on, signed_payload json, payload_hash, key_id, supersedes_id FK, replaces_id FK, status (CredentialStatus); unique (credential_no, revision)
digital_badges	MD-09	A Digital Badge for a verified module, imported from UCreds or issued by MCMS (FR-138, FR-115)	module_achievement_id FK (unique), holder_id FK, source (BadgeSource), badge_ref, badge_url, issued_on, public_code (unique), status (CredentialStatus)
credential_events	MD-09	The history of a credential: issue, correction, replacement, revocation (FR-158, FR-160)	credential_id FK, action, reason, actor_id FK, external_ref, from_status, to_status, linked_credential_id FK, occurred_at
verification_logs	MD-09	One row per public verification request (FR-156)	occurred_at, credential_id FK (null when not found), outcome (VerifyOutcome), channel (page or api), network_address, code_fingerprint

5.4.5 Quality Verification and Quality Assurance

Table 58: *Tables for QV, CQI and monitoring*

Table	Module	Purpose	Key columns
qv_records	MD-10	One QV cycle of a SAMC (FR-163, FR-171)	micro_credential_id FK, mc_version_id FK, cycle_no, qv_reference, verified_credit, verified_mqf_level, valid_from, valid_until, setting_version_id FK, status (QvStatus)
qv_submissions	MD-10	Each submission package sent to QVC and MQA (FR-169, FR-170)	qv_record_id FK, package_file_id FK, submitted_on, submitted_by FK, submission_ref, decision_on
qv_evidence	MD-10	Items of the QV evidence checklist (FR-165, FR-166)	qv_record_id FK, checklist_key, offering_id FK, file_id FK, accepted, accepted_by FK, accepted_at
cqi_records	MD-14	A quality review of an offering (FR-243, FR-248)	reference (for example CQI-DS-2026-1, unique), offering_id FK, owner_id FK, indicators json, due_on, submitted_at, confirmed_by FK, status (CqiRecordStatus)
cqi_actions	MD-14	Improvement actions of a CQI record (FR-244, FR-245)	cqi_record_id FK, description, responsible_id FK, due_on, effectiveness_note, closed_at, status (CqiActionStatus)
periodic_reviews	MD-14	Scheduled review of a micro-credential and its outcome (FR-249, FR-250)	micro_credential_id FK, due_on, completed_on, decided_by FK, outcome (ReviewOutcome)
suspensions	MD-14	Suspension or termination of an offering or micro-credential, with the transition plan (FR-251, FR-253)	micro_credential_id FK, offering_id FK, kind, reason, effective_from, transition_plan, notified_at, decided_by FK, lifted_at
feedback_responses	MD-14	A participant's course feedback, stored without a name on reports (FR-239)	offering_id FK, enrolment_id FK, answers json, submitted_at
quality_domain_checks	MD-14	The result of each quality domain check of a CQI record (FR-241)	cqi_record_id FK, domain, finding, met, checked_by FK, checked_at

5.4.6 Credit Bank, Transfer and Pathways

Table 59: *Tables for ACR, ABC, transfer and pathways*

Table	Module	Purpose	Key columns
acrs	MD-11	The Academic Credit Record, generated from verified records (FR-179, FR-180)	reference (MCMS-A-..., unique), owner_id FK, achievement_record_id FK, mc_version_id FK, qv_record_id FK (SAMC), credits, mqf_level, completion_date, assessment_basis, payload json, confirmed_by FK, confirmed_at, supersedes_id FK, status (AcrStatus)
acr_checks	MD-11	The eligibility checks run before an ACR is prepared (FR-178)	acr_id FK, check_key, passed, detail, checked_at
consents	MD-11	The participant's consent decisions for ABC deposit (FR-185, FR-186)	acr_id FK, decision (ConsentDecision), template_version_id FK (wording shown), decided_at, decided_by FK
abc_batches *	MD-11	A batch of ACR sent to ABC together (FR-187, FR-188)	reference (for example ABC-FTMK-2026-03, unique), ptj_id FK, export_file_id FK, format_version, sent_at, prepared_by FK, status (AbcBatchStatus)
abc_deposits	MD-11	The deposit of one ACR in ABC (FR-189)	acr_id FK (unique), abc_batch_id FK, abc_reference, deposit_date, credits, credit_expiry, setting_version_id FK (validity period used), recorded_by FK, deposit_status (DepositOutcome), credit_status (CreditStatus)
credit_ledger_entries	MD-11	Append-only history of the credit status of a deposit: deposit, provisional redemption, redemption, release, expiry (FR-196)	abc_deposit_id FK, event, from_status, to_status (CreditStatus), credits, redemption_event_id FK (null for deposit and expiry), recorded_by FK, occurred_at
redemption_events	MD-11	A step of the Single Redemption Principle for one transfer line: provisional, final, released (FR-213, FR-214, FR-215)	abc_deposit_id FK, transfer_line_id FK, event, evidence_ref, abc_updated_on, abc_update_ref (FR-220), recorded_by FK, occurred_at
transfer_applications	MD-12	A credit transfer case (FR-198, FR-206)	reference (TR-..., unique), user_id FK, programme_id FK, applied_at, status (TransferStatus)
transfer_decisions	MD-12	The academic decision on an application, by the receiving deputy dean (FR-206, FR-207)	transfer_application_id FK, outcome (TransferStatus), decided_by FK, decided_on, authority, decision_ref, conditions, notes
transfer_lines	MD-12	One ACR offered against one target course (FR-199, FR-205)	transfer_application_id FK, acr_id FK, mc_version_id FK, parent_course_id FK, credits_requested, credits_approved, decision (TransferStatus), conditions
equivalence_assessments	MD-12	The outcome-by-outcome comparison of a line (FR-202, FR-203)	transfer_line_id FK, source_outcome_id FK, target_outcome_ref, match (MatchLevel), evaluator_id FK, note
pathways	MD-13	A mapping of micro-credentials to a programme, approved by Senate (FR-222, FR-224)	reference (for example PW-BITZ-1, unique), programme_id FK, version_no, approval_ref, approved_at, status
pathway_items	MD-13	A micro-credential mapped to a course, with its credit (FR-223, FR-226)	pathway_id FK, micro_credential_id FK, parent_course_id FK, credits, plo_map json
stackable_plans	MD-13	An individual plan towards a degree (FR-228, FR-232)	reference (SP-..., unique), user_id FK, pathway_id FK, programme_id FK, utem_credit_percent, residency_semesters, setting_version_id FK, decided_by FK, conditions, status (PlanSuitability)
plan_requirements	MD-13	The checks of a plan: home credit share, residency, credit-bearing, learning-outcome mapping, remaining requirements (FR-229, FR-230, FR-231)	stackable_plan_id FK, requirement_key, met, detail, checked_at
admissions	MD-12	Programme admission recorded by Pejabat Pendaftar, kept apart from transfer (FR-233, FR-211)	user_id FK, programme_id FK, intake, recorded_by FK, recorded_at, evidence_ref

5.4.7 Audit, Notifications and Integration

Table 60: *Tables for audit, messages and integration*

Table	Module	Purpose	Key columns
audit_events	MD-15	The append-only audit trail with a hash chain (Section 5.6) (FR-254)	seq (unique, monotonic), occurred_at, actor_id FK, actor_role, action, subject_type, subject_id, before json, after json, reason, external_ref, context json, request_id, prev_hash, hash

Table	Module	Purpose	Key columns
`audit_chain_heads` *	MD-15	One row holding the latest sequence and hash, locked while an audit row is written; also the last verified point	`chain` (unique, value `main`), `last_seq`, `last_hash`, `verified_seq`, `verified_hash`, `verified_at`
`corrections`	MD-15	A correction made under the Guide procedure: original, reason, action, approver, date (FR-256, FR-257)	`target_type`, `target_id`, `original_ref`, `reason`, `requested_by` FK, `approved_by` FK, `executed_at`, `status` (RequestStatus)
`discrepancies`	MD-15	A mismatch between records found by a check (FR-257)	`subject_type`, `subject_id`, `description`, `found_at`, `resolved_by` FK, `correction_id` FK
`retention_holds`	MD-15	A hold that stops disposal of records (FR-263)	`scope_type`, `scope_ref`, `reason`, `placed_by` FK, `placed_at`, `released_at`
`report_runs`	MD-15	A generated report with scope, filters and file (FR-260, FR-261)	`report_type`, `scope` json, `filters` json, `file_id` FK, `requested_by` FK, `status` (RunStatus)
`disposal_records`	MD-15	An approved disposal of records after the retention period (FR-262)	`category`, `criteria`, `record_count`, `approved_by` FK, `executed_at`, `hold_check_passed`
`notification_preferences`	MD-16	The user's choice of channel and digest per notice group (FR-267)	`user_id` FK, `notice_group`, `channel`, `digest`; mandatory groups cannot be switched off
`adapter_configs`	MD-16	The mode (Stub, Manual, Live) and non-secret settings of each adapter (FR-270, FR-25)	`adapter` (unique), `mode`, `settings` json, `updated_by` FK; secrets stay in environment configuration
`notifications`	MD-16	A message to a user, stored before it is sent (AD-15) (FR-265, FR-268)	`user_id` FK, `template_key`, `template_version_id` FK, `locale`, `subject`, `body`, `link`, `mandatory`, `read_at`, `sent_at`, `email_status` (EmailStatus), `digest_group`
`integration_jobs`	MD-16	One logical exchange with an external system, for example one deposit export or one payment (FR-270)	`adapter`, `operation`, `direction`, `idempotency_key` (unique), `subject_type`, `subject_id`, `request_summary` json, `last_error`, `status` (JobStatus)
`integration_calls` *	MD-16	One row per adapter call. With `notifications` it is the outbox: the scheduler re-dispatches rows that are pending or due for retry (FR-271, IR-17)	`integration_job_id` FK, `adapter`, `operation`, `direction`, `request_ref`, `attempts`, `next_attempt_at`, `last_error_ref`, `status` (CallStatus)
`job_runs`	MD-16	Progress of a long background job such as an import, an issue run or a report (NFR-11)	`job_class`, `queue`, `idempotency_key` (unique), `subject_type`, `subject_id`, `total`, `processed`, `failed_count`, `started_at`, `finished_at`, `error_ref`, `status` (RunStatus)
`alert_log`	MD-16	One row per record, alert kind and day offset, so that a timed alert is sent once and is caught up after a missed run (FR-173, FR-194)	`subject_type`, `subject_id`, `alert_kind`, `offset_days`, `sent_at`, `notification_id` FK; unique (subject_type, subject_id, alert_kind, offset_days)

Background work runs on six named queues: `default`, `notifications`, `imports`, `credentials`, `reports` and `integration` (AD-05). `job_runs.queue` holds the queue name. Screens read progress from `job_runs` ("issuing 14 of 27"), not from Redis.

5.4.8 Tables Added to the Shared List

Seven tables are not named in any fact sheet of Section 4 (marked with an asterisk above). Each is needed for a stated requirement. The design also names `template_versions` in Section 5.4.1, which the fact sheet of MD-00 does not list.

Table 61: *Additional tables*

Table	Owner	Reason
`reference_sequences`	MD-00	Gap-free yearly counters for reference numbers (Section 5.5)
`role_assignment_micro_credentials`	MD-01	A scope of "listed micro-credentials" (FR-12) needs a real relationship, not a JSON list (DR-09)
`assessment_component_outcomes`	MD-02	The many-to-many alignment of outcomes and assessments (FR-40)
`template_versions`	MD-00	Versions of a template, which FR-22 requires
`abc_batches`	MD-11	The prototype and FR-187 group ACR into deposit batches sent to ABC together
`audit_chain_heads`	MD-15	Serialises audit writes and holds the last verified point of the chain (AD-11)
`integration_calls`	MD-16	The outbox row of each adapter call, with attempts and next attempt (FR-271)

The SRS entities DIGITAL_BADGE, REDEMPTION_EVENT, AUDIT_LOG, SETTING, READINESS_CHECK, EQUIVALENCE_ASSESSMENT and TRANSFER_DECISION map to `digital_badges`, `redemption_events`, `audit_events`, `settings` with

`setting_versions`, `readiness_items` with `readiness_confirmations`, `equivalence_assessments` and `transfer_decisions`. The credit status history of a deposit is kept in `credit_ledger_entries`, and each redemption step in `redemption_events`. One `credentials` row is the certificate together with its Digital Credential: they share one number, one public code and one signed payload, and the PDF and the JSON file are two renderings of it. The separate recognition stages stay separate (DR-01): the approval decision, the credential, the QV record, the ACR, the deposit, the transfer decision and the redemption each have their own table and status.

5.5 Identifiers

Table 62: *Identifier types*

Identifier	Form	Generation	Where used
Primary key	ULID, 26 characters	Generated by the application (`HasUlids` trait) when the model is created	Every table. Never shown to a participant and never used as a public link
Human reference	A prefix, the year and a number, for example `MCMS-C-2026-000098`, `MCMS-A-2026-000102`, `TR-2026-0012`	The `ReferenceNumbers` service takes the next value of `reference_sequences` for (kind, scope, year)	Credentials, ACR, transfer cases, enrolments, plans, batches
Public verification code	10 characters of Crockford base32, for example `3HXP92LDKQ`	Random, from a cryptographically secure source (50 bits). Not derived from any ID, number or name	`credentials.public_code`, QR code, verification link (DR-07, FR-146)

Rules for human references:

- The format of each kind is held in code, not in settings. The prefix, the width of the number and whether the number restarts each year are fixed per kind. Table 63 lists them.
- The generator opens a transaction, selects the `reference_sequences` row for (kind, scope, year) with `lockForUpdate`, adds one, and writes the record in the same transaction. A rollback undoes the increment, so numbers have no gaps. The year is the year of creation in Asia/Kuala_Lumpur.
- A reference has a unique index on its table. A reference is never reused or edited. Human references carry no secret; access is always decided by a policy and never by knowing the number (SR-03).

Table 63: *Human reference formats*

Kind	Example	Scope key	Restarts yearly
Credential	`MCMS-C-2026-000098`	none	Yes
ACR	`MCMS-A-2026-000102`	none	Yes
Credit transfer case	`TR-2026-0012`	none	Yes
Stackable plan	`SP-2026-003`	none	Yes
Enrolment	`E-2026-0644`	none	Yes
ABC batch	`ABC-FTMK-2026-03`	PTj code	Yes
Offering	`DS-2026-2`	micro-credential code	Yes (cohort number in the year)
CQI record	`CQI-DS-2026-1`	offering reference	No (derived from the offering)

The prototype shows verification codes as eight characters in two groups (`3HXP-92LD`). The design uses ten characters, as AD-09 states. The page groups them for reading (`3HXP9-2LDKQ`). On input, the service removes hyphens and spaces, upper-cases the text and maps the letters I and L to 1 and O to 0, as Crockford base32 allows. The alphabet has no I, L, O or U. A new code is generated until the unique index accepts it.

5.6 Audit Design

The audit trail satisfies FR-254, FR-255, SR-10 and AD-11. It records every approval and decision, credential action, result change, ACR, deposit, transfer and redemption status change, role and setting change, correction and disposal. A single `AuditLog::record()` call writes a row. Business code never writes `audit_events` directly.

5.6.1 Columns

Table 64: *Columns of `audit_events`*

Column	Type	Content
`id`	char(26)	ULID of the row
`seq`	bigint, unique	Position in the chain, starting at 1, increased by one for each row. Taken from `audit_chain_heads.last_seq`

Column	Type	Content
`occurred_at`	timestamptz	Time of the event, from the application clock (the demo clock is never used in production, AD-18)
`actor_id`	char(26), FK	The user who acted. Null for the system, with `actor_role` set to `system`
`actor_role`	string	The role the actor acted in (FR-11)
`action`	string	A dotted name such as `results.verified`, `credential.revoked`, `setting.changed`
`subject_type`, `subject_id`	string, char(26)	The record affected. A soft reference, so that the row outlives a disposed record
`before`, `after`	json	Values before and after. Only changed columns are stored. Identity numbers and marks are masked in these values
`reason`	text	The reason given in the dialog, required for corrections, revocations and rejections
`external_ref`	string	A Senate minute reference, an ABC reference or a decision reference
`context`	json	Approver, setting version used, job id and similar facts needed to explain the action
`request_id`	string	The request or job identifier, to join with application logs (NFR-41)
`prev_hash`	char(64)	The `hash` of the row with `seq` minus one. The first row holds 64 zeros
`hash`	char(64)	SHA-256 of the row (below)

5.6.2 Hash Chain and Transaction Rule

- **Hash.** `hash = SHA-256(prev_hash + canonical_json(row))`. The canonical JSON has the columns in fixed order (`seq`, `occurred_at` in UTC ISO 8601 with microseconds, `actor_id`, `actor_role`, `action`, `subject_type`, `subject_id`, `before`, `after`, `reason`, `external_ref`, `context`), keys sorted, no spaces, UTF-8.
- **Same transaction.** `AuditLog::record()` must run inside the database transaction of the change it describes. If no transaction is open it throws. The row and the change commit together or not at all.
- **Serialising.** The call locks the single `audit_chain_heads` row (`lockForUpdate`), reads `last_seq` and `last_hash`, writes the new row, and updates the head, all in the caller's transaction. This uses only Eloquent features and so works on PostgreSQL and MySQL (AD-03). Audit writes are therefore serial. At the Release 1 volume (the peak of NFR-05 is 2,000 registrations in 2 hours) a short lock is acceptable. Section 9 holds the load figures.
- **Database role.** The application connects as `mcms_app`, which has `INSERT` and `SELECT` on `audit_events` and no `UPDATE`, `DELETE` or `TRUNCATE`. Migrations run as a different role, `mcms_migrator`. A trigger that raises an error on `UPDATE` and `DELETE` of `audit_events` is added in a migration as a second barrier (SR-10).
- **Integrity check job.** A scheduled job (JB-18, `VerifyAuditChainJob`) runs every night at 02:00. It reads rows after `audit_chain_heads.verified_seq` in order, recomputes each hash and checks the link to the previous row, then moves `verified_seq` and `verified_hash` forward. It also recomputes the whole chain once a month. A mismatch writes a security log entry and an alert to the system administrator (SR-19). The audit screen (CODL-10) and the audit export (DOC-16) show an integrity line with the last verified sequence and time.
- **Reading.** The audit views filter by subject, actor and date. Staff see only entries of records in their scope (FR-255).

5.7 Versioned Settings

A setting is a key with a typed value. Its history is kept in `setting_versions` (FR-20, FR-21, AD-12). A row of `setting_versions` is never updated or deleted. A change inserts a new row with the new value, an `effective_from` time (now or in the future), the user and the reason, and points to the row it supersedes.

- **Resolving a value.** The value in force at time *t* is the row of the setting with the largest `effective_from` not after *t*. The `SettingResolver` takes the key and the time, defaulting to now, and returns the value and the id of the version.
- **Decisions store the version.** A decision that depends on a setting stores `setting_version_id` and uses the value of that version for ever (DR-06). Table 65 lists them.
- **Calculated alerts** (for example QV expiry at 180, 90 and 30 days) use the value in force on the day the scheduler runs. They are not decisions.
- **Audit.** A setting change writes an audit row with the old and new values and the reason. Confirmation is required in the dialog (IR-06).

Table 65: *Decisions that store a setting version*

Decision	Column	Setting keys used
QV validity set on a QV result	`qv_records.setting_version_id`	`qv.max_validity_years`, `qv.min_completed_cohorts`
Credit expiry set on an ABC deposit	`abc_deposits.setting_version_id`	`credit.abc_validity_years`
Stackable plan suitability	`stackable_plans.setting_version_id`	`stackable.min_home_credit_percent`, `stackable.residency_semesters`

Decision	Column	Setting keys used
Learning-time checks when a version is submitted	`mc_versions.setting_version_id` (added to the table)	`credit.slt_hours_per_credit`, `slt.min_hours_samc`, `slt.min_hours_umc_whole`, `slt.min_hours_module_credit_model`
Public verification fields on a credential page	not stored (current value is used)	`verification.public_fields`

Settings are cached per request. The cache key includes the latest `setting_versions.id` of the setting, so a new version is used at once on every node (NFR-10).

5.8 Bilingual Content

Text that a participant sees is stored in Bahasa Melayu and English (DR-10, NFR-49, AD-13). Each such field is a pair of columns with the suffixes `_ms` and `_en`.

- **Fields.** Names and descriptions of micro-credentials versions, learning outcomes, modules and assessment components, programme and PTj names, readiness item labels, the recognition statement, and the subject and body of every template version.
- **Draft state.** While a version is a draft, one language may be empty. The form marks the missing language (FR-08). The submit check and the publish check for templates both require both languages. A template version cannot be published with a missing language.
- **Reading.** The `HasBilingualAttributes` trait gives `$model->title` in the locale of the user, and falls back to the other language only on staff screens, marked as a fallback. Participant documents and credentials show both languages, with BM beneath EN (FR-145).
- **Locale.** `users.locale` holds `ms` or `en`. A notification stores its rendered subject and body in the recipient's locale at creation, and the template version used.
- **Search.** The full-text vector of a bilingual field concatenates both languages, using the `simple` text search configuration, so that no stemming of one language damages the other.

5.9 No-Delete and Correction Policy

Records are not deleted and not changed in place where the SRS forbids it (DR-03, DR-04, BR-38). The policy has three parts.

5.9.1 Which Tables Are Append-Only

Table 66: *Mutability of record classes*

Class	Tables	Rule
Append-only	`audit_events`, `setting_versions`, `credential_events`, `credit_ledger_entries`, `redemption_events`, `assessment_result_revisions`, `approval_decisions`, `transfer_decisions`, `consents`, `verification_logs`, `corrections`, `disposal_records`	Only `INSERT` and `SELECT` are granted to the application role
Frozen after a milestone	`mc_versions` and its child tables (after Approved), `assessment_results` (after Verified), `credentials` payload columns (after issue), `acrs` payload (after Verified), `qv_records` result columns (after a result)	The model blocks the update. A database trigger rejects an update of the frozen columns. Status columns that the workflow changes are not frozen
Mutable with audit	Reference and working data: `users`, `role_assignments`, `ptjs`, `programmes`, `parent_courses`, drafts, `offerings` before opening, `readiness_items`, `cqi_*`, `stackable_plans` before decision	Updates are allowed and write an audit row with before and after values
No `DELETE`	All tables except the working tables `notifications` (read ones, after the retention period) and `import_row_errors` of a rejected file	The application role has no `DELETE` grant. Disposal runs as a separate role, `mcms_disposal`, through the approved disposal process

5.9.2 Correction Patterns

A correction follows FR-256: the original is kept, the reason, the actor and the approver are recorded, and linked records are updated in a controlled way. Table 67 gives the pattern for each record.

Table 67: *How each record is corrected*

Record	Correction	Result in the data
Approved version	A minor or material change (FR-77, FR-76)	A new `mc_versions` row with `version_no` plus one and `supersedes_id` set. The old row becomes Superseded. A version with participant records is never deleted (BR-39)
Credential, wrong detail	Correct (FR-158)	A new `credentials` row with the same `credential_no`, `revision` plus one, a new `public_code` and a newly signed payload. The old row becomes Corrected. A `credential_events` row of action Corrected links both
Credential, replaced	Replace (FR-159)	A new row with a new `credential_no` and `replaces_id` set. The old row becomes Replaced
Credential, withdrawn	Revoke (FR-160)	No new row. `credentials.status` becomes Revoked, with a `credential_events` row holding the reason and the authorised body's decision reference. The row stays and the public page shows Revoked
Verified result	Change after verification (FR-125)	A `result_change_requests` row, and when approved an `assessment_result_revisions` row with the old and new score. The verified `assessment_results` row is not edited. The effective score is the latest revision
ACR	Correction needed (FR-180)	A new `acrs` row with `supersedes_id`. The old row stays with its status and a `superseded_at` time
Enrolment	Withdrawal, cancellation	The status changes and an audit row is written. A re-registration creates a new row with `attempt_no` plus one
Setting	Change	A new `setting_versions` row (Section 5.7)
Any other record	Correction by an authorised party (FR-256, FR-257)	A `corrections` row holds the original reference, reason, action, approver and date. The update is allowed only for mutable tables, with an audit row. EV-60 tells the owning module to apply it

The `credentials.status` column is a projection of the latest `credential_events` row of that credential. Only the `CredentialService` changes it, in the same transaction that inserts the event. The signed payload, public code, number, holder and revision never change after issue (FR-161). A revoked or replaced credential is never erased and still answers a verification request with its status.

5.9.3 Migrated Records

Records imported from earlier spreadsheets (DR-12) carry `is_migrated = true` on `offerings` and `enrolments`. No ACR or credit claim is created from them unless eligibility is verified (DR-13). The ACR service checks the flag before it generates an ACR.

5.10 Retention and Archiving

Retention periods are open (A-08, OI-07). The design stores one setting per record category, `retention.<category>`, with the value TBD until OI-07 is closed (FR-262, NFR-37). Only the ACR rule is fixed: ACR are kept indefinitely unless the owner or a recognised authority directs otherwise (G 12.8).

Table 68: *Retention by record category (periods are placeholders)*

Record category	Tables	Setting key	Period
Development and approval records	`micro_credentials`, `mc_versions` and children, `reviews`, `approval_decisions`, `change_requests`	`retention.curriculum`	TBD (OI-07). Versions linked to participants are kept while those records are kept
Registration records	`offerings`, `enrolments`, `module_enrolments`, `payments`	`retention.registration`	TBD (OI-07)
Assessment results	`assessment_results`, `assessment_result_revisions`, `import_row_errors`	`retention.assessment`	TBD (OI-07)
Achievement Record and credentials	`achievement_records`, `module_achievements`, `credentials`, `digital_badges`, `credential_events`	`retention.credentials`	TBD (OI-07). Verification stays possible for the life of the credential
QA, monitoring and CQI	`cqi_*`, `periodic_reviews`, `suspensions`	`retention.quality`	TBD (OI-07)

Record category	Tables	Setting key	Period
QV documents and decisions	`qv_records`, `qv_evidence`	`retention.qv`	At least the QV validity period plus the records requirement
ACR	`acrs`, `consents`	not configurable	Indefinite (G 12.8)
ABC deposit and credit	`abc_batches`, `abc_deposits`, `credit_ledger_entries`	`retention.abc`	Per ABC rules
Credit transfer decisions	`transfer_*`, `equivalence_assessments`	`retention.transfer`	TBD (academic regulations)
Pathways and plans	`pathways`, `pathway_items`, `stackable_plans`, `plan_requirements`, `admissions`	`retention.pathways`	Programme lifetime plus review period
Audit trail	`audit_events`	not configurable	At least as long as the longest linked record. Audit rows are not disposed of in Release 1
Verification logs	`verification_logs`	`retention.verification_logs`	TBD (OI-07)
Uploaded files	`stored_files`, `export_files`	`stored_files.retain_until` per row	Follows the owning record

- **Archiving.** Archiving means the record leaves default lists and searches. A record past its active period is shown only through the traceability and report screens (Section 4, MD-15). Release 1 does not move rows to another table or database. Table partitioning of `audit_events` and `verification_logs` by year is a later change if volume requires it (NFR-12).
- **Disposal.** A disposal request names a category and a cut-off date. The job lists the records past the period, excludes any under a `retention_holds` row or linked to a record that is still retained, and waits for approval. After approval, the `mcms_disposal` role deletes them and a `disposal_records` row and an audit row are written. Every deletion is thus approved and logged (FR-262, DR-04).
- **Files.** Object storage keeps a file while its `stored_files` row exists. Disposal removes the object after the row is marked. Backups follow the backup schedule in Section 9 (SR-11).
- **Personal data.** A participant's data request is answered from the tables above (SR-14). Data is not removed while a record that must be kept refers to it (BR-39).

5.11 Key Indexes

Every foreign key has an index. Table 69 lists the other indexes that matter. Full-text indexes and partial indexes are created in migrations; the queries that use them go through Eloquent scopes (AD-03). Search always filters by the user's scope first, then ranks (AD-16, NFR-02).

Table 69: *Key indexes*

Table	Index	Purpose
`users`	unique `entra_oid`, unique `institution_id`, GIN on `search_vector`	Sign-in lookup, merge of identities (FR-09), name search
`role_assignments`	(`user_id`, `role`, `valid_to`)	Authorisation check on every request (SR-03)
`setting_versions`	(`setting_id`, `effective_from` descending)	Resolve the value in force
`micro_credentials`, `mc_versions`	GIN on `search_vector`; (`micro_credential_id`, `version_no`) unique; (`status`, `ptj_id`)	Catalogue and staff search, version history, queues
`offerings`	unique `reference`; (`status`, `start_date`)	Public catalogue and open offerings (FR-92)
`enrolments`	unique (`offering_id`, `user_id`, `attempt_no`); (`user_id`, `status`)	One active registration, "my learning"
`assessment_results`	(`module_enrolment_id`, `assessment_component_id`, `status`)	Results of a cohort and bulk verification (FR-124)
`credentials`	unique `public_code`; unique (`credential_no`, `revision`); (`holder_id`, `status`)	Verification within 1 second (NFR-03); holder wallet
`acrs`, `abc_deposits`	unique `reference`; unique `acr_id`; (`credit_status`, `credit_expiry`)	ACR lookup; expiry alerts (FR-194)
`credit_ledger_entries`	(`abc_deposit_id`, `occurred_at`)	Status history of a deposit
`audit_events`	unique `seq`; (`subject_type`, `subject_id`, `seq`); (`actor_id`, `occurred_at`)	Trace a record or a person (FR-258)

Table	Index	Purpose
notifications	(user_id, read_at, created_at descending)	Bell count and list
integration_jobs, integration_calls	unique idempotency_key (jobs); (status, next_attempt_at) (calls)	Outbox re-dispatch, failed list for the system administrator
job_runs, alert_log	unique idempotency_key; unique (subject_type, subject_id, alert_kind, offset_days)	Idempotent jobs and alerts
verification_logs	(occurred_at); (network_address, occurred_at)	Rate-limit review and abuse checks
reference_sequences	unique (kind, scope_key, year)	Number generation

Capacity at registration (NFR-06) is protected by locking the offerings row (lockForUpdate) while the places taken are counted and the enrolment row is inserted. No separate counter column is kept.

5.12 Migrations and Seed Data

5.12.1 Migration Order

Migrations live in the module that owns the table (layout in Section 3). Their file names start with a timestamp, and a migration that adds a foreign key runs after the migration of the table it points to. They are forward-only in staging and production and are applied by the deployment pipeline (NFR-32). Destructive changes follow expand, migrate, contract over separate releases.

Table 70: *Migration order*

Step	Module	Tables created
1	MD-00 Platform	settings, setting_versions (the user FK is added in step 3), templates, template_versions, stored_files, reference_sequences
2	MD-15 Records	audit_chain_heads, audit_events, so that every later table can be audited from its first row
3	MD-01 Identity	ptjs, programmes, parent_courses, users, role_assignments, delegations, conflict_declarations, notification_preferences (from MD-16); then the deferred foreign keys of steps 1 and 2 to users
4	MD-02 Curriculum	micro_credentials, mc_versions, learning_outcomes, mc_modules, assessment_components, assessment_component_outcomes, proposal_collaborators, change_requests; then role_assignment_micro_credentials
5	MD-03 Approval	review_assignments, reviews, review_comments, approval_decisions, version_transition_rules
6	MD-10 Quality verification	qv_records, qv_evidence, qv_submissions (before offerings, which point to QV)
7	MD-04 Offering	offerings, readiness_items, readiness_confirmations; then the foreign key of qv_evidence to offerings
8	MD-05 Enrolment	enrolments, module_enrolments, payments, pdpa_acknowledgements
9	MD-06 Learning platform	import_files, import_row_errors, export_files
10	MD-07 Assessment	assessment_results, assessment_result_revisions, integrity_cases, result_change_requests, result_appeals
11	MD-08 Achievement record	achievement_records, module_achievements
12	MD-09 Credentials	credentials, digital_badges, credential_events, verification_logs
13	MD-11 Credit bank	acrs, acr_checks, consents, abc_batches, abc_deposits, redemption_events, credit_ledger_entries
14	MD-12 Transfer	transfer_applications, transfer_lines, transfer_decisions, equivalence_assessments, admissions; then the foreign key of redemption_events to transfer lines
15	MD-13 Pathways	pathways, pathway_items, stackable_plans, plan_requirements
16	MD-14 Quality and change	cqi_records, cqi_actions, periodic_reviews, suspensions, feedback_responses, quality_domain_checks
17	MD-15 Records	corrections, discrepancies, retention_holds, disposal_records, report_runs
18	MD-16 Integration hub	notifications, adapter_configs, integration_jobs, integration_calls, job_runs, alert_log
19	MD-00 Platform	Database roles and grants (mcms_app, mcms_migrator, mcms_disposal), the triggers on append-only and frozen tables, and the full-text indexes. This step is the only place with PostgreSQL-specific SQL (AD-03)

A cycle exists between micro_credentials and mc_versions only if a pointer to the current version is stored. The design stores no such pointer: the current version is the latest version in Approved status, found by query. Likewise qv_records and offerings point at each other, so the first foreign key is added in step 7.

5.12.2 Seed Data

There are two seeders. `ReferenceSeeder` is safe for production: it creates the settings and their first versions (the defaults of the prototype, for example 40 learning hours per credit, QV validity of 3 years, credit validity of 10 years, 30 percent home credit, 30-minute idle timeout), the templates in both languages, and the PTj list. `SampleWorldSeeder` loads the prototype's sample world for development, test and staging. It checks `APP_ENV` and refuses to run in production (AD-18). The prototype's "today" is 26 October 2026; the seeder sets the demo clock to that date.

The sample world is written through the same services as the application, not by direct inserts, so that audit rows, hash chains, reference numbers and signed credentials are valid. It is the data on which the Pest and Playwright tests run (AD-17).

Table 71: *Prototype sample data and the tables it seeds*

Prototype key (data.js and screen files)	Seeded into	Content
`people`, `users`	`users`, `role_assignments`	Amir Hakimi (participant), Dr. Aina Zulkifli and Dr. Rosli Hamzah (lecturers), Hakim Rahman and Dr. Farah Nadia Ismail (deputy deans), Farid Hashim (system administrator), Ir. Siti Aminah Yusof (industry expert), and 18 further participants
`ptj`, `programmes`, `courses`	`ptjs`, `programmes`, `parent_courses`	FTMK, FKE, FPTT, FTKIP, CODL; BITZ and BITD; five courses
`mcs`	`micro_credentials`, `mc_versions`, `mc_modules`, `learning_outcomes`, `assessment_components`	Ten micro-credentials in the three categories at different statuses (Draft, In review, Pending approval, Approved)
`cohorts`	`offerings`, `readiness_items`	Offerings from Not ready to Ended, including DS-2026-2 with 28 learners (weights 40, 30, 30, pass mark 50, one failing learner)
`enrolments`	`enrolments`, `module_enrolments`, `payments`	Amir's four enrolments, and the learners of each cohort
`credentials`	`credentials`, `credential_events`, `digital_badges`	MCMS-C-2026-000098, 000209, 000418 and the revoked 000377, signed with the test key
`acrs`, `acrBatches`	`acrs`, `consents`, `abc_deposits`, `abc_batches`, `credit_ledger_entries`, `redemption_events`	The deposited ACR 000102, the verified ACR 000214 awaiting consent, and batch ABC-FTMK-2026-03
`transfers`, `admissions`, `plans`, `pathways`	`transfer_applications`, `transfer_lines`, `equivalence_assessments`, `admissions`, `stackable_plans`, `pathways`, `pathway_items`	TR-2026-0012 and TR-2026-0009, plans SP-2026-003 and 004, pathway PW-BITZ-1
`reviews`, `ddComments`	`reviews`, `review_comments`	Four reviews and their comments
`cqi`	`cqi_records`, `cqi_actions`	CQI-DS-2026-1 and CQI-CND-2026-1
`settings`, `settingHistory`	`settings`, `setting_versions`	Sixteen settings and the change of `alerts.qv_expiry_days` of 2 September 2026
`imports`	`import_files`, `export_files`, `import_row_errors`	Four UCreds files, including results-CND-2026-2.csv with two rejected rows
`notifs`	`notifications`	The notifications of each persona
`audit`	`audit_events`	Written by the services above as they run; not copied
`tasks`, `adapters`	not stored	The task queue is computed from the data. Adapter status comes from configuration and `integration_jobs`

Tests find seeded rows by code or reference (for example `DS-2026-2`), never by ID. The seeder uses a fixed random seed so that references and public codes are the same on every run.

6. Interface Design

This section describes how MCMS meets its users and how it exchanges data with other systems. Screen-level layouts are in the UI/UX Specification v1.1 and are not repeated. The external interfaces implement SRS Section 5 (IR-14, IR-15, IR-16): each is a PHP interface with Stub, Manual and Live implementations chosen by configuration (AD-08), and every call is recorded in `integration_calls` (Section 5).

6.1 User Interface

6.1.1 Route Conventions

MCMS has one web application with three front ends that share one design: the public pages, the participant portal and the staff portal (IR-01). The prototype uses hash routes such as `#/dd/verify/DS-2026-2`. In MCMS the same path is a real URL: `/dd/verify/{cohort}`.

Table 72: *Route conventions*

Rule	Convention
Role prefix	A staff or participant screen starts with the role prefix of the UI/UX Specification: `/st` (participant), `/lec` (lecturer), `/ext` (industry expert), `/dd` (deputy dean), `/codl` (CODL administrators). A user with several roles (FR-11) sees one menu and may open any prefix that a role allows
Shared routes	`/profile`, `/notifications`, `/help`, `/search` serve every signed-in role. `/signin`, `/signin/mfa`, `/signin/noaccount` and, outside production, `/signin/stub` serve sign-in
Public routes	No prefix: `/` (home), `/catalogue`, `/offering/{reference}`, `/pathways`, `/how`, `/faq`, `/verify`, `/verify/{code}`, `/privacy`, `/terms`, `/accessibility`
Path words	The words follow the prototype: `/dd/queue`, `/dd/proposal/{id}`, `/dd/senate/{id}`, `/dd/readiness/{cohort}`, `/dd/verify/{cohort}`, `/dd/completion/{cohort}`, `/dd/cred/{id}`, `/dd/acr/{id}`, `/dd/deposit/{batch}`, `/dd/transfer/{id}`
Route keys	A path parameter is the human reference of the record (DS-2026-2, MCMS-C-2026-000098, TR-2026-0012), not the ULID. Models define `getRouteKeyName()` as `reference`. The reference is not a secret: the policy decides access (SR-03)
Printable documents	`/doc/{kind}/{id}` returns the HTML document. The same path with `.pdf` returns the PDF and with `.csv` or `.json` the machine-readable file where one exists (Section 6.2.9)
Route names	Dot names that mirror the path: `dd.verify`, `st.transfer.show`. Each page root carries `data-screen="DD-09"` with its UI/UX screen ID, which the Playwright tests use (AD-17)
Middleware	`web`, `auth`, `locale`, `idle-timeout` and `role:{prefix}`. The `role` middleware only checks that the user holds the role. The Policy of the record decides everything else (SR-03, AD-07)
Not found and forbidden	A record outside the user's scope returns the same "not found" page as a missing record, so existence is not revealed
Query state	Search, filter, sort, page and rows per page of a table are query-string parameters, so a list can be bookmarked and the back button works
Language	The locale comes from the user's profile. A signed-out visitor uses the `locale` cookie, and the switch changes the page in place (FR-08, NFR-25)

6.1.2 Page Anatomy

Every signed-in page is rendered by the shell component and has the same parts (IR-03, UI/UX Specification Section 4):

- **Top bar** with the search box, the language switch, the notification bell with the unread count and the profile menu.
- **Side navigation** in navy, built from the role's menu. Counts on menu items come from the same query as the task queue.
- **Page head** with the title, a short subtitle and the primary action. Records show a stage track for the recognition stage they belong to (IR-04).
- **"Needs you" panel** on list pages, listing the items the user must act on first.
- **Panels and tables** holding the content. A table is the `DataTable` Livewire component with search, sort, filters and pagination (5, 10, 25 or 50 rows).
- **Dialogs, drawers and toasts** for every state change, as UI/UX Specification Sections 4.1 to 4.3 define. A state-changing button opens a confirmation dialog first (IR-06). The dialog calls a Livewire action. The action calls the application service of the owning module, inside a database transaction that also writes the audit row (AD-11).
- **Long jobs** (credential issue, imports, large reports) show a `JobProgress` component that polls the matching `job_runs` row every two seconds and shows "issuing 14 of 27" (NFR-11).

The shared Blade and Livewire components are listed in Section 3. Pages are full-page Livewire components, one per screen. A page that is only a view with no interaction may be a Blade view served by a controller.

6.1.3 Screens by Role

The 114 screens of the approved prototype are listed in UI/UX Specification Section 6.6. Table 73 summarises them by role and names the owning modules. Screen IDs are used in tests and in Section 4.

Table 73: *Screens by role (UI/UX Specification Section 6.6)*

Role	Prefix	Home route	Screens	Main areas (examples)	Modules
Common and sign-in	none	`/signin`	10	Sign-in, MFA check, account not recognised, profile, notifications, help, search, error (CMN-01 to CMN-10)	MD-01, MD-16
Public	none	`/`	11	Catalogue and offering (PUB-02, PUB-03), pathways (PUB-04), home, how recognition works and FAQ (PUB-01, PUB-05, PUB-06), privacy, terms and accessibility (PUB-09, PUB-10, PUB-11; static Blade pages), verify a credential (PUB-07, PUB-08)	MD-04, MD-09, MD-13
Participant (P8)	`/st`	`/st/home`	14	Find and register (ST-02, ST-03), registrations and payments (ST-04, ST-05), Achievement Record (ST-06), credentials (ST-07, ST-08), credit ledger and ACR (ST-09, ST-10), transfer (ST-11 to ST-13), stackable plan (ST-14)	MD-05, MD-08, MD-09, MD-11, MD-12, MD-13
Lecturer (P1)	`/lec`	`/lec/home`	18	Proposals and versions (LEC-03 to LEC-06), reviews (LEC-07, LEC-08), cohorts and results (LEC-09, LEC-10), offering and readiness (LEC-11, LEC-12), CQI (LEC-13, LEC-14), QV evidence (LEC-15, LEC-16), reports	MD-02, MD-03, MD-04, MD-07, MD-10, MD-14
Industry expert (X2)	`/ext`	`/ext/home`	2	Review invitations and comments (EXT-01, EXT-02)	MD-03
Deputy dean (P2)	`/dd`	`/dd/queue`	29	Task queue (DD-01), proposals and Senate decision (DD-02 to DD-05), readiness (DD-06, DD-07), verify results and confirm completion (DD-08 to DD-10), credentials (DD-11, DD-12), QV, CQI, review and suspension (DD-13 to DD-17), ACR and deposits (DD-18 to DD-20), transfer (DD-21, DD-22), stackable pathways (DD-23 to DD-25), reports and traceability (DD-26 to DD-29)	MD-03, MD-04, MD-07, MD-08, MD-09, MD-10, MD-11, MD-12, MD-13, MD-14, MD-15
CODL administrator (S1, P5)	`/codl`	`/codl/users`	13	Users and roles (CODL-01, CODL-02), reference data, settings, templates (CODL-03 to CODL-06), UCreds exchange, adapter status, test mailbox (CODL-07 to CODL-09), audit trail, reports, retention (CODL-10 to CODL-13)	MD-00, MD-01, MD-06, MD-15, MD-16
Printable documents	`/doc`	`/doc/index`	17	DOC-01 to DOC-17 (Section 6.2.9)	MD-00 and the owner of each record

The CODL screens use the prototype's wording "System administrator" and "CODL operator" for the two CODL roles. CODL roles never reach an academic action screen (FR-15, AD-07). The deputy dean screens that record a decision show the CODL role only as a viewer where the SRS allows it.

6.2 External Interfaces

Table 74 lists the external interfaces. All of them run through the integration hub (MD-16). Outbound calls are placed on the `integration` queue and retried with increasing delay (FR-271). A call that still fails is listed on the adapter status screen (CODL-08) and does not block the business process that created it. Retry follows JB-16 (`RetryAdapterCallJob`): back-off of 1, 5, 15 and 60 minutes, then Failed and EV-63. These values are configuration.

Table 74: *External interfaces, adapters and modes in Release 1*

Interface	PHP interface	Protocol	Direction	Release 1 mode	Requirements
Microsoft Entra ID	`IdentityProvider`	OpenID Connect, authorisation code with PKCE	In (sign-in)	Live in production, Stub elsewhere	FR-01, IR-18
UCreds	`LearningPlatformGateway`	CSV files over UTF-8	Out (enrolments), in (results, badges)	Manual (CSV)	FR-111, FR-112, IR-19
Payment	`PaymentGateway`	HTTPS and JSON	Out (request), in (callback)	Stub (dummy service)	FR-101, IR-17
Email	`MailTransport`	SMTP with TLS	Out	Live	IR-23, FR-268

Interface	PHP interface	Protocol	Direction	Release 1 mode	Requirements
ABC	`AbcGateway`	CSV and PDF files	Out (deposit), in (reference recorded by hand)	Manual	FR-188, FR-220
QVC and MQA	`QvGateway`	Zip package	Out (package), in (decision recorded by hand)	Manual	FR-169, FR-170
Credential signing	`SigningKeyStore`	Key store call (sign request)	Out	Live (key vault of CODL)	SR-08, IR-20
Public verification	none (a route)	HTTPS, HTML and JSON	In, no login	Live	FR-153, IR-21
Open Badges credential	none (a file)	JSON-LD, W3C Verifiable Credential	Out (download)	Live	FR-142, IR-20
Printable documents	none (a service)	HTML to PDF	Out (download)	Live	IR-07

The student information system and Release 2 and 3 live connectors (UCreds, ABC, QVC and MQA, payment) have no Release 1 implementation. Each interface above already has the Stub and Manual classes, so a Live class is added without changing a business module (FR-117).

6.2.1 Microsoft Entra ID (OpenID Connect)

Sign-in uses Laravel Socialite with the Microsoft provider, set to the UTeM tenant (AD-06, IR-18). The flow is the authorisation code flow with PKCE. MCMS stores no password (FR-02).

1. The user opens `/signin` and chooses to sign in. MCMS redirects to the Entra authorisation endpoint with scopes `openid profile email`, a random `state`, a `nonce` and the PKCE challenge.
2. Entra authenticates the user, applies Conditional Access (MFA for staff) and redirects to `/auth/callback` with a code.
3. MCMS exchanges the code for tokens, checks the ID token (signature against the tenant keys, `iss`, `aud`, `exp`, `nonce`), and reads the claims in Table 75.
4. MCMS finds the user by `oid` (FR-09). If none exists it applies the first-sign-in rules (FR-03); an account that is not known and not pre-registered gets the "Account not recognised" page (CMN-04).
5. A session is started. Idle timeout is 30 minutes, from the setting `auth.idle_timeout_minutes`, with a warning two minutes before (FR-06).

Table 75: *ID token claims used*

Claim	Use	Handling
`oid`	The stable identity of the person in the tenant	Stored in `users.entra_oid`. The key of the account. Never shown
`upn` or `email`	Sign-in name and contact email	Stored in `users.upn` and `users.email`. Refreshed at each sign-in. Not used as the key, because a name can change
`name`	Display name	Shown read-only on the profile. The name printed on certificates is a separate editable field (FR-07)
`amr`	How the user authenticated	A staff role (any role other than participant) requires the value `mfa`. If it is missing the session is not opened (below)
`groups`	Directory group identifiers	Used only to decide the account type (staff or student) on first sign-in. Groups never grant a role. Roles come only from `role_assignments` (AD-07)

The staff or matric number (`users.institution_id`) is read from a directory attribute delivered in the token. The attribute name is a configuration value that CODL confirms (DA to be added, see the report to the lead).

Table 76: *Sign-in error handling*

Situation	Behaviour
`state`, `nonce` or signature invalid	The callback is refused with HTTP 400 and the sign-in page. A security log entry is written (SR-18)
Entra returns an error or is unreachable	The sign-in page shows a plain message with an ERR reference and a retry button. Users with a live session are not affected
Account not recognised	CMN-04 explains how external learners register at Pejabat Pendaftar or HEPA. No account is created (FR-02)
Account deactivated	The same page as above, with a message that the account is inactive. A security log entry is written

Situation	Behaviour
Staff role and no `mfa` in `amr`	CMN-03 "Verify it is you". MCMS sends the user back to Entra with `prompt=login`. If `amr` still lacks `mfa`, the session is refused and the event is logged (SR-02)
Participant	The second step is not required (FR-01)
Stub provider	Available only when `APP_ENV` is not `production`. The application refuses to start in production when the stub is enabled (AD-06, FR-05)

6.2.2 UCreds: CSV Export of Enrolments

CODL exports the enrolments of an offering to a CSV file and loads it into UCreds by hand (CODL-07, FR-111). A lecturer may also download the file from the cohort screen (LEC-10). Only participants whose enrolment status is Enrolled are included. The job (JB-03, `ExportUcredsEnrolmentJob`) runs on the `imports` queue, writes an `export_files` row, and stores the file in object storage (FR-116).

Table 77: *Enrolment export, template `enrol v3` (file name `enrol-{offering}.csv`)*

Column	Type	Content and rule
`learner_id`	string	Matric number or participant identifier (`users.institution_id`). Required
`full_name`	string	Name as in the official record. Required. Written in full including bin, binti, a/l and a/p (NFR-45)
`enrolment_id`	string	The enrolment reference, for example E-2026-0644. Required. Unique in the file
`offering`	string	Offering reference, for example DS-2026-2
`cohort`	string	The cohort label of the offering, for example Cohort 2026/2
`start_date`	date	ISO 8601, `YYYY-MM-DD`
`end_date`	date	ISO 8601, `YYYY-MM-DD`

The file is UTF-8 (with no byte-order mark), comma-separated, with a header row, quoted fields where needed and line ends of `CRLF` (IR-19). It carries no identity card number, email address or marks. The same export of the same offering and template version produces the same bytes in the same order (sorted by `learner_id`), so a repeated export does not create a conflicting file (IR-17). The template version is stored on the `import_files` row and printed on the export document (DOC-15).

6.2.3 UCreds: CSV Import of Results

UCreds results are loaded from a CSV file (CODL-07, FR-112). Imported results are stored as Provisional and create no achievement, badge or credential until the deputy dean verifies them (FR-114, BR-20). CODL cannot verify.

Table 78: *Results import, template `results v3` (file name `results-{offering}.csv`)*

Column	Type	Content and rule
`learner_id`	string	Must match a participant enrolled in the offering. Required
`enrolment_id`	string	Must match the enrolment of that learner. Required
`offering`	string	Must equal the offering chosen on the screen
`progress_percent`	integer	0 to 100
`modules_completed`	integer	0 up to the number of modules of the version
`{csv_key}` (one column for each assessment component, for example `module1_mark`, `module2_mark`, `project_mark`)	decimal	A score from 0 to the component's `max_mark`, with a decimal point and at most two decimals. Required and not empty. The set of columns is fixed by the version's `assessment_components.csv_key` and published with the template version

Table 79: *Badge import, template `badges v3` (file name `badges-{offering}.csv`)*

Column	Type	Content and rule
`learner_id`	string	Must match an enrolled participant. Required
`enrolment_id`	string	Must match. Required
`module_no`	integer	The module sequence number within the version
`badge_id`	string	The UCreds badge identifier. Required
`badge_url`	URL	HTTPS link to the badge
`badge_date`	date	ISO 8601

A badge row is matched to the verified module achievement of that learner and stored in `digital_badges` (EV-27 BadgeImported). A badge without a verified achievement is flagged on the import report and not accepted (FR-115).

Table 80: *Validation rules and staging*

Level	Rule	Result
File	Extension `.csv`, size within the upload limit (SR-09), scanned for malware (AD-14), valid UTF-8, one header row, comma delimiter	
File	The header set equals exactly one accepted template version, and the version is the one chosen on the screen. A missing column is named, for example "column module3_mark is missing, this file uses template v2, v3 is required"	
File	The offering exists and is Running or Ended	
File	Any file-level error above	The whole file is rejected. No row is staged. The report says why (FR-113)
Row	`learner_id` and `enrolment_id` match an enrolment of the offering. Otherwise "unknown matric number"	
Row	Every mark column is present and numeric, within range, with at most two decimals. Otherwise "project_mark is empty" or "score out of range"	
Row	Dates are ISO 8601 (IR-25)	
Row	The learner appears once in the file. A second row is rejected as a duplicate	
Row	A row for a result that is already Verified is rejected with the reason "already verified". Correction goes through FR-125	
Row error	A row with an error is listed in a downloadable report (DOC-14) with the row number, the learner and the reason. The report does not show marks	
Staging	Valid rows are written to `assessment_results` with status Provisional, with the `import_file_id`. Rejected rows are stored in `import_row_errors` with the reason only (FR-114)	
Same file twice	The SHA-256 of the file is compared with earlier imports of the offering. If equal, the screen warns and staging again replaces the Provisional rows of the earlier import, with no duplicates (IR-17)	
Size and time	5,000 rows validated and staged within 2 minutes (NFR-07). Validation is JB-02 (`ImportResultsJob`) on the `imports` queue. It reads the file in a stream, in chunks	
Notice	The lecturer is told which marks are missing. The deputy dean is told that results are staged. Each import writes an audit row and an `import_files` row (FR-116), and raises EV-25 (ResultsStaged) or EV-26 (ImportRejected)	

6.2.4 Payment Service (Stub)

The `PaymentGateway` interface has two methods: `createPayment(PaymentRequest)` and `getStatus(string $providerRef)`. The Release 1 implementation is a dummy service inside the application, selected by configuration, with a Live class added later (R3, FR-101).

Table 81: *Payment request, result and callback*

Item	Content
Request	JSON: `idempotency_key` (the enrolment reference plus the attempt number), `amount` (decimal string), `currency` (`MYR`), `enrolment_ref`, `description`, `payer_name`, `return_url`, `callback_url`. No identity number is sent
Response	JSON: `provider_ref`, `status` (`pending`, `paid`, `failed`), `redirect_url` when the participant must pay on the provider's page, `failure_reason` when failed
Callback	The provider calls `POST /integrations/payment/callback` with the JSON result and an HMAC-SHA256 signature header over the body. MCMS checks the signature with a shared secret from configuration, then updates the `payments` row and the enrolment
Idempotency	`payments.idempotency_key` is unique. A repeated `createPayment` with the same key returns the stored result. A repeated callback for a finished payment changes nothing (IR-17)
Failure	A failed result leaves the enrolment at Pending payment and shows the reason with a retry button. A retry makes a new attempt with attempt number plus one and a new key. No receipt is issued for a failed payment
Timeout or no answer	The call is marked failed with a retry. A reconciliation job calls `getStatus` for payments that stay pending and for calls that gave no answer, so a payment made at the provider is never lost
Receipt	When the status is `paid`, the enrolment moves to Enrolled in the same transaction and the receipt document (DOC-01) becomes available (FR-99)
Stub behaviour	The stub returns `paid` or `failed` by a rule on the amount or the payer name, so that tests can force each path. It is disabled in production (AD-18)

6.2.5 Email (SMTP)

The `MailTransport` interface is implemented with Laravel Mail over the UTeM SMTP relay, with TLS (IR-23, AD-15). Mail is sent from a monitored UTeM address set in configuration.

- **Outbox.** A notification is first inserted into `notifications` (the user sees it in the bell at once). JB-05 (`SendNotificationJob`) on the `notifications` queue then sends the email. The send is idempotent per notification: a notification with `email_status` Sent is never sent again.
- **Content.** The subject and body come from the published template version in the recipient's language (FR-266). A message names the record and links to the page. It carries no marks, identity number or credential payload.
- **Mandatory notices** (decisions on the user's records, credential actions, consent requests, enrolment changes) are always sent. Other notices follow preferences, and lecturers and deputy deans may choose a daily digest (FR-267, FR-269).
- **Failure.** A transient SMTP error (for example 4xx) sets the call to retry with increasing delay. A permanent error (5xx) or a bounce marks the notification Failed and lists it for the system administrator. The notification stays visible in the bell. Failures do not roll back the business action (NFR-18).
- **Test mailbox.** Outside production the transport writes mail to the test mailbox (CODL-09), not to the relay.

6.2.6 ABC Deposit Export

The deputy dean prepares a batch of ACR that have consent and have been confirmed (FR-187). The export is a CSV file and a PDF document (DOC-10), for staff to send to ABC by an approved channel (IR-26). The format is not yet agreed with ABC (OI-11); the columns below follow the ACR fields of Appendix F3 and are the proposed format (FR-188). JB-13 (`ExportAbcDepositJob`) builds the file. The batch can be exported again; the file for a given batch and format version has the same content.

Table 82: *ABC deposit export, proposed format `abc v1` (file name `{batch}.csv`)*

Column	Source	Note
`acr_reference`	`acrs.reference`	For example MCMS-A-2026-000102
`created_on`	`acrs.created_at`	ISO 8601 date
`learner_id`	`users.institution_id`	Participant identity. Names, identity card numbers, emails and marks are not in the file
`issuer`	UTeM and the PTj	Provider of the credential
`credential_name`	`mc_versions.name_en` and code	Course or credential name
`learning_outcomes`	`learning_outcomes.text_en`	Joined with a semicolon
`credits`	`acrs.credits`	Verified credit only, decimal
`mqf_level`	`acrs.mqf_level`	Integer 1 to 8
`completion_date`	`acrs.completion_date`	ISO 8601
`assessment_basis`	`acrs.assessment_basis`	Text
`verified_by`, `verified_on`	`acrs.confirmed_by`, `acrs.confirmed_at`	Verification metadata
`consent_date`	`consents.decided_at`	Only rows with consent Given are exported

The prototype export has seven columns (reference, learner ID, micro-credential, credits, MQF level, completion date and consent date). The extra columns above come from FR-179 and are to be reduced or renamed when OI-11 is closed. The format version is stored in `abc_batches.format_version`.

- **Direction in.** ABC returns the ABC reference by a channel outside MCMS. The deputy dean records, for each ACR, the deposit date, the ABC reference, the credits and the outcome (Successful, In process, Correction needed) (FR-189). A reference range for a batch (for example 4391 to 4419) is entered once and expanded by the application.
- **Idempotency.** One ACR has at most one deposit row (unique `acr_id`). Recording the same ACR again is refused and shows the earlier record (IR-17).
- **Errors.** An ACR whose consent is missing, declined or withdrawn is excluded, and the screen says which. A batch that is empty cannot be exported.
- **Later status.** For every credit status change that must also be made in ABC, the user records whether it was done, with date and reference (FR-220).

6.2.7 QVC and MQA Submission Package

For a SAMC, staff generate the submission package from the QV record (DD-14, LEC-16, FR-169). JB-14 (`BuildQvPackageJob`) on the `reports` queue builds a zip file, stores it in object storage and links it to `qv_submissions.package_file_id`. The package is for staff to submit to QVC and MQA by an approved channel. Staff then record the submission date, the QVC or MQA reference and the person who submitted in `qv_submissions` (FR-170). The decision is recorded by hand (FR-171).

Table 83: *Contents of the QV submission package*

File in the zip	Content
`00-cover-and-declaration.pdf`	The QV submission package document (DOC-11): cover, declaration, signatory
`01-evidence-index.pdf` and `.csv`	The checklist items, what evidence answers each, file names and SHA-256 hashes

File in the zip	Content
`02-summary.pdf`	Summary of the micro-credential: category, version, credit, MQF level proposed, learning outcomes, assessment
`03-cohort-results.csv`	Aggregate results of completed cohorts: counts, pass rate, mark distribution. No names or identity numbers
`04-quality-reviews.pdf`	CQI records of the cohorts, with actions and their effectiveness
`evidence/`	The accepted evidence files, in folders by checklist item
`manifest.json`	Package version, QV record reference, generation time, and the SHA-256 of every file

The package is generated again on request and each generation is stored as a new file; the one recorded as submitted is kept. A package cannot be generated while required checklist items are not accepted (FR-168). A file that fails the malware scan is excluded and listed on the screen. The package is generated by a job, so a long build does not hold a web request (NFR-11).

6.2.8 Public Verification Page and JSON API

Anyone can check a credential without signing in (FR-153). The page is reached by the code, a link or a QR code that opens the link (IR-11). The same data is served as JSON for other systems (IR-21). Responses follow NFR-03: 95 percent within one second.

Table 84: *Verification endpoints*

Endpoint	Method	Returns
`/verify`	GET	The HTML form to enter a code (PUB-07)
`/verify/{code}`	GET	The HTML result page (PUB-08). With the header `Accept: application/json` it returns the JSON below
`/api/v1/verify/{code}`	GET	The JSON result. Read-only, no authentication, rate limited
`/issuer`	GET	The issuer profile (name, URL) as JSON-LD, linked from every credential
`/issuer/keys/{key_id}`	GET	The public verification key as a Multikey document, so a credential can be checked without calling MCMS (IR-20, SR-08)

Rules for the response:

- **Fields.** Only the holder's name, the credential name, the issuer, the issue date and the status (`valid`, `corrected`, `replaced`, `revoked`), plus any field listed in the setting `verification.public_fields` (FR-154). No identity card or passport number, matric number, email address or result is ever returned.
- **Lookup.** The code is normalised (Section 5.5) and looked up by the unique `public_code`. A revoked credential still answers with status `revoked`. A credential that was corrected or replaced answers with that status and no link to the new credential.
- **Not found.** An unknown or malformed code returns HTTP 404 with the same neutral body in every case: no hint whether a similar code exists (FR-157). The code in the URL is never echoed in the body.
- **Rate limit.** Verification and catalogue routes are limited by network address with Laravel's rate limiter (SR-17). A stricter limit applies to not-found answers, which are the signature of code guessing. Above the limit the response is HTTP 429 with a `Retry-After` header. The limits are configuration values (a DA is proposed in the report).
- **Logging.** Each request writes a `verification_logs` row with the time, the credential (when found), the outcome and the network address. A not-found request stores a hash of the code, not the code (FR-156).
- **Headers.** `Cache-Control: no-store` so a revocation is visible at once. The page has no cookies and no scripts from other sites.
- **File check.** A verifier may upload a Digital Credential file. The page checks its signature against the published key and compares it with the stored payload hash (FR-155). The file is not stored.

```
GET /api/v1/verify/3HXP92LDKQ        200 OK
{
  "status": "valid",
  "holder_name": "Amir Hakimi",
  "credential": "Data Structures",
  "issuer": "Universiti Teknikal Malaysia Melaka",
  "issued_on": "2026-03-03"
}

GET /api/v1/verify/ZZZZZZZZZZ        404 Not Found
{
  "status": "not_found",
  "message": "No credential matches this code."
}
```

6.2.9 Open Badges 3.0 Credential

A certificate is issued with a Digital Credential: a JSON-LD document that follows the Open Badges 3.0 and W3C Verifiable Credentials data models and is signed with Ed25519 using a Data Integrity proof of cryptosuite `eddsa-rdfc-2022` (AD-09, FR-142). The document is built on the `credentials` queue by JB-22 (`IssueCredentialJob`, started by JB-01 after EV-31 CompletionConfirmed), signed by the `SigningKeyStore`, and stored in `credentials.signed_payload`. A module badge is a separate object in `digital_badges` and is imported from UCreds or issued by MCMS (EV-29, EV-33). The application never holds the private key: it sends the canonical hash to the key store and receives the signature (SR-08). Blockchain is not used. The example is shortened; the host name is a placeholder.

```
{
  "@context": [
    "https://www.w3.org/ns/credentials/v2",
    "https://purl.imsglobal.org/spec/ob/v3p0/context-3.0.3.json"
  ],
  "id": "https://mcms.example.my/verify/3HXP92LDKQ",
  "type": ["VerifiableCredential", "OpenBadgeCredential"],
  "issuer": {
    "id": "https://mcms.example.my/issuer",
    "type": ["Profile"],
    "name": "Universiti Teknikal Malaysia Melaka"
  },
  "validFrom": "2026-03-03T02:00:00Z",
  "name": "Data Structures",
  "credentialSubject": {
    "type": ["AchievementSubject"],
    "identifier": [{
      "type": "IdentityObject",
      "identityType": "name",
      "hashed": false,
      "identityHash": "Amir Hakimi"
    }],
    "achievement": {
      "id": "https://mcms.example.my/achievements/DS-v2",
      "type": ["Achievement"],
      "achievementType": "MicroCredential",
      "name": "Data Structures",
      "description": "Unbundled Micro-Credential (UMC) from BITS 1123",
      "creditsAvailable": 3,
      "criteria": { "narrative": "Completed all modules and passed assessment." }
    }
  },
  "proof": {
    "type": "DataIntegrityProof",
    "cryptosuite": "eddsa-rdfc-2022",
    "created": "2026-03-03T02:00:05Z",
    "verificationMethod": "https://mcms.example.my/issuer/keys/2026-1",
    "proofPurpose": "assertionMethod",
    "proofValue": "z3FXQ... (base58 signature)"
  }
}
```

- **Content.** The credential carries the verified credit only when it is verified and valid. A SAMC adds the QV claim (QV status, MQF level) with the QV record reference (DR-05). Names and descriptions are given in both languages by language maps in the stored version.
- **Identity.** The document contains the holder's name as in the official record. It contains no identity card number, matric number or email address.
- **Key id.** The `verificationMethod` names the key. A new key can be used for later credentials, and old keys stay published so earlier credentials still verify (SR-08).
- **Download.** The holder downloads the signed JSON and a PDF rendering (DOC-04) (FR-147).

6.2.10 Printable Documents

Every printable document is a Blade A4 template with a print bar, rendered to PDF by headless Chromium (Spatie Browsershot) by JB-04 (`RenderPdfJob`) on the `reports` queue, except the certificate and Digital Credential, which are issued by JB-01 and JB-22 on the `credentials` queue (AD-10, IR-07). Templates are bilingual and versioned in `templates` and `template_versions` (FR-22). The template version used is stored with the generated file. Certificates and Digital Credentials are rendered again from the stored signed payload, never from live data, so a reprint always matches what was issued.

Table 85: *Documents, template keys and outputs*

ID	Document	Template key	Output	Source of data
DOC-01	Payment receipt	`doc.receipt`	PDF	`payments`, `enrolments`
DOC-02	Enrolment confirmation	`doc.enrolment`	PDF	`enrolments`, `offerings`
DOC-03	Certificate	`doc.certificate`	PDF	`credentials` (signed payload)
DOC-04	Digital credential	`doc.credential`	JSON, PDF	`credentials` (signed payload)
DOC-05	Achievement record	`doc.record`	PDF	`achievement_records`, `module_achievements`, `digital_badges`
DOC-06	Academic Credit Record	`doc.acr`	PDF	`acrs`, `abc_deposits`
DOC-07	Proposal print	`doc.proposal`	PDF	`mc_versions` and children
DOC-08	Senate decision pack	`doc.pack`	PDF	`mc_versions`, `reviews`, `approval_decisions`
DOC-09	Report (ten kinds)	`doc.report`	PDF, CSV	Report queries (Section 4, MD-15)
DOC-10	ABC deposit export	`doc.abc`	PDF, CSV	`abc_batches`, `acrs`
DOC-11	QV submission package	`doc.qvpack`	PDF, zip	`qv_records`, `qv_evidence`
DOC-12	Credit transfer decision letter	`doc.transfer`	PDF	`transfer_applications`, `transfer_lines`
DOC-13	Suspension notice	`doc.suspend`	PDF	`suspensions`
DOC-14	Import validation report	`doc.import`	PDF, CSV	`import_files`, `import_row_errors`
DOC-15	UCreds enrolment export	`doc.export`	PDF, CSV	`export_files`, `enrolments`
DOC-16	Audit trail export	`doc.audit`	PDF, CSV	`audit_events` (with integrity line)
DOC-17	Documents and PDFs	none (index page)	HTML	List of the above

Each page has a header with the UTeM name, the document title, the reference and the date, and a footer with the reference, the page number, the generation time and "Generated by MCMS". Participant documents are bilingual, with BM beneath EN. Documents that leave UTeM show no identity number, email or marks. A PDF is generated by a job and the user is notified when it is ready; small documents are returned directly.

6.3 Internal Event Catalogue

Modules communicate by domain events as well as by service interfaces (AD-01). An event is a PHP class named in the past tense. It is dispatched after the database transaction that caused it commits (`afterCommit`), and its listeners that do slow work run on a queue. The numbering EV-01 to EV-63 is owned by Section 4 and is reused here without change; Table 86 repeats the catalogue of Section 4.19.1 so that Section 6 can be read alone. Jobs (JB-01 to JB-22) are listed in Section 4.19.2 and are named in the interface descriptions above.

Table 86: *Domain event catalogue (EV numbers from Section 4)*

ID	Event	Published by	Consumed by	Effect
EV-01	`RoleAssigned`	MD-01	MD-15, MD-16	Audit entry. User is notified of the role and scope.
EV-02	`RoleRevoked`	MD-01	MD-15, MD-16	Audit entry. Open tasks of the user are listed for reassignment.
EV-03	`ConflictDeclared`	MD-01	MD-03, MD-16	The reviewer is removed from the review task. The person who assigned the reviewer is notified.
EV-04	`ReferenceDataChanged`	MD-01	MD-02, MD-13	Linked UMC versions and pathway mappings are flagged for review when a parent or target course changes.
EV-05	`SettingChanged`	MD-00	MD-15	New setting version stored and audited. Later decisions use the version in force on their date.
EV-06	`ProposalSubmitted`	MD-02	MD-03, MD-16	Version locked. Endorsement task created for P2, who is notified.
EV-07	`MaterialChangeRecorded`	MD-02	MD-03, MD-04, MD-10	New version in Draft goes through approval. Open offerings of the old version are held. A Quality-Verified SAMC gets a re-verification task.

ID	Event	Published by	Consumed by	Effect
EV-08	`MinorChangeLogged`	MD-02	MD-03, MD-14	P2 gets an acknowledgement task. The change is added to CQI.
EV-09	`ReviewerAssigned`	MD-03	MD-16	Each reviewer is notified with the due date.
EV-10	`ReviewRecorded`	MD-03	MD-16	P2 is notified. The review summary is rebuilt.
EV-11	`ProposalReturned`	MD-03	MD-02, MD-16	Version returns to Returned for amendment and unlocks. P1 is notified.
EV-12	`ProposalEndorsed`	MD-03	MD-16	Category verified. The version moves to Pending approval when sent to Senate.
EV-13	`ProposalNotSupported`	MD-03	MD-02, MD-16	Version ends as Not approved. P1 is notified.
EV-14	`SenateDecisionRecorded`	MD-03	MD-02, MD-04, MD-12, MD-13, MD-16	Version becomes Approved, Not approved or Returned for amendment. An approved version becomes selectable for offerings. An approved material change flags earlier equivalences.
EV-15	`VersionSuperseded`	MD-02	MD-16	Earlier Approved version becomes Superseded. Enrolled participants are notified with the transition rule.
EV-16	`TransitionRuleRecorded`	MD-03	MD-05, MD-16	Enrolments are bound to the version by the rule. Participants are notified (mandatory).
EV-17	`ReadinessConfirmed`	MD-04	MD-16	Offering becomes Ready. Conditions are shown to P1.
EV-18	`OfferingOpened`	MD-04	MD-16	Offering is listed in the public catalogue and registration opens.
EV-19	`OfferingHeld`	MD-04	MD-16	Offering is held or cancelled. Registered participants are notified.
EV-20	`StudentRegistered`	MD-05	MD-16	Participant is notified. Payment is requested when a fee applies.
EV-21	`PaymentConfirmed`	MD-05	MD-05, MD-16	Pending payment becomes Enrolled. Receipt is rendered.
EV-22	`EnrolmentConfirmed`	MD-05	MD-07, MD-08, MD-10, MD-16	Achievement record and result rows are created. Confirmation with access instructions is sent.
EV-23	`EnrolmentWithdrawn`	MD-05	MD-08, MD-16	Record is closed as withdrawn. No credential is issued.
EV-24	`ImportExportLogged`	MD-06	MD-15	Export file, hash and count are logged.
EV-25	`ResultsStaged`	MD-06	MD-07, MD-16	Valid rows become Provisional results. A verification task is created for P2.
EV-26	`ImportRejected`	MD-06	MD-16	P5 is notified with the row error report. Nothing is staged for a file-level error.
EV-27	`BadgeImported`	MD-06	MD-09	Badge is matched to a verified achievement or flagged.
EV-28	`IntegrityCaseOpened`	MD-07	MD-16	Affected results become Held. P2 is notified.
EV-29	`ResultsVerified`	MD-07	MD-08, MD-09, MD-16	Achievements and module status are created. A badge is issued for a verified module if the design includes badges. Participants are notified.
EV-30	`ResultChangeApproved`	MD-07	MD-08, MD-09, MD-16	A new result revision applies. If a credential exists, a correction task is created for P2.
EV-31	`CompletionConfirmed`	MD-08	MD-05, MD-09, MD-10, MD-14, MD-16	Enrolment becomes Completed. JB-01 issues the certificate and Digital Credential after the pre-issuance checks.
EV-32	`CredentialIssued`	MD-09	MD-08, MD-11, MD-16	Credential is added to the Achievement Record. MD-11 prepares an ACR for an eligible UMC or QV-valid SAMC, never for an MCE. Participant is notified.
EV-33	`BadgeIssued`	MD-09	MD-08	Badge is added to the Achievement Record. It carries no credit wording.
EV-34	`IssuanceFailed`	MD-09	MD-16	P2 is told which participants failed and which check failed.
EV-35	`CredentialCorrected`	MD-09	MD-16	New revision is current. Participant is notified (mandatory).
EV-36	`CredentialReplaced`	MD-09	MD-16	Old credential shows the new ID. Participant is notified (mandatory).
EV-37	`CredentialRevoked`	MD-09	MD-16	Credential shows Revoked on verification. Participant is notified (mandatory).
EV-38	`QvSubmitted`	MD-10	MD-16	Status becomes Submitted. P1 and P2 are notified.
EV-39	`QvVerified`	MD-10	MD-04, MD-11, MD-16	QV record becomes the source for QV, credit and MQF wording. SAMC becomes ACR-eligible.
EV-40	`QvNotVerified`	MD-10	MD-16	Record returns to Preparing. Findings are listed to P1.
EV-41	`QvReverificationRequired`	MD-10	MD-16	Re-verification flag set after a material change.

ID	Event	Published by	Consumed by	Effect
EV-42	`QvLapsed`	MD-10	MD-04, MD-09, MD-11, MD-14, MD-16	Catalogue and new credentials drop QV wording. No new ACR for the SAMC. A task is created for the PTj.
EV-43	`AcrPrepared`	MD-11	MD-16	P2 is notified to confirm.
EV-44	`AcrConfirmed`	MD-11	MD-16	Participant gets the consent request.
EV-45	`ConsentGiven`	MD-11	MD-11, MD-16	ACR enters the deposit queue.
EV-46	`ConsentDeclined`	MD-11	MD-16	ACR stays unqueued. Participant can consent later.
EV-47	`DepositRecorded`	MD-11	MD-15, MD-16	Ledger credit becomes Available with expiry at deposit date plus 10 years. Participant is notified that a deposit is not a transfer.
EV-48	`CreditExpired`	MD-11	MD-12, MD-16	Credit becomes Expired. An affected application is flagged for re-evaluation.
EV-49	`CreditRedeemed`	MD-11	MD-15, MD-16	Credit becomes Redeemed and is final.
EV-50	`CreditReleased`	MD-11	MD-16	Credit returns to Available, or becomes Expired if validity ended.
EV-51	`TransferApplied`	MD-12	MD-16	Application is under evaluation. Evaluator is notified.
EV-52	`TransferDecided`	MD-12	MD-11, MD-16	An approved outcome sets the credit Provisionally redeemed. Not approved changes nothing in ACR, credentials or credit.
EV-53	`AdmissionRecorded`	MD-12	MD-13	Admission fact is available to the stackable plan.
EV-54	`PathwayPublished`	MD-13	MD-16	Mapping is shown on the public pathway page.
EV-55	`PathwayFlaggedForReview`	MD-13	MD-16	P1 and P2 are notified of a changed target curriculum.
EV-56	`StackablePlanDecided`	MD-13	MD-16	Plan status is updated. Participant is notified.
EV-57	`CqiRecordConfirmed`	MD-14	MD-15	CQI record is confirmed and audited.
EV-58	`PeriodicReviewDue`	MD-14	MD-16	Lecturer and PTj are reminded.
EV-59	`SuspensionDecided`	MD-14	MD-02, MD-03, MD-04, MD-05, MD-16	Version becomes Suspended or Terminated. Offerings are held. Affected participants are notified (mandatory). Credentials and ACR are untouched.
EV-60	`CorrectionRecorded`	MD-15	MD-09, MD-11, MD-16	The target module applies the revision. The participant is notified when a credential, ACR, deposit or credit is affected.
EV-61	`DisposalApproved`	MD-15	MD-15	Disposal record is written and rows are removed only after the approved disposal process.
EV-62	`NotificationFailed`	MD-16	MD-15	Failure is logged after the last retry.
EV-63	`AdapterCallFailed`	MD-16	MD-16	P5 is notified. The failed call appears in the failed-call queue.

Rules for events: an event holds record identifiers and a few facts, not whole records. A listener must be safe to run twice (the same event delivered again does nothing new). A listener never changes a table of another module; it calls that module's service. Timed alerts (QV expiry at 180, 90 and 30 days, credit expiry at 365 and 90 days, an overdue decision after 14 days) are not events. They come from the scheduler and are made idempotent by `alert_log` (Section 5).

7. Process Design

This section shows how the modules of Section 4 work together in the main business processes. Each process has a sequence diagram, a table of checks, confirmations, events and failure paths, and the requirements it satisfies. Behaviour follows the approved prototype acts and the SRS functional requirements. Event names are `PastTense` class names. The numbered event list (EV-xx) is owned by Section 4, and these names are used here without numbers so that the two sections stay consistent. Job names are class names for the same reason (JB-xx numbers are in Section 4).

Conventions that apply to every process:

- Every action on a record is authorised by a Laravel Policy for role, scope and separation of duties (AD-07), before any change is made.
- Every state change and its audit row are written in one database transaction (AD-11). Domain events are dispatched after commit.
- A confirmation dialog is shown for every irreversible action (IR-06). The dialog names the effect. The server also checks that the confirmation token was posted.
- Notifications are stored first and sent by a queued job (AD-15). A failed notification never rolls back the business change (FR-271).
- Dates and times are stored in UTC and shown as Malaysia time (IR-24). The demo clock is available only outside production (AD-18).

7.1 Propose, Review, Endorse and Record the Senate Decision

A lecturer prepares a proposal for a new micro-credential. The deputy dean of the PTj assigns reviewers, a colleague reviews after declaring any conflict of interest, and the deputy dean endorses. The Senate does not use MCMS, so the deputy dean records the Senate decision with its meeting reference (D-11).

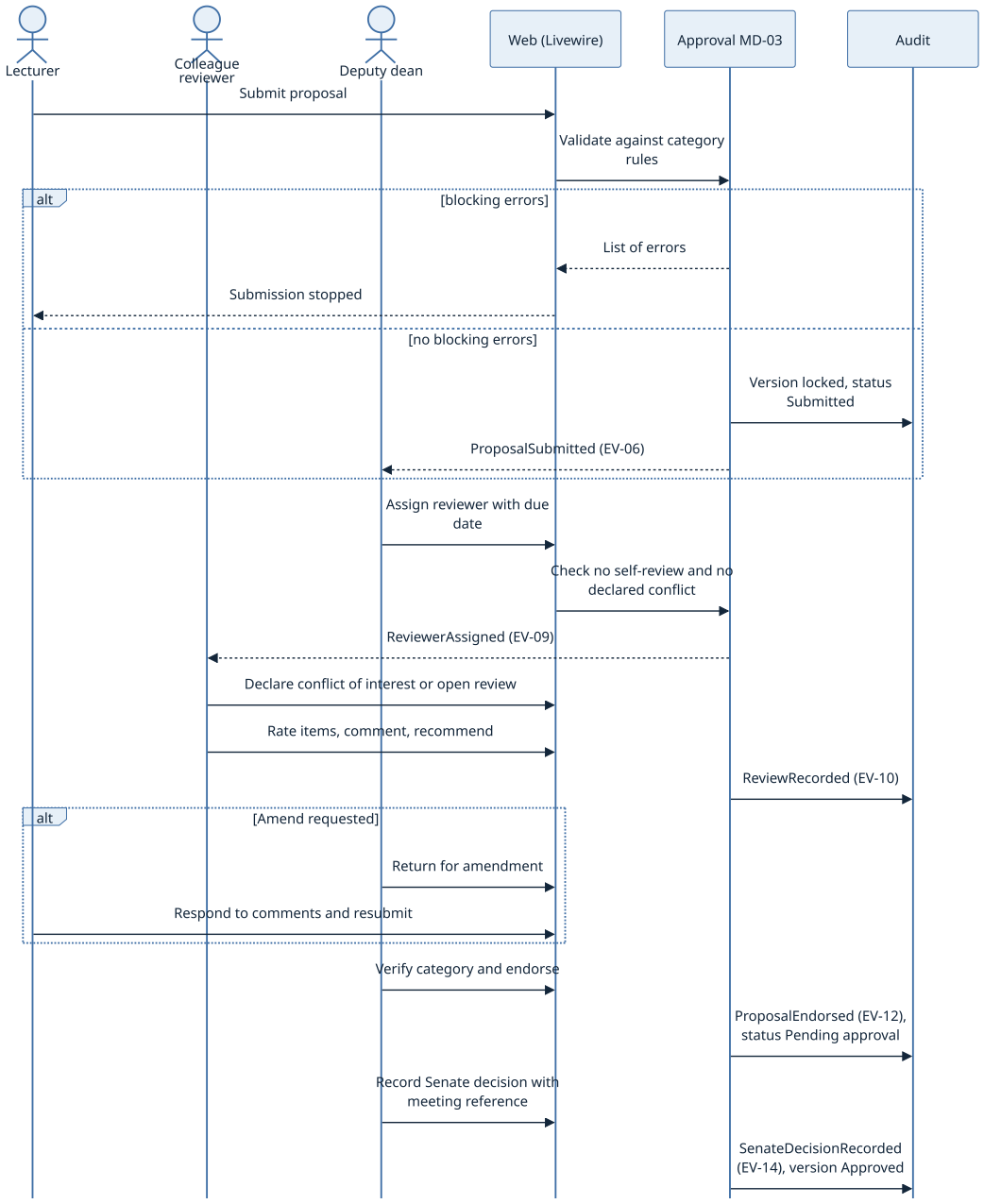


Figure 28: *Propose, review, endorse and record the Senate decision.*

Table 87: *Checks, confirmations, events and failures for process 7.1*

Step	Checks and rules	Confirmation and event	Failure path
Submit	Category rules, weightings total 100 percent, UMC rules (parent accredited, module SLT equals parent SLT), IP declaration present, warnings answered in writing	Confirm submit. Content locked. ProposalSubmitted (EV-06)	Blocking errors stop submission and are listed. Warnings need a written response
Assign reviewer	Reviewer is not the proposer or a co-developer, no declared conflict, due date set	ReviewerAssigned (EV-09)	Self-review is refused with the SoD message
Declare conflict	Declaration must come before the review form opens	Confirm. ConflictDeclared (EV-03)	Reviewer is removed and the assigning deputy dean is notified
Review	Item ratings Satisfactory or Needs amendment, recommendation Support, Amend or Not support	Confirm. ReviewRecorded (EV-10)	Review cannot be saved without a recommendation
Respond to comments	Comment status open, answered or resolved. A blocking comment that is unresolved stops endorsement	ReviewCommentAnswered (audit entry)	Endorse button disabled with the reason shown
Return and resubmit	Return goes to Returned for amendment. Resubmission repeats the submit checks	ProposalReturned (EV-11), ProposalSubmitted (EV-06)	A returned proposal is unlocked only by this return

Step	Checks and rules	Confirmation and event	Failure path
Endorse	Category verified separately from the proposed category, no unresolved blocking comment	Confirm. ProposalEndorsed (EV-12). Status Endorsed then Pending approval	Not supported ends the proposal: ProposalNotSupported (EV-13)
Record Senate decision	Decision, date, role, comments and meeting reference. Approved only for the exact version	Confirm. SenateDecisionRecorded (EV-14)	Missing meeting reference blocks saving. Overdue decisions are reminded after the setting `alerts.decision_overdue_days`

Requirements satisfied: FR-34, FR-48, FR-47, FR-60, FR-17, FR-14, FR-61, FR-62, FR-63, FR-64, FR-65, FR-66, FR-67, FR-68, FR-69, FR-70, FR-71, BR-39, BR-42.

7.2 Material Change to an Approved Version

An approved version is never edited. A change creates a new version as a copy. A minor change is logged and acknowledged. A material change repeats the review and approval path of process 7.1 and does not take effect until approved.

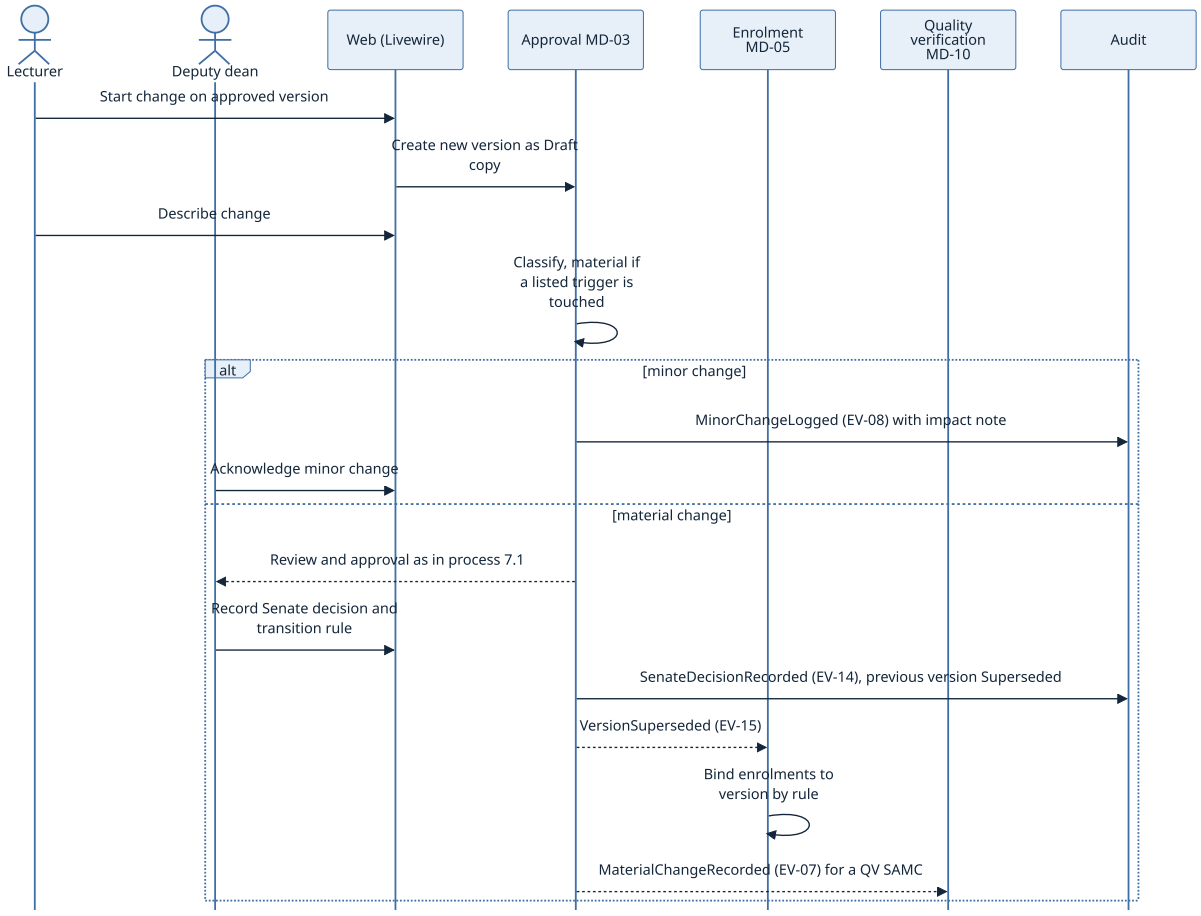


Figure 29: *Material and minor change to an approved version.*

Table 88: *Checks, confirmations, events and failures for process 7.2*

Step	Checks and rules	Confirmation and event	Failure path
Start change	Only an Approved version can be copied. The old version stays unchanged and referenced	VersionDraftCreated (audit entry)	Edit of a locked version is refused
Classify	Listed triggers (outcomes, SLT, credit, MQF, QV, core content, assessment, mode, credential, partner) set the class to material and the proposer cannot lower it	Confirm for material. MaterialChangeRecorded (EV-07)	A minor change needs a description and an impact note, otherwise it is not saved
Approval	Same checks as process 7.1. The new version is not effective until Approved	SenateDecisionRecorded (EV-14)	If not approved, the old version stays in force
Transition rule	Deputy dean records one rule: finish on old, move to new, or equivalence	Confirm. TransitionRuleRecorded (EV-16)	The new version cannot take effect without a rule

Step	Checks and rules	Confirmation and event	Failure path
Notify participants	Mandatory notice with change, date and rule. It cannot be switched off	ParticipantNoticeQueued (audit entry)	Failed email is retried. The in-system notice remains
QV SAMC	A re-verification task is created and the QV status becomes Re-verification required in the QV record	QvReverificationRequired (EV-41)	A task is open until QV is recorded again
Offering hold	An offering with a pending material change is held until the version is approved	OfferingHeld (EV-19)	Registration stays closed while held

Requirements satisfied: FR-75, FR-76, FR-77, FR-78, FR-79, FR-80, FR-81, FR-74, FR-88, BR-18, BR-39.

7.3 Readiness and Opening an Offering

An offering (cohort) is created only for an Approved version. The lecturer and the deputy dean complete the readiness checklist. Registration opens when the deputy dean confirms Ready, or Ready with conditions when the conditions are recorded.

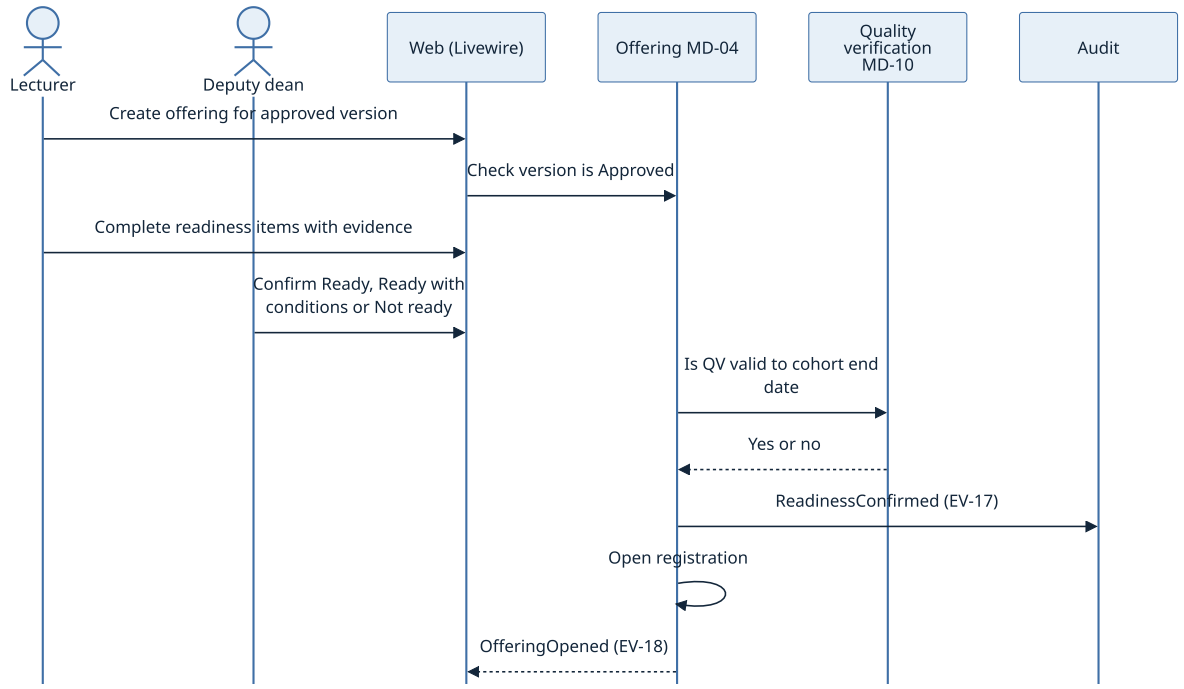


Figure 30: *Readiness and opening an offering.*

Table 89: *Checks, confirmations, events and failures for process 7.3*

Step	Checks and rules	Confirmation and event	Failure path
Create offering	Version status is Approved. Out-of-scope PTj is refused. The offering is not public	OfferingCreated (audit entry)	A version that is not Approved cannot be selected
Readiness items	Each item Yes, Not yet or Not applicable, with an evidence note	ReadinessItemSaved (audit entry)	Items can be saved incomplete. They gate confirmation only
Confirm readiness	Ready with conditions requires the conditions in text. Not ready keeps the offering at Not ready	Confirm. ReadinessConfirmed (EV-17)	Without a confirmation the registration window does not open
QV check	For a SAMC, QV must be valid to the cohort end date. If not, the QV claim, credit and MQF are hidden in the catalogue	Warning to the lecturer	The offering still opens, but without the claim
Wording check	Recognition statements come from templates. Free text is scanned for forbidden claims	–	Publishing is blocked with the offending phrase shown
Open	Window start reached. Capacity and deadlines set	OfferingOpened (EV-18). Catalogue shows it	The scheduler opens at window start, and a late run opens it on the next tick
Recheck for new cohort	Checklist pre-filled from the last cohort. A material change puts the offering on hold	Confirm. OfferingHeld (EV-19)	Hold stays until the version is approved

Requirements satisfied: FR-83, FR-84, FR-85, FR-86, FR-87, FR-88, FR-89, FR-90, FR-91, FR-93, FR-94, FR-151, BR-17, BR-02.

7.4 Register and Pay

A student registers while the window is open and capacity remains. A fee leads to Pending payment and a payment call through the payment adapter (a stub in Release 1). Capacity is checked under a row lock so that the last seat cannot be given twice. Payment calls carry an idempotency key.

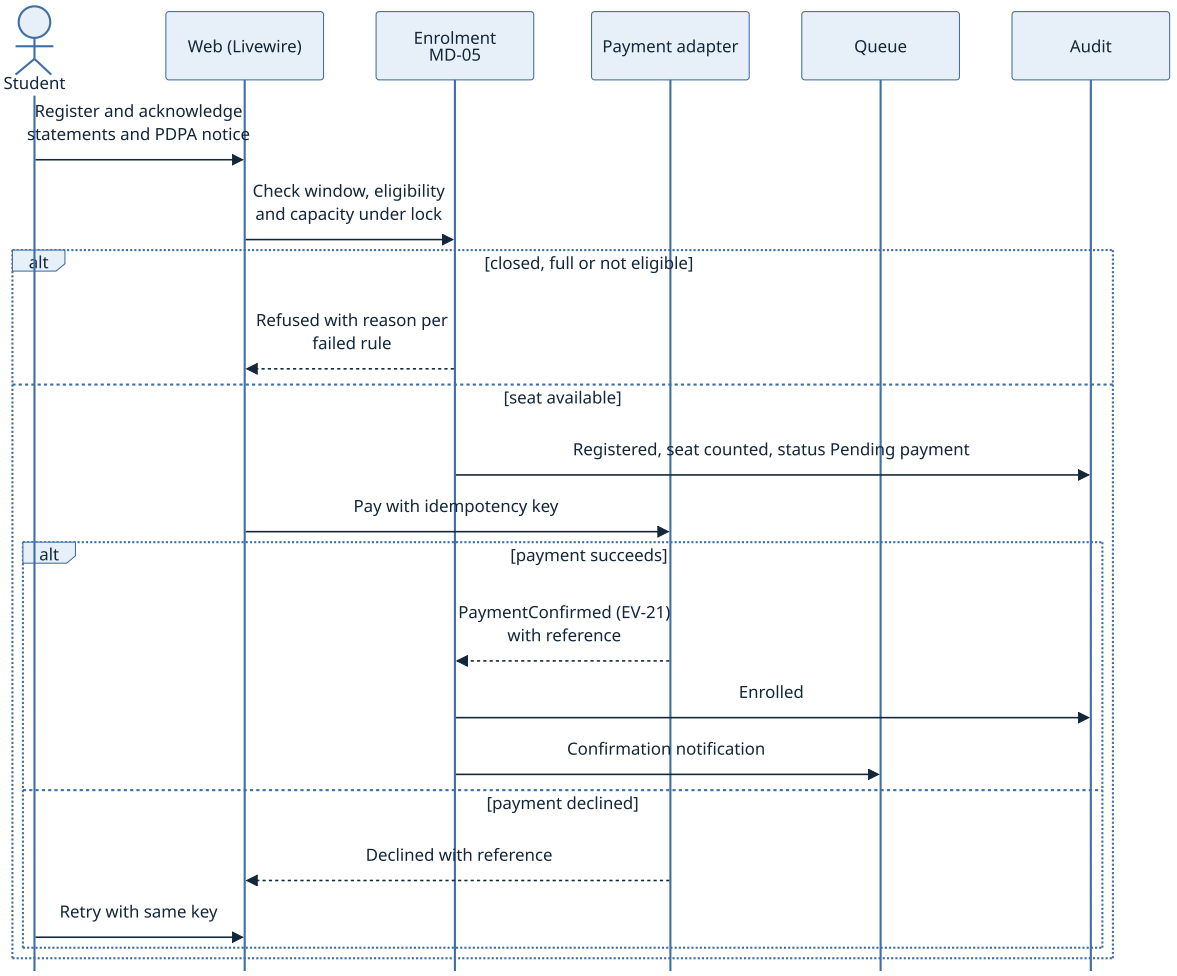


Figure 31: *Register and pay, including declined payment and capacity full.*

Table 90: *Checks, confirmations, events and failures for process 7.4*

Step	Checks and rules	Confirmation and event	Failure path
Acknowledge	Information and recognition statements, text version stored with the enrolment. PDPA notice at first registration	Confirm. StatementsAcknowledged (audit entry)	Registration is blocked until both are acknowledged
Eligibility	Group and prerequisites checked, each failed rule explained	–	Registration refused with the reason for each failed rule
Capacity	Count of Registered, Pending payment and Enrolled seats is read after locking the offering row. At the last seat the offering becomes Full	StudentRegistered (EV-20)	Full offering refuses the request. A second request for the last seat waits for the lock and is then refused
Pay	No fee leads directly to Enrolled. A fee leads to Pending payment and a payment call. The attempt record stores the key, amount and reference	Confirm. PaymentConfirmed (EV-21) or PaymentDeclined (audit entry)	A declined payment shows the reference and the status stays Pending payment. The student can retry until the window ends
Idempotency	The key is made from enrolment ID and attempt number. A repeated call with the same key returns the first result and creates no second charge or second enrolment	–	A timeout is treated as unknown, and the status is read back before any retry
Window end	Pending payment older than the window end is cancelled by the scheduler and the seat is freed	EnrolmentCancelled (audit entry)	Late payment confirmation for a cancelled enrolment is logged for follow-up and creates no enrolment

Step	Checks and rules	Confirmation and event	Failure path
Confirmation	Notification with access instructions in the recipient language	EnrolmentConfirmed (EV-22)	Failed email is retried. The in-system notice remains
Withdraw	Allowed while Enrolled and before the offering deadline. Refunds are outside MCMS	Confirm. StudentWithdrew	After the deadline the action is not offered

> **Note:** The SRS state diagram has no path from Pending payment to Withdrawn. This design lets the student withdraw only from Enrolled, and an unpaid enrolment ends as Cancelled at window end. A seat is counted from Registered, which is a design assumption to be listed in Section 11.

Requirements satisfied: FR-97, FR-98, FR-99, FR-100, FR-101, FR-103, FR-105, FR-109, FR-271, IR-17, IR-15, NFR-05, NFR-06, NFR-20.

7.5 UCreds Import, Rejected Rows and Lecturer Marks

CODL exports the enrolment list to UCreds and later imports the results file. The import validates the file and each row. Valid rows are staged as Provisional. Rejected rows are reported. The lecturer then completes marks and sends results for verification.

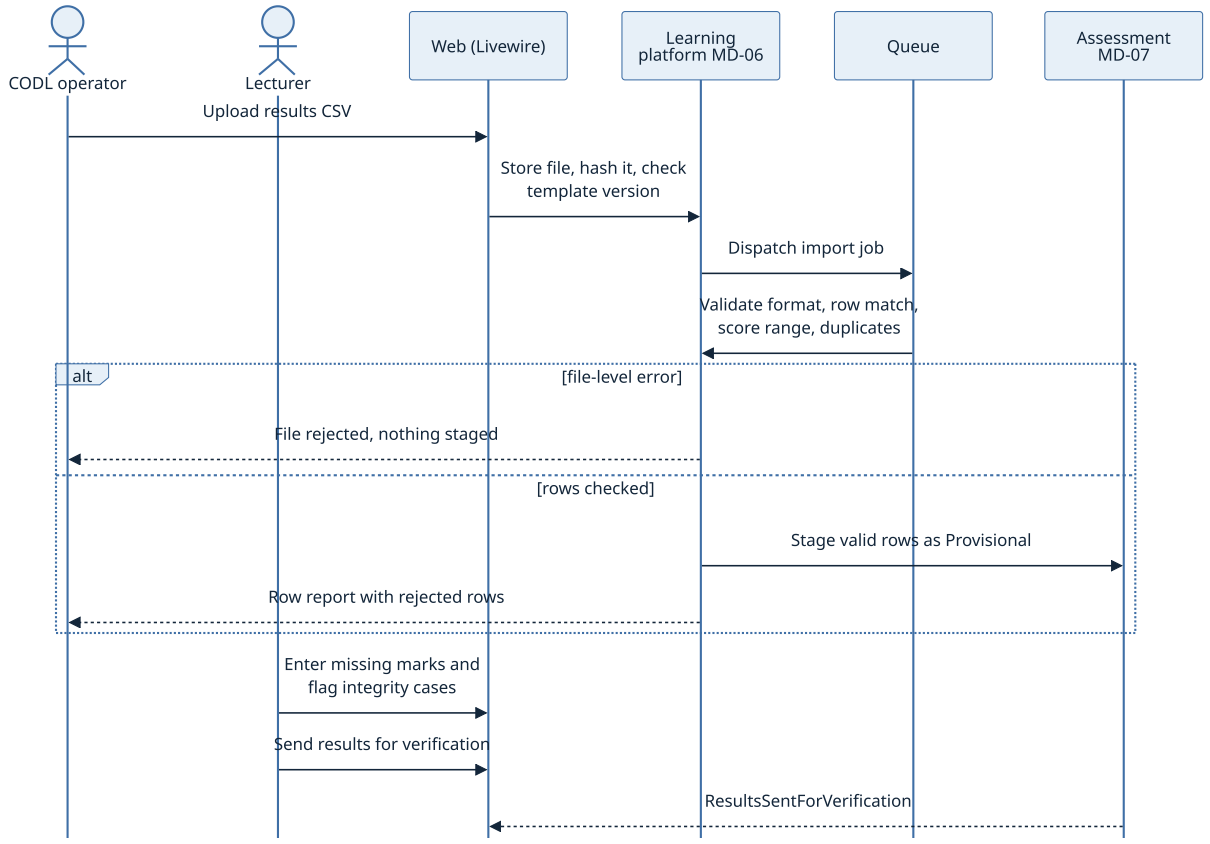


Figure 32: *UCreds import with rejected rows, then lecturer completes marks.*

Table 91: *Checks, confirmations, events and failures for process 7.5*

Step	Checks and rules	Confirmation and event	Failure path
Export	Enrolment list in the UCreds template. File name, hash and count are logged	ImportExportLogged (EV-24)	Adapter failure is retried and then listed for CODL
Upload	Template version, file hash, duplicate file check. The original file is kept	Confirm. ResultsFileUploaded (audit entry)	A file already imported (same hash and cohort) is refused as a duplicate
Validate	Format, matching of each row to an enrolment, score range, duplicate rows	–	A file-level error rejects the whole file. Row errors reject only those rows
Stage	Valid rows are stored as Provisional. No achievement and no badge exists yet	ResultsStaged (EV-25)	A rerun of the same file does not create duplicate staged rows because the natural key is enrolment and component

Step	Checks and rules	Confirmation and event	Failure path
Row report	Downloadable report with row number, reason and value	–	Rejected rows are fixed in the file and uploaded again as a new import
Enter marks	Totals and pass or fail computed against the pass standard of the approved version. Attendance alone cannot give a pass	MarksSaved (audit entry)	Out-of-range score is refused on the form
Integrity flag	A flag holds the result so that it cannot be verified	Confirm. IntegrityCaseOpened (EV-28)	The hold stays until the case is closed
Send for verification	The lecturer sends results to the deputy dean	Confirm. ResultsSentForVerification	Blank mandatory component keeps the result In progress
Badge import	Badge ID, link and date are matched to a verified achievement	BadgeImported (EV-27)	A badge with no verified achievement is flagged and not shown as credit

Requirements satisfied: FR-110, FR-111, FR-112, FR-113, FR-114, FR-115, FR-116, FR-120, FR-121, FR-122, IR-19, NFR-07, BR-20.

7.6 Verify Results, Confirm Completion and Issue

The deputy dean verifies results, which creates achievements. The deputy dean then confirms completion of the micro-credential. The confirmation starts the issuing job, which runs the pre-issue checks, signs each credential, renders the PDF and sends the notifications. The deputy dean does not issue by hand (UI/UX spec §2.2).

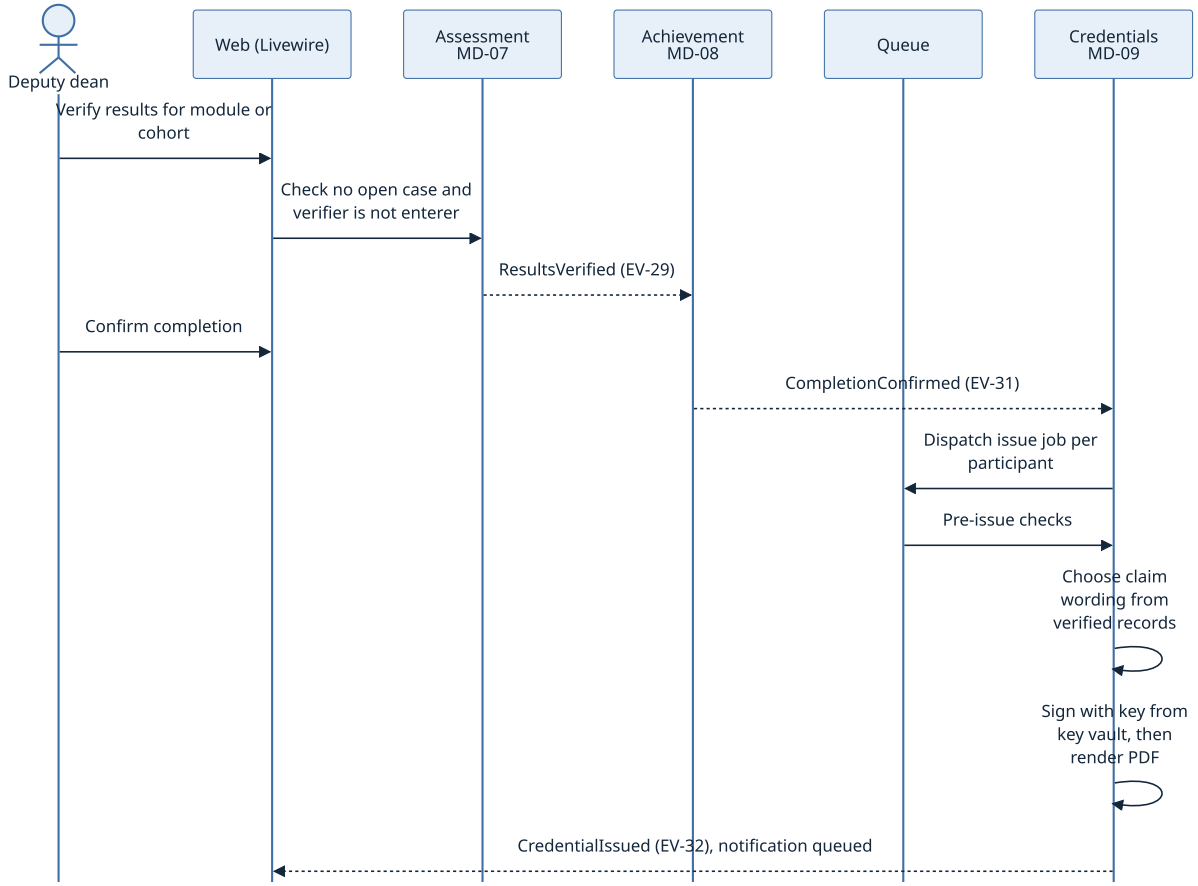


Figure 33: *Verify results, confirm completion and the issuing job.*

Table 92: *Checks, confirmations, events and failures for process 7.6*

Step	Checks and rules	Confirmation and event	Failure path
Verify results	Verifier recorded with date and identity evidence link. The user who entered a result cannot be its only verifier. An open integrity case blocks verification	Confirm. ResultsVerified (EV-29). Module achievement recorded when all outcomes and assessments are met	Blocked rows are listed and the rest of a bulk verification proceeds only if the user confirms the listed exceptions

Step	Checks and rules	Confirmation and event	Failure path
Confirm completion	UMC complete when all modules and requirements are met. A SAMC or MCE is complete when outcomes, SLT, assessment and conditions are met	Confirm. CompletionConfirmed (EV-31)	Confirmation is blocked until every requirement is met
Pre-issue checks	Completion confirmed, results verified, no open case, no earlier active credential of the same kind, template present	–	A failing check stops that participant and lists the failing check. Others continue
Claim wording	Credit and MQF are shown only when verified. For a UMC, ACR eligibility follows accreditation. For an MCE, no credit and no ACR. For a SAMC, the QV claim and ACR eligibility need a QV that is valid at the issue date	–	A SAMC without valid QV is issued with no QV claim and gets no ACR. A forbidden claim blocks issue
Sign	Open Badges 3.0 JSON-LD signed with Ed25519, key ID stored in the credential. The key stays in the vault	–	Vault unavailable leaves the participant in the queue for retry
Render PDF	Bilingual A4 template, version recorded	–	PDF failure is retried. The signed credential exists and the PDF is attached when it succeeds
Notify	Participant receives results verified and credential issued notices	CredentialIssued (EV-32), ResultsVerifiedNoticeQueued (audit entry)	Failed email is retried
Report	The issuance queue shows each participant as Issued or failed with the reason	–	Failed participants are listed to the deputy dean and can be retried

Requirements satisfied: FR-123, FR-124, FR-126, FR-131, FR-134, FR-130, FR-137, FR-138, FR-140, FR-141, FR-142, FR-143, FR-144, FR-146, FR-150, FR-151, FR-152, FR-177, BR-19, BR-21, BR-07, BR-22, IR-20, NFR-08, SR-08.

7.7 Credential Correction, Replacement and Revocation

Credentials are never deleted. A confirmed error is corrected as a new revision, a replacement has a new ID and is linked to the old one, and a revocation is recorded with a stated reason. The deputy dean of the owning PTj records the decision. CODL has no part in these actions (D-12).

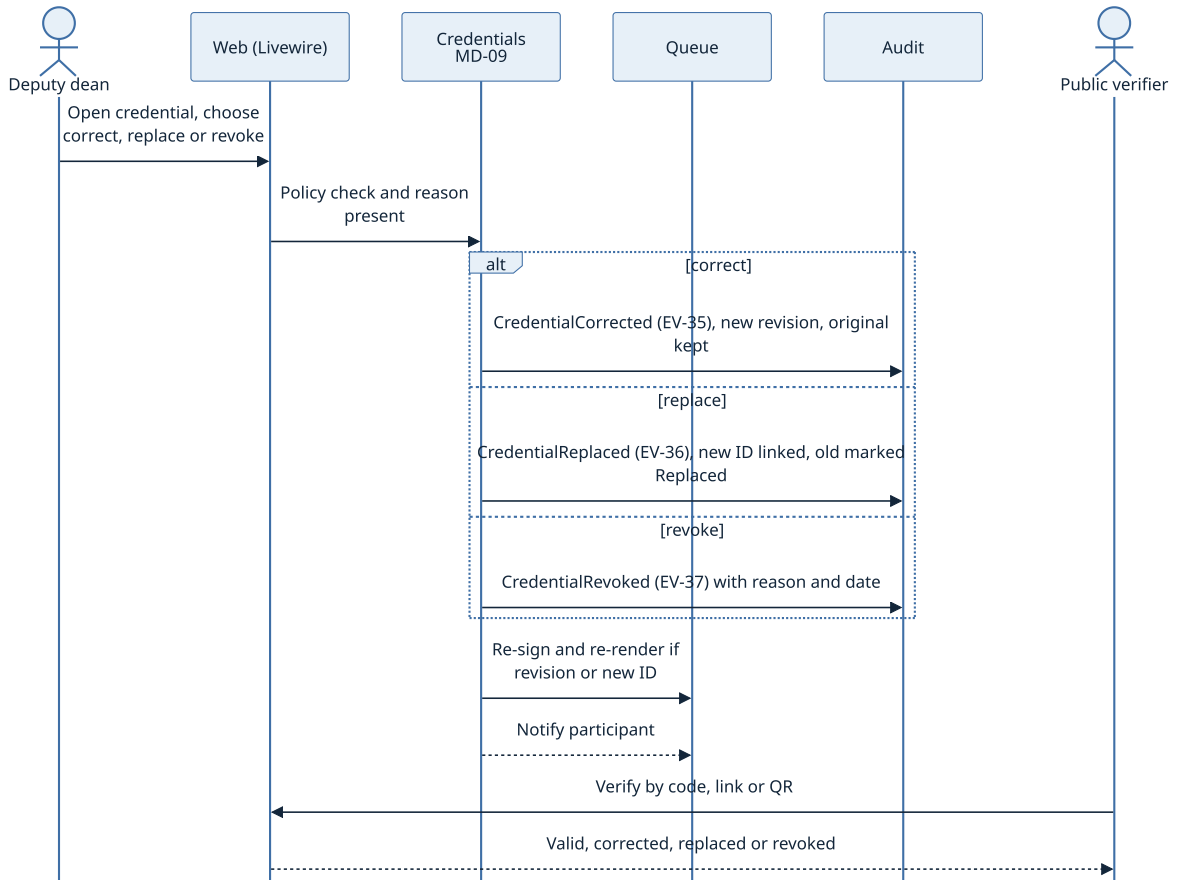


Figure 34: *Credential correction, replacement and revocation.*

Table 93: *Checks, confirmations, events and failures for process 7.7*

Step	Checks and rules	Confirmation and event	Failure path
Trigger	A result change after issue creates a correction task. A discrepancy or a reported error can also start it	CredentialCorrectionTaskCreated (audit entry)	The task stays open until resolved
Correct	Reason required. Original revision kept. The change is checked against the source record	Confirm. CredentialCorrected (EV-35)	Without a reason the form is not saved
Replace	New credential with a new ID, old marked Replaced and showing the new ID	Confirm. CredentialReplaced (EV-36)	Replacement is not offered for a Revoked credential
Revoke	Recorded decision with stated reason and date	Confirm. CredentialRevoked (EV-37)	Revoked is final, with no undo. A mistaken revocation is handled by a replacement
Notify	Mandatory notice to the participant	CredentialNoticeQueued (audit entry)	Failed email is retried
Delete	No delete action exists. The database role has no delete right on credentials	–	Not applicable

Table 94: *What the public verification page shows by credential status*

Status	Shown to the public	Not shown
Issued	Valid, credential title, issuer, issue date, category, credit and MQF only if verified, QV statement only if valid at issue date, signature check result	Identity number, matric number, email, results
Corrected	Corrected, plus the same fields as Issued with the date of correction	The old values
Replaced	Replaced, with the ID of the new credential	The old values
Revoked	Revoked with date. The reason is shown only if the setting `verification.public_fields` allows it	Any claim of credit or QV
Code not found	A neutral not-found message identical for every unknown code	Any similar code

Requirements satisfied: FR-158, FR-159, FR-160, FR-161, FR-162, FR-125, FR-153, FR-154, FR-156, FR-157, FR-256, BR-38, DR-04, DR-03.

7.8 ACR Confirmation, Consent, Export and Deposit

An ACR is generated for a completed UMC (accredited) or a SAMC with valid QV. The deputy dean confirms it. The participant then gives or declines consent. Consented ACRs enter the deposit queue. The deputy dean exports the deposit data and records the deposit after the ABC accepts it. Deposit is storage only and never a transfer (BR-26).

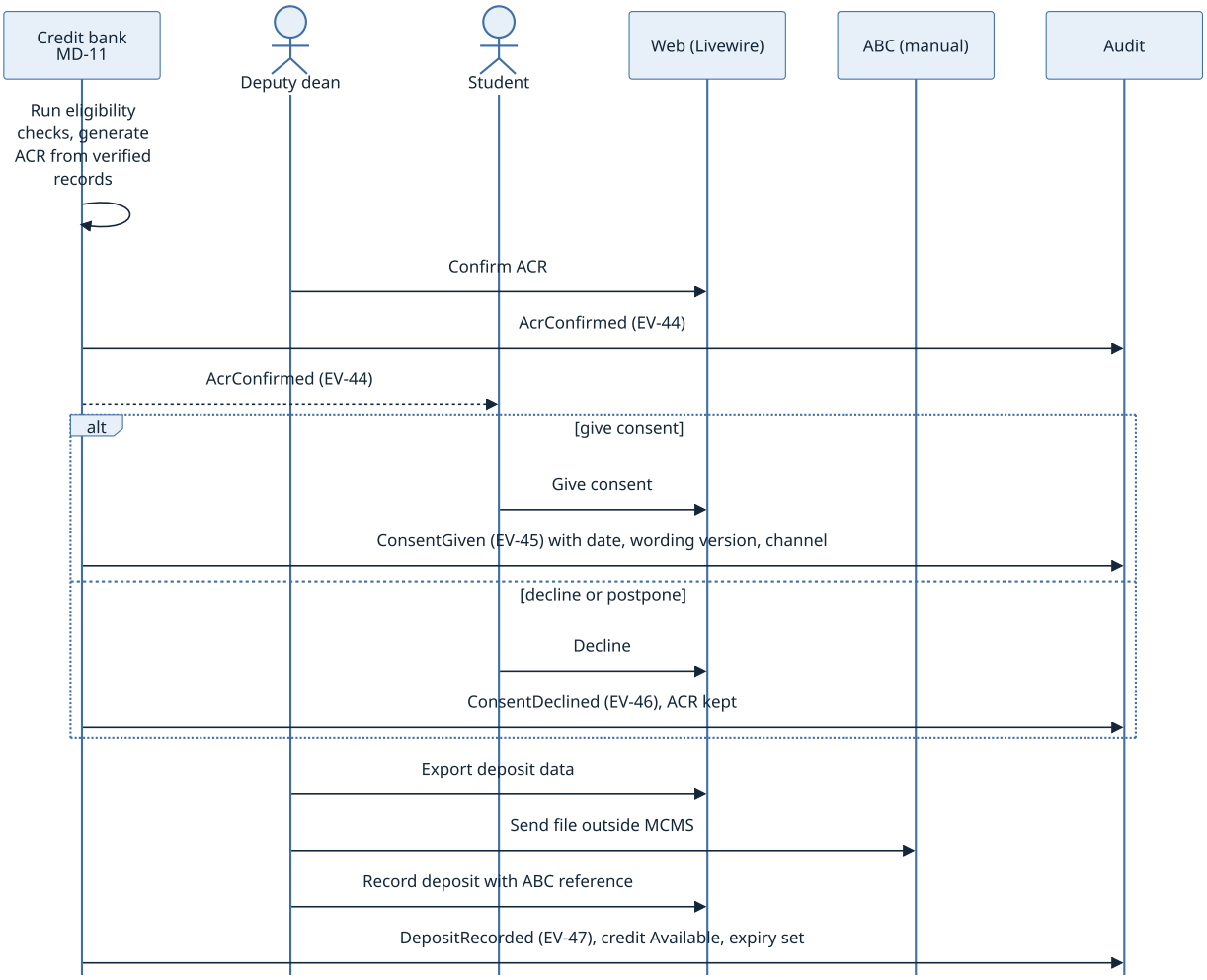


Figure 35: *ACR confirm, consent and deposit recording.*

Table 95: *Checks, confirmations, events and failures for process 7.8*

Step	Checks and rules	Confirmation and event	Failure path
Eligibility	Only an accredited UMC or a SAMC with valid QV. Blocked for MCE, SAMC without QV, module, badge and certificate. Checks recorded	AcrChecksRecorded (audit entry)	A failing check blocks generation and the reason is shown
Generate	Fields come from verified records and cannot be typed	AcrPrepared (EV-43)	Missing data stops generation with the field named
Confirm	Deputy dean reviews the read-only ACR. Label Academic Credit Record shown	Confirm. AcrConfirmed (EV-44)	Confirmation is blocked if any record changed after generation
Consent	Explicit consent with date, wording version and channel. Withdrawal is recorded as a new event before deposit	Confirm. ConsentGiven (EV-45), ConsentDeclined (EV-46), ConsentWithdrawn	Without consent the ACR is not in the queue. Decline keeps the ACR and a later request is possible
Queue and export	Only consented and confirmed ACRs. Only an ACR can be selected. Export is a CSV file in the agreed format	DepositExportCreated (audit entry)	Export format is open (OI-11). A format change affects only the export mapper
Record deposit	Date, ABC reference, credits, status (Successful, In process, Correction needed) and recorder	Confirm. DepositRecorded (EV-47)	Without a reference the form is not saved. A status of Correction needed keeps the ACR in the queue
Expiry	Expiry is deposit date plus `credit.abc_validity_years`. The participant sees it	CreditAvailable	Credit expiry alerts run from the scheduler (section 7.12) using `alerts.credit_expiry_days`

Step	Checks and rules	Confirmation and event	Failure path
Notice	Participant is told that deposit is not a credit transfer	DepositNoticeQueued (audit entry)	Failed email is retried

Requirements satisfied: FR-177, FR-178, FR-179, FR-180, FR-181, FR-182, FR-185, FR-186, FR-187, FR-188, FR-189, FR-190, FR-191, FR-193, FR-194, FR-196, BR-24, BR-26, BR-27, BR-25, SR-12.

7.9 Credit Transfer Decision, Provisional and Final Redemption

A transfer is decided case by case by the deputy dean of the faculty that owns the degree. Equivalence is judged on learning outcomes. An approval makes the credits Provisionally redeemed. The deputy dean later records completion (Redeemed) or release (Available again, or Expired).

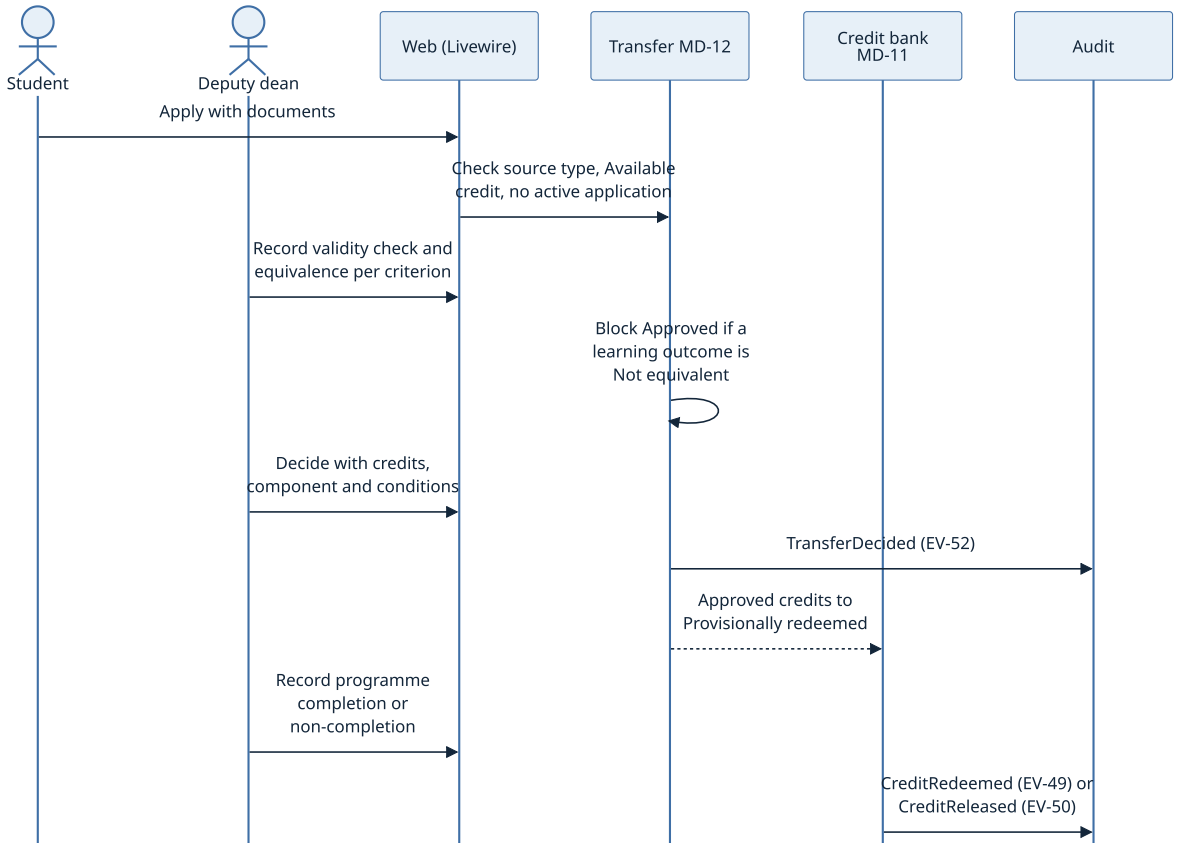


Figure 36: *Credit transfer decision, provisional and final redemption.*

Table 96: *Checks, confirmations, events and failures for process 7.9*

Step	Checks and rules	Confirmation and event	Failure path
Apply	Source is a UTeM UMC, another eligible UMC or a SAMC with verifiable QV. Only Available credit is selectable. A second application on the same credits is blocked	Confirm. TransferApplied (EV-51)	MCE is refused with an explanation. Redeemed or Provisionally redeemed credit is not selectable. Expired credit is selectable only with a re-evaluation flag
Validity check	Authenticity, ABC status and QV or accreditation status	Confirm. ValidityChecked (audit entry)	Not valid leads to Not approved
Equivalence	Each criterion has a finding and a result. Credit or SLT alone is not evidence	Confirm. EquivalenceEvaluated (audit entry)	A criterion with no finding blocks saving
Limits	Programme transfer limit and non-transferable components are checked. A partly completed UMC or a badge set is not full credit unless flagged separately recognisable	Warning shown to the decision maker	Limit warnings do not block. A learning-outcome mismatch and a partial UMC do block
Decide	Outcomes are Approved, Partly approved, Not approved and More evidence needed. Only the configured authority can decide, using the setting version in force	Confirm. TransferDecided (EV-52)	Unauthorised user is denied. More evidence needed returns the application

Step	Checks and rules	Confirmation and event	Failure path
Provisional redemption	Approved credits become Provisionally redeemed and are linked to the programme and the record	TransferDecided (EV-52)	Not approved changes no ACR, badge, credential or credit status
Final redemption	Authorised user records programme completion with an evidence reference	Confirm. CreditRedeemed (EV-49)	No evidence reference blocks the form
Release	Withdrawal or non-completion with an evidence reference. Credit returns to Available if still valid or becomes Expired	Confirm. CreditReleased (EV-50)	Release does not change the ACR
Admission	Admission is a separate record. An approval does not imply admission	Confirm. AdmissionRecorded (EV-53)	Not applicable

Requirements satisfied: FR-198, FR-199, FR-200, FR-201, FR-202, FR-203, FR-204, FR-205, FR-206, FR-207, FR-208, FR-209, FR-211, FR-213, FR-214, FR-215, FR-216, FR-217, FR-218, FR-219, FR-220, BR-29, BR-30, BR-31, BR-32, BR-33, BR-34.

7.10 QV Expiry Alerts and Lapse

The scheduler raises alerts before a QV expires. When the QV expires, is suspended or is withdrawn, the QV claim stops, new ACR eligibility stops and a task is created for the deputy dean. A QV result is recorded by the deputy dean (UI/UX spec §2.2).

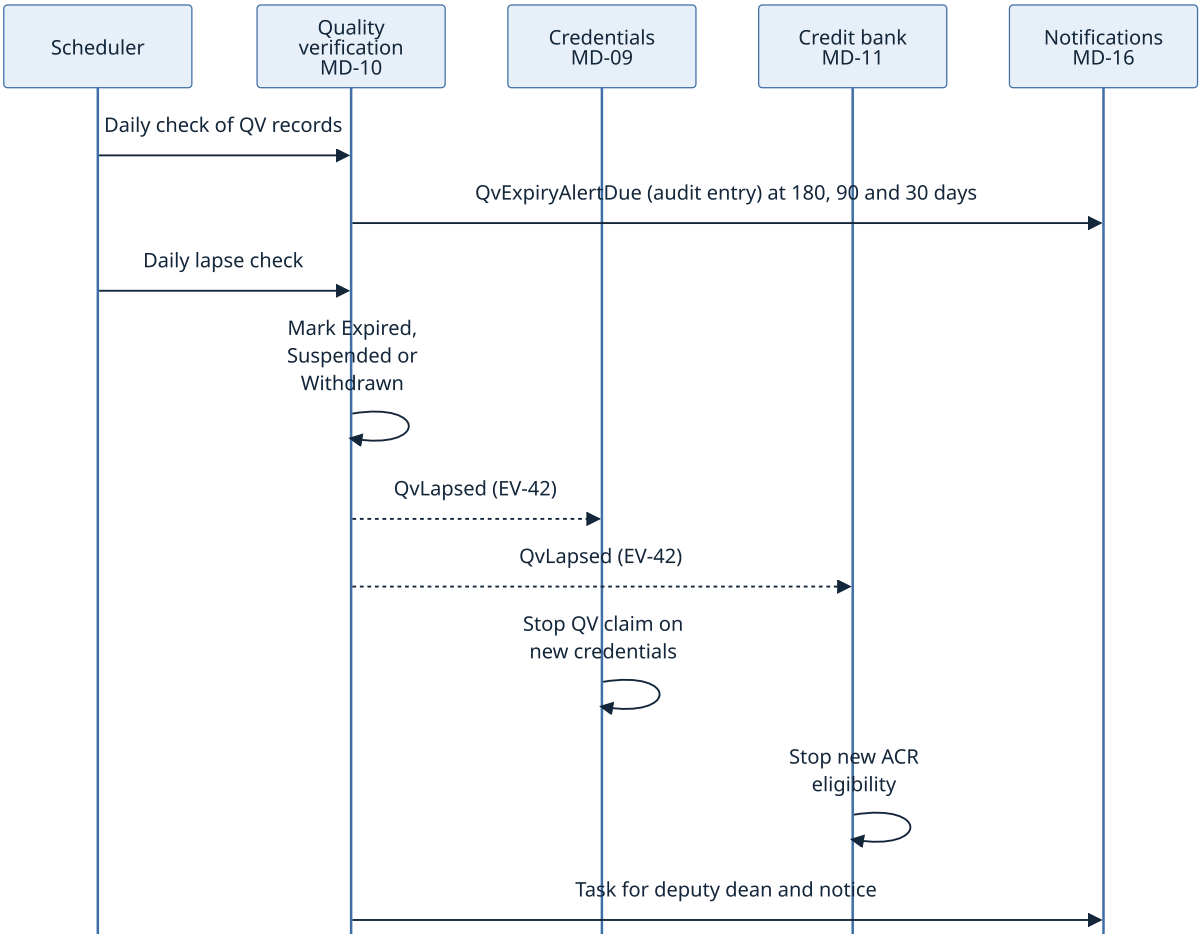


Figure 37: *QV expiry alerts and lapse.*

Table 97: *Checks, confirmations, events and failures for process 7.10*

Step	Checks and rules	Confirmation and event	Failure path
Alert	Alert days come from `alerts.qv_expiry_days`. One alert per record and day offset, kept in an alert log	QvExpiryAlertDue (audit entry)	A missed run is caught up on the next run because the rule is "offset reached and not yet sent"
Lapse	Expiry date passed, or suspension or withdrawal recorded. Status changes in the QV record, the sole source	QvLapsed (EV-42)	The lapse run is idempotent because a record already lapsed is skipped

Step	Checks and rules	Confirmation and event	Failure path
Effect on credentials	New credentials get no QV claim. Earlier credentials keep the status they had at issue date	–	Not applicable
Effect on offerings	Catalogue hides the QV claim, credit and MQF for the affected SAMC	–	Not applicable
Effect on ACR	A new ACR is not allowed. Existing ACRs remain	–	Not applicable
Task	A task for the deputy dean to renew or withdraw the claim	QvLapseTaskCreated (audit entry)	Task stays open until the QV is renewed or the state is acknowledged
Record QV result	Expiry cannot be later than `qv.max_validity_years` after the start date	Confirm. QvVerified (EV-39) or QvNotVerified (EV-40)	Out-of-range date is refused

Requirements satisfied: FR-172, FR-171, FR-173, FR-174, FR-175, FR-152, FR-89, BR-02, BR-23, NFR-43.

7.11 Stackable Plan Check

An individual stackable plan is checked against the home credit share and the residency rule. The plan decision "Suitable to proceed" needs admission to be recorded first. The mapping used by the plan was approved by the Senate.

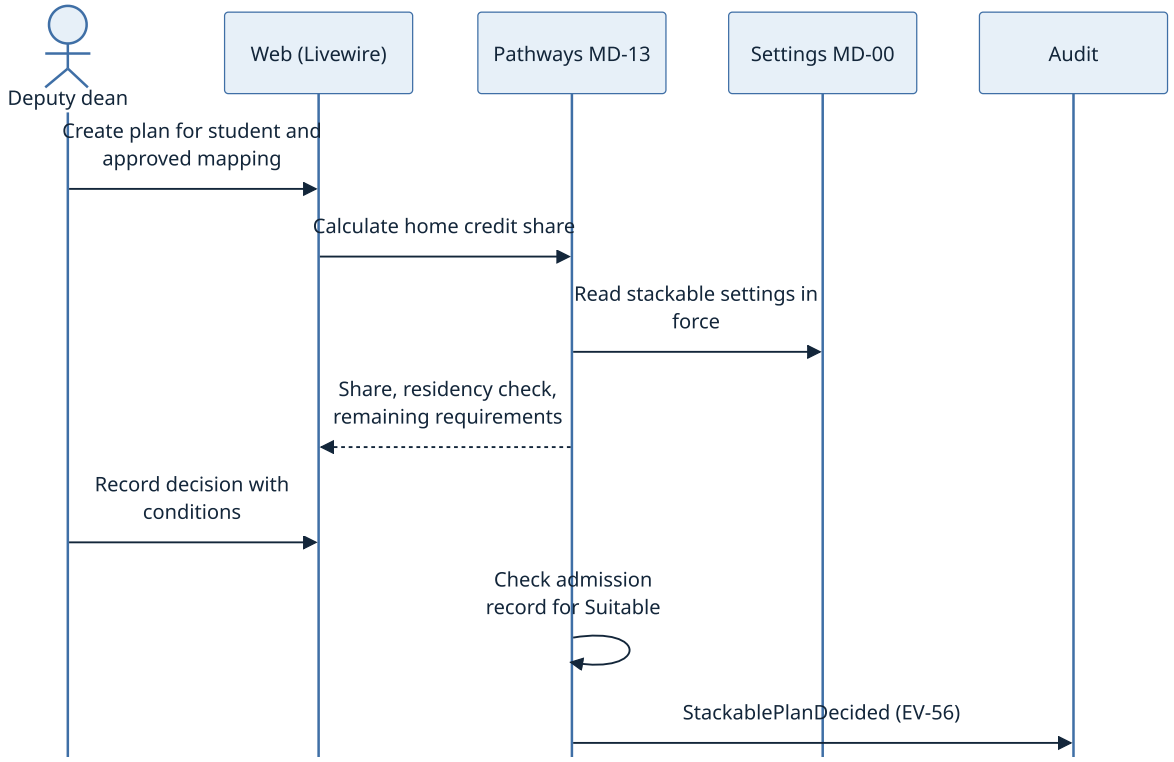


Figure 38: *Stackable plan check.*

Table 98: *Checks, confirmations, events and failures for process 7.11*

Step	Checks and rules	Confirmation and event	Failure path
Mapping	Only credit-bearing credentials can be mapped, so an MCE is blocked. An approved mapping transfers no credit	SenateDecisionRecorded (EV-14)	An unapproved mapping cannot be published
Home credit share	Credit earned at UTeM divided by total credit. Below `stackable.min_home_credit_percent` the plan is marked not meeting. The screen states that 30 percent does not mean 70 percent is transferable	–	Plan decision is limited to Not yet suitable or Not suitable
Residency	Residency semesters against `stackable.residency_semesters` for conventional delivery. ODL scope is flagged for review	–	Missing semesters leave the plan at Needs review

Step	Checks and rules	Confirmation and event	Failure path
Decision	Suitable, Needs review or Not yet suitable, with conditions and follow-up	Confirm. StackablePlanDecided (EV-56)	Suitable needs the admission record, otherwise the action is blocked
Award	Recorded from programme records after admission and requirements. MCMS does not award	Confirm. AwardRecorded (audit entry)	Blocked without admission and requirements
Curriculum change	A material change to the target curriculum flags the mapping for review	MappingFlaggedForReview	The flag stays until the mapping is reviewed

Requirements satisfied: FR-221, FR-222, FR-223, FR-224, FR-225, FR-227, FR-228, FR-229, FR-230, FR-231, FR-232, FR-233, FR-234, BR-05, BR-06, BR-37.

7.12 Job Designs

All background work runs as Laravel queued jobs on Redis under Horizon (AD-05, NFR-11). The scheduler starts timed jobs (NFR-43). Rules common to all jobs:

- A job is idempotent. It records its natural key before it acts, and a second run with the same key returns the first result.
- A job is retried with a growing delay (for example 1, 5, 15 and 60 minutes) up to a set number of tries. When tries are exhausted the job moves to the failed-job list, and an alert is sent to CODL (FR-271).
- A job never holds a database transaction while it calls an external service. It reads, calls, then writes in a short transaction.
- A job writes progress to a `job_runs` row (total, done, failed, status). The screen polls that row.
- A failure of one item does not stop the others. Failures are listed with a reason.

Table 99: *Job designs*

Job (class)	Trigger and queue	Idempotency key	Retries and failure handling	Progress
IssueCredentialsBatchJob (JB-01) and IssueCredentialJob (JB-22)	CompletionConfirmed (EV-31). Queue `credentials`. One parent job fans out one job per participant	Participant, micro-credential and credential kind. A unique index stops a second active credential	5 tries with backoff. Vault or PDF failure leaves the item Queued for retry. A pre-issue check failure is a result, not an exception, and is listed with the reason	Parent job counts total, issued and failed. Issuance queue screen shows them
ImportResultsJob (JB-02)	Upload by CODL. Queue `imports`	File hash and cohort. Rows keyed by enrolment and component	3 tries on infrastructure errors only. A validation error is not retried. The file stays stored	Rows read, staged and rejected. Row report is built at the end
GenerateReportJob (JB-15) and RenderPdfJob (JB-04)	User request for a report, export or document. Queue `reports`	Document type, record ID, record revision and template version	3 tries. Headless Chromium has a time limit per document. A timeout fails that file only. The user can request it again	Status Waiting, Running, Ready or Failed. The download is a signed short-lived URL
AlertSchedulerJobs (QvExpiryAlertsJob (JB-06), CreditExpiryAlertsJob (JB-07), DecisionOverdueRemindersJob (JB-10), CompletionWindowRemindersJob (JB-11), PeriodicReviewRemindersJob (JB-12))	Scheduler, daily. Queue `default`	Record, alert kind and day offset in an alert log	A missed day is caught up because the rule is "offset reached and not yet sent". Sending is a separate notification job with its own retries	Count of alerts raised is logged. Failed sends appear in the notification log
OpenCloseRegistrationJob (JB-21) and CancelUnpaidEnrolmentsJob (JB-20)	Scheduler, every 5 minutes. Queue `default`	Offering and window edge. Enrolment and state	A late run applies the same state change. A row already changed is skipped	Count logged
VerifyAuditChainJob (JB-18)	Scheduler, nightly. Queue `default`	Date of the run	A broken link raises a security alert at once and is never auto-repaired	Rows checked and result

> **Note:** Job class names above are working names. The JB-xx numbers of Section 4 apply once that section is final (this section was written before Section 4 and uses class names only).

Requirements satisfied: NFR-08, NFR-07, NFR-04, NFR-11, NFR-43, NFR-18, IR-17, FR-144, FR-116, FR-173, FR-194, FR-249, FR-271, FR-268.

8. Security and Privacy Design

This section describes how MCMS meets the security requirements SR-01 to SR-21 and the privacy requirements under the Personal Data Protection Act 2010 (PDPA). The target is OWASP ASVS Level 2 (SR-16). Terms of the architecture decisions AD-06, AD-07, AD-09, AD-11 and AD-14 apply.

8.1 Authentication and Sessions

- **Sign-in.** Staff and students sign in with Microsoft Entra ID through OpenID Connect and Laravel Socialite (AD-06, FR-01). MCMS has no password store and no public sign-up (FR-02). The authorisation-code flow with PKCE is used. The `state` and `nonce` values are checked, and the ID token signature, issuer, audience and expiry are validated.
- **First sign-in.** A profile is created from the token claims and the Participant role is assigned (FR-03). Outside applicants are recognised by an attribute or an import (FR-04).
- **MFA.** Multi-factor authentication for staff roles is enforced by Entra Conditional Access (SR-02). MCMS also checks the `amr` claim at sign-in. If a user holds a staff role and `amr` does not contain an MFA method, the session is created without the staff roles and the user is told to complete MFA. The check is repeated when a staff role is assigned.
- **Session.** A server-side session is stored in Redis with a rotated session ID at sign-in. The cookie has the `Secure`, `HttpOnly` and `SameSite=Lax` attributes. The idle timeout is 30 minutes and is the setting `auth.idle_timeout_minutes` (FR-06). The session also has an absolute lifetime. Sign-out ends the session on the server.
- **Stub identity provider.** The stub sign-in (FR-05) is registered only when `APP_ENV` is not `production`. A boot check refuses to start the application in production if the stub is enabled, and a test in the pipeline fails if the stub route exists in a production configuration.
- **Deactivation.** A deactivated account cannot sign in and its sessions are removed. Linked records are kept (FR-23).
- **Transport.** All traffic uses TLS 1.2 or higher (SR-06, IR-22), with HSTS.

8.2 Authorisation Design

Every action is checked on the server (SR-03, FR-10). Hiding a button is a convenience and not a control. Authorisation is built from three rules, applied by one Laravel Policy per record type (AD-07).

1. **Role.** The user holds a role that may take the action (FR-11).
2. **Scope.** The role assignment carries a scope: the whole University, a PTj, or a list of micro-credentials (FR-12). A record outside the scope is not shown and not found by search (AD-16).
3. **Separation of duties on the record.** The policy loads the record and compares it with the actor (FR-14): a proposer cannot review or decide their own proposal, a person who entered a result cannot be its only verifier, and a reviewer who declared a conflict cannot open the review (FR-17).

CODL roles (S1 and P5) never pass an academic policy, and there is no super-user bypass for an academic action (D-12, FR-15). The approving roles for each decision type come from the setting `approval.route.*`, which is versioned (FR-16, AD-12). Role assignment needs a reason and an effective date and is audited (FR-13). Least privilege applies to database roles and to staff accounts (SR-04), and roles are reviewed on a schedule (SR-05).

Table 100 is the policy matrix. It follows UI/UX spec §2.2. "In scope" means the role assignment covers the PTj or micro-credential. "SoD" means the separation-of-duties rule for the record also applies. A dash means the action is not available to the role.

Table 100: *Policy matrix of roles and key actions*

Action	Student	Lecturer	Deputy dean	CODL S1 and P5	Industry expert	Public
Browse catalogue and verify a credential	Yes	Yes	Yes	Yes	Yes	Yes
Register, pay and withdraw	Own	–	–	–	–	–
Propose and edit proposal	–	In scope	–	–	–	–
Review a colleague's proposal	–	Assigned, SoD, conflict declared	–	–	–	–
Comment on an invited proposal	–	–	–	–	Invited	–
Assign reviewers, verify category, endorse	–	–	In scope, SoD	–	–	–
Record Senate decision	–	–	In scope	–	–	–
Confirm readiness and open offering	–	Items only	In scope	–	–	–
Import UCreds results	–	–	–	Yes	–	–
Enter results and send for verification	–	In scope	–	–	–	–

Action	Student	Lecturer	Deputy dean	CODL S1 and P5	Industry expert	Public
Verify results and confirm completion	–	–	In scope, SoD	–	–	–
Issue credentials	–	–	System job after confirmation	–	–	–
Correct, replace or revoke a credential	–	–	In scope, recorded decision	–	–	–
Record QV result	–	–	In scope	–	–	–
Confirm ACR and record deposit	–	–	In scope	–	–	–
Give or decline consent to deposit	Own	–	–	–	–	–
Apply for credit transfer	Own	–	–	–	–	–
Decide credit transfer, redeem or release	–	–	Faculty that owns the degree	–	–	–
Define and approve pathway mapping and plan	–	In scope (draft)	In scope	–	–	–
Edit settings, templates and reference data	–	–	–	Yes	–	–
Assign roles and scopes	–	–	–	Yes	–	–
View audit trail and reports	–	Own records	In scope	University	–	–
Suspend or terminate an offering	–	–	In scope	–	–	–

Policies are tested. For each row of the matrix there is a Pest test that proves the allowed role passes and every other role is refused, named after the requirement (FR-10, FR-14, FR-15, AD-17).

8.3 Input and Output Protection

The controls follow OWASP ASVS Level 2 (SR-16) and are listed in Table 101.

Table 101: *Input and output controls*

Area	Control	Where	SRS key
CSRF	Laravel CSRF token on every state-changing request, including Livewire calls. `SameSite=Lax` cookies	Middleware	SR-16
XSS	Blade double-brace escaping is the default. Raw output (`{!! !!}`) is banned by a lint rule except in one reviewed helper. Content-Security-Policy header with no inline script except nonces. Rich text is limited to a small whitelist and cleaned on input	Blade, headers	SR-16
Validation	Every request is validated by a Form Request or a Livewire rule on the server. Allow-lists for enumerations, size and format limits for text. Business rules are in domain services, so a UI bypass does not skip them	Request layer	SR-16
SQL	Eloquent and the query builder with bindings only. No string-built SQL. Full-text search uses bound parameters and is scoped	Data layer	SR-16
Mass assignment	Models declare fillable fields. Status fields are changed only through actions	Models	SR-16
File upload	Type by content and extension allow-list, size limit, random storage name, ClamAV scan before the file is available, object storage outside the web root, signed short-lived download URLs. An infected file is deleted from quarantine and the event is logged	Platform MD-00	SR-09
CSV import	Template version, row limits, treated as data only. Cells starting with `=`, `+`, `-` or `@` are prefixed with an apostrophe in exports to stop spreadsheet formula injection	Learning platform MD-06, Records MD-15	SR-16
Headers	HSTS, `X-Content-Type-Options`, `Referrer-Policy`, `Permissions-Policy`, frame ancestors none	Nginx and middleware	SR-06
Errors	Generic message to the user, details to the log. No stack traces in production	Exception handler	NFR-19
PDF rendering	Templates are server-owned. User text is escaped before it reaches headless Chromium. The browser has no network access except for the template assets, and it runs as an unprivileged user	Platform MD-00	SR-16
API	The public verification API is read-only, rate-limited, and returns the same fields as the page (SR-07)	Credentials MD-09	SR-07

Area	Control	Where	SRS key
Dependencies	Lock files, automated advisories in the pipeline, and an update review before each release	CI	SR-21

8.4 Secrets and Key Management

- **Application secrets** (database, Redis, SMTP, OIDC client secret, storage keys) are environment variables supplied at deployment from a secret store. They are not in the repository and not in logs. Each environment has its own set.
- **Credential signing keys.** Credentials are signed with Ed25519 (AD-09). The private key is held in a key vault owned by CODL, and the application calls the vault to sign and never sees the private key in plain text (SR-08, IR-13). If the vault is not available, issuing waits and nothing is signed with another key.
- **Key identifiers.** Each credential stores the key ID of the key that signed it, as the verification method in its proof. The public keys are published at a fixed URL as a controller document so that anyone can verify a credential.
- **Rotation.** A new key pair is created on a schedule or when compromise is suspected. New credentials use the new key from its start date. Old public keys stay published so that earlier credentials still verify. A compromised key is marked revoked with a date, and the affected credentials are listed by key ID for a decision on replacement.
- **Separate keys per environment.** Test and staging keys are different from production keys, and credentials signed in those environments say so in the issuer metadata.
- **Backups** of data are encrypted, and access to them is limited (SR-11). Encryption at rest is used for the database volume and the object store.

8.5 Audit Integrity

The audit trail is an append-only table (AD-11, FR-254, SR-10, DR-03).

- The application database role has INSERT and SELECT only on `audit_events`. It has no UPDATE and no DELETE. A migration role used only at deployment owns the table.
- Each row stores the hash of the previous row and its own hash, computed over the previous hash and the canonical row content (actor, action, record type and ID, before and after summary, reason, time). Rows are chained in one global sequence.
- The audit row is written in the same transaction as the change it records, so a change without an audit row cannot be committed (BR-38).
- A verification job (VerifyAuditChainJob (JB-18), section 7.12) recomputes the chain every night. A break raises a security alert at once (SR-19), and the job never repairs the chain.
- The head hash is copied each day to a store the application cannot write to, such as a log server, so that deletion of the tail can be seen.
- Corrections are new rows. Earlier rows are never changed (FR-256, D-20). Viewing the trail is limited to the user's scope (FR-255).

8.6 Public Surfaces

The public surfaces are the catalogue and the credential verification page and API (FR-153).

- **Minimum fields.** The page shows only the permitted fields (FR-154), set by `verification.public_fields`: status, credential title, issuer, issue date, category, and credit, MQF and QV only when verified. It never shows identity numbers, matric numbers, email or results.
- **No enumeration.** The verification code is a random 10-character string from a cryptographic generator, so codes cannot be guessed from sequence or from other data (DR-07, DR-07). Internal IDs are ULIDs and are not used on public pages.
- **Not found.** An unknown or malformed code returns the same message, the same status code and similar response time (FR-157). It never suggests a similar code.
- **Rate limiting.** Requests are limited per IP address by Laravel's rate limiter, with a stricter limit on failed lookups, so that bulk guessing is impractical (SR-17). Limited requests get a neutral message.
- **Logging.** Each lookup logs time, code and outcome, and the IP address only (FR-156). No other personal data is logged.
- **Availability.** The page reads from a small projection so it stays fast (NFR-03, NFR-14).
- **Caching.** The page is not cached by shared caches, so a revocation shows at once.

8.7 Personal Data under PDPA 2010

MCMS holds personal data of participants and staff (SR-12, NFR-36). The design handles it as in Table 102.

Table 102: *PDPA handling*

Principle	Design	SRS key
General and notice	A PDPA notice is shown and acknowledged at first registration. The acknowledged text version is stored	FR-100

Principle	Design	SRS key
Minimisation	Only data needed for each purpose is collected. Staff and matric numbers are optional on the profile. Public pages and exports show the minimum	SR-13
Consent	Consent to deposit is explicit, with date, wording version and channel. It can be declined, postponed or withdrawn before deposit, and each change is a new row	FR-185
Disclosure	Sharing a credential is the participant's own action. MCMS sends data to the ABC only for a consented ACR	SR-15
Access and correction	A participant sees their own record and requests a correction through the party with authority. The correction is a new revision with the reason and the approver	SR-14
Security	Controls in Sections 8.1 to 8.6 and encrypted backups	SR-06
Retention	Retention periods are settings per record category. Disposal needs approval, is logged, and is blocked by a hold. Audit rows and credentials are kept	FR-262
Data location and transfer	The system runs on UTeM-controlled hosting (AD-18). Email goes through the university relay (AD-15). No analytics or third-party scripts are loaded on pages	NFR-36
Test data	Non-production environments use seed data and never a copy of production personal data	NFR-44

Note: The retention periods are not yet decided (A-08, OI-07). The design keeps them as settings, so the decision does not change the code.

8.8 Security Logging and Alerting

Security events are written to the application log in a structured form, separate from the audit trail (SR-18, NFR-41). The events are:

- Sign-in success and failure, MFA failure, role-less staff sessions, and sign-out.
- Authorisation denials (policy refusals), with actor, action and record type.
- Role assignment and revocation, and setting changes.
- Failed verification lookups and rate-limit hits.
- File upload rejections and virus detections.
- Key vault use and failures, and key rotation.
- Audit chain check results.
- Adapter failures and demo-clock or stub flags found active in a non-production check.

The log never contains passwords, tokens, signing material, or results. Logs are sent to a central store with limited access. Alerts go to CODL for a broken audit chain, a spike of denials or failed sign-ins, a spike of failed verification lookups, a virus detection, a vault failure and a stub flag active in production (SR-19, NFR-40). An incident follows the response procedure of CODL (SR-20). A penetration test is done before the Release 1 go-live, and its findings are fixed or accepted in writing (SR-21).

8.9 Threat Table

Table 103 lists the main threats, their impact and the mitigations in this design.

Table 103: *Threats and mitigations*

Threat	Impact	Mitigation	SRS key
Stolen or guessed staff credentials	Unauthorised academic action	Entra sign-in, MFA by Conditional Access and `amr` check, idle timeout, audit of role use	SR-02
Stub identity provider left on in production	Anyone signs in as any role	Stub only when not production, boot refusal, pipeline test	FR-05
Privilege escalation, or a CODL user acting academically	False approvals or credentials	Policies with role, scope and separation of duties, no bypass, tests per matrix row	SR-03
Insider acts on own record	Conflict of interest, false verification	Self-review block, enterer is not verifier, conflict declaration, audit	FR-14
IDOR (changing an ID in a URL)	Reading another record	Policy and scope check on every record load, ULIDs, scoped queries	SR-03
Cross-site request forgery	Action done in a user's name	CSRF tokens, SameSite cookies	SR-16
Cross-site scripting	Session abuse, defacement	Blade escaping, CSP, input cleaning	SR-16
SQL injection	Data theft or change	Bindings only, no raw SQL	SR-16
Malicious upload	Malware, stored XSS	Content type allow-list, ClamAV, quarantine, signed URLs	SR-09

Threat	Impact	Mitigation	SRS key
Spreadsheet formula injection in CSV	Code run on a staff device	Cell prefixing in exports	SR-16
Forged or altered credential	Loss of trust in credentials	Ed25519 signature, key IDs, public verification, vault-held key	SR-08
Signing key compromise	Forged credentials	Vault, rotation, revocation of the key, list of affected credentials	SR-08
Audit trail tampering	Loss of evidence	Append-only role, hash chain, nightly verification, external head hash	SR-10
Enumeration of credentials	Disclosure of the existence of credentials	Random 10-character codes, identical not-found, rate limits	SR-17
Over-disclosure on the public page	PDPA breach	Minimum fields, no identity data, setting-controlled fields	FR-154
Personal data leak from backups or logs	PDPA breach	Encrypted backups, no personal data in logs, limited access	SR-11
Replay of a payment or import	Double charge or duplicate records	Idempotency keys, file hash check, natural keys	IR-17
Race for the last seat	Overbooking	Row lock on capacity check	NFR-06
False claim of credit, MQF or QV	Misleading participants	Wording from verified records and templates, forbidden-claim scan	FR-151
Denial of service on public pages	Verification unavailable	Rate limiting, read projection, separate worker pools	NFR-14
Vulnerable dependency	Remote exploit	Advisories in the pipeline, update review, pen test	SR-21

9. Quality Attributes

This section states how the design meets the non-functional requirements of SRS Section 7. Load figures come from the indicative sizing of A-09 and are not binding commitments (D-21). Where a figure is a value the sources do not fix, it is marked as an assumption for confirmation.

9.1 Performance

Table 104: *Performance targets and how the design meets them*

Requirement	Target	How it is met	How it is checked
NFR-01	95% of page requests within 2 seconds at the server	Server-rendered pages with eager loading (no N+1 queries), indexes on scope and status columns, paginated lists (default 10 rows), OPcache, Redis cache for settings and reference data	Request timing in logs; a load test on the test environment
NFR-02	First page of search results within 2 seconds for 95% of requests	PostgreSQL full-text search with GIN indexes on search columns, always filtered by the user's scope first (AD-16); results paginated	Load test with the sample world and ten times its volume
NFR-03	Public verification page within 1 second for 95% of requests	Lookup by indexed random code; one query; the page is a small view with no sign-in and no session write; the signature is checked at issue and stored, and rechecked in the background rather than on each view	Load test of the public route
NFR-04	Standard report for one PTj and one year within 10 seconds; larger reports in the background	Reports read through the read-only query contracts with indexed date and PTj columns; a report that exceeds the limit is queued on the `reports` queue and the user is notified when ready (FR-260)	Timing on seed data of one year
NFR-05	2,000 registrations in 2 hours with at least 100 concurrent users, no errors or loss	Registration is one short transaction; the offering row is locked only for the capacity check; payment and email are not inside the transaction; PHP-FPM sized for 100 concurrent requests	Load test script of the registration flow on the test environment
NFR-06	Never over capacity	Row lock (`SELECT ... FOR UPDATE` through Eloquent `lockForUpdate`) on the offering during the capacity check, plus a database check on the counter	Concurrency test in Pest
NFR-07	5,000 result rows validated and staged within 2 minutes	Streamed CSV read, validation in chunks of 500, bulk insert into staging, run as a job on the `imports` queue (FR-112, FR-114)	Timing with a 5,000-row file
NFR-08	500 credentials in 10 minutes	One issue job per credential on the `credentials` queue, run by several workers; PDF rendering on its own queue; signing is local with the vault key service (FR-144); jobs are idempotent	Timing with a 500-learner cohort and stub vault
NFR-09	Average CPU under 70% at peak	Heavy work is queued (NFR-11); capacity plan in Table 105	Load test measurement

> **Note:** As the PRD states for Release 1, performance targets are measured once on the test environment and reported; they are not tuned beyond Release 1 needs.

9.2 Scalability

The sizing of A-09 is 200 offerings and 20,000 participants a year, with peaks of 2,000 registrations within a few hours. The design must allow ten times the volume without a change of architecture (NFR-12).

Table 105: *Scalability measures*

Concern	Measure	Basis
Application servers	Stateless: sessions, cache and locks in Redis; files in MinIO; the code and the web process hold nothing on local disk. A second web node is added behind a load balancer without code change.	NFR-10
Workers	Each queue has its own worker count in Horizon configuration; credential issuing and PDF rendering can be given more workers during a cohort release.	AD-05

Concern	Measure	Basis
Database	Indexes by scope, status and date; no table scans in lists; audit and integration logs are partitioned by month once they pass an agreed row count (assumption DA to be set by Section 11); a read replica can serve reports without changing code because reads use the Laravel connection.	AD-03
Search	Stays in PostgreSQL. If volume outgrows it, a search service replaces one repository class behind the search contract.	AD-16
Storage	Object storage grows by disk addition; files are addressed by key, not by path.	AD-14
Initial deployment	Production starts with one web VM, one worker VM, one database VM and one VM for Redis and MinIO (assumption; to be sized by the load test).	AD-18, D-21

9.3 Availability

The targets are 99.5% availability during working hours, 08:00 to 18:00 Monday to Friday (NFR-13), and 99.5% for the public verification page at all hours (NFR-14).

- **Single points of failure.** In Release 1 the database and Redis are single instances. Failure of either stops the system until it is restored; the recovery time is that of NFR-17. A standby database is a later option and is not needed to meet 99.5%.
- **Verification page.** It reads one table and can be served by a second web node and a database read connection. It does not depend on Redis for a session, and the rate limiter falls back to allowing requests if Redis is down, so verifiers are not locked out by a cache failure.
- **External systems.** A failed adapter never stops a function that does not use it. Dependent jobs wait and retry (NFR-18).
- **Maintenance.** Planned maintenance is announced 48 hours ahead through the banner and kept outside working hours (NFR-15). A deployment uses maintenance mode for the migration step only, and migrations are written to be backward compatible with the previous release where practicable (expand, then contract).
- **Errors.** Users see a clear message with a reference code; the detail is logged (NFR-19).

9.4 Backup and Restore

Table 106: *Backup and restore design*

Item	Design	Basis
Database	Daily full backup with continuous archiving of the write-ahead log, so that restore to a point in time is possible. Encrypted at rest.	NFR-16, SR-07
Files	Daily copy of the MinIO buckets to separate storage; object versioning on.	NFR-16
Signing keys	Held in the CODL key vault with its own backup procedure owned by CODL; never in the application backup.	AD-09, SR-08
Retention of backups	At least 30 days.	NFR-16
Location and access	Stored separately from the production system; access limited to named administrators.	SR-11
Recovery point objective	24 hours (assumption of the SRS, to be confirmed by CODL and UTeM ICT). Log archiving can reduce it.	NFR-16
Recovery time objective	8 working hours (assumption of the SRS, to be confirmed).	NFR-17
Restore test	At least twice a year, a restore to a clean environment with a check that the audit hash chain verifies and the sample counts match.	NFR-17, SR-10

> **Note:** RPO 24 hours and RTO 8 working hours are the SRS values, which are themselves assumptions awaiting UTeM ICT. They are flagged for confirmation in Section 11.

9.5 Accessibility

The interface conforms to WCAG 2.1 level AA (NFR-21, IR-10). The rules below are build rules for developers; they are applied once, in the shared components, and checked in tests.

1. Every control works by keyboard. Dialogs and drawers trap focus, close on Escape and return focus to the control that opened them. Table rows that open records are focusable and respond to Enter and Space.
2. Every field has a visible label tied to the input. Placeholders never stand in for labels. Errors are linked to the field with `aria-describedby` and announced in a live region.
3. Status is never shown by colour alone: `x-ui.status-tag` always prints the word. The stage track gives each stage a text state.
4. Text contrast is at least 4.5:1, including white and light-blue text on the navy side navigation. Tokens that fail the ratio are not added to `tokens.css`; a CI step checks the token pairs.
5. Images have text alternatives. Icons are decorative unless they carry the only label, in which case they have an accessible name.

6. The layout reflows to one column below 760 px; tables scroll inside their frame; the page does not scroll sideways (IR-01, NFR-24).
7. Toasts use a polite live region. Idle timeout warnings use an alert dialog with a safe default.
8. Pages for participants stay under 1.5 MB without media where practicable (NFR-26).
9. Generated PDFs are tagged with the document language and have a logical reading order where Chromium supports it.

Checks: automated axe checks run in the Playwright suite on every main screen in both languages; a keyboard-only walk of the hand-off scenario is done by hand before each release.

9.6 Bilingual Operation

Every screen, notice and credential exists in Bahasa Melayu and English (AD-13, IR-02). The language switch keeps the user on the same page with entered data (NFR-25). Tests compare the two language files for missing keys and render each main screen in both languages. Bilingual content fields mark a missing language and a record cannot be issued as a credential in a language whose required fields are empty. Fonts are bundled so that long Malaysian names print in full (NFR-45).

9.7 Observability

Table 107: *Logging, metrics and health checks*

Area	Design	Basis
Application log	Structured JSON lines with time, level, request identifier, user identifier and module. Kept at least 90 days. A log processor removes identity card numbers, tokens and secrets.	NFR-41
Error reference	Each unhandled error gets a reference code shown to the user and stored with the full detail.	NFR-19
Security log	Sign-in, failed authorisation, role change, key use and personal-data export are written to a separate channel that UTeM ICT security monitoring can read.	SR-18
Queues	Laravel Horizon shows throughput, wait time, failed jobs and worker load. Failed jobs are listed for administrators with a retry action.	NFR-40
Scheduler	Each timed job writes a run record with its result; a missed run raises an alert.	NFR-43
Integration log	Every adapter call is logged and visible to administrators (FR-25, FR-116).	AD-08
Health checks	A `/health` endpoint (protected by token and network) checks database, Redis, MinIO, ClamAV, queue lag, disk and certificate expiry. The public site has a separate minimal `/up` check.	NFR-40
Metrics	Request time, error rate, queue length, job failures, disk use and CPU are exported to the monitoring system chosen by UTeM ICT (to be confirmed).	NFR-40
Audit	The audit trail is a business record, not a log; its integrity check can be run from the administration screen and by a scheduled job (SR-10).	AD-11

9.8 Maintainability

- **Modularity.** Modules and their allowed dependencies are enforced by Deptrac and architecture tests (NFR-27).
- **Conventions.** Laravel conventions, PSR-12, strict types and static analysis at level 8 (NFR-28).
- **Tests.** Every business rule has at least one automated test, and the suite runs on every change (NFR-29). Test names carry the requirement key, which gives a traceability list for free.
- **Portability.** Queries use Eloquent and the query builder; vendor-specific features appear only in migrations and are listed in the release notes (NFR-30).
- **Configuration.** Environment values are configuration and business values are settings (NFR-31, FR-20).
- **Documentation.** The SDD, adapter interfaces, CSV templates and operating procedures are updated with each release (NFR-34).
- **Administration without the database.** Users, roles, settings, templates, reference data and adapter modes are managed in the interface (NFR-42).

10. Deployment and Operations

This section describes where MCMS runs, how a change reaches production, and how it is watched. The hosting is Linux virtual machines at the UTeM data centre, which is an assumption awaiting confirmation (AD-18, A-09, NFR-33).

10.1 Environments

Table 108: *Environments and their settings*

Setting	Development	Test	Staging	Production
Purpose	Developer work	Automated tests, user acceptance	Release rehearsal on production-like data shape	Real use
`APP_ENV`	`local`	`testing`	`staging`	`production`
Identity provider	Stub	Stub	Live Entra ID test tenant (assumed available), or Stub	Live Entra ID only
Payment	Stub	Stub	Stub	Stub in Release 1
UCreds	Manual, sample files	Manual, sample files	Manual	Manual
ABC and QVC / MQA	Manual	Manual	Manual	Manual
SMTP	Mailpit	Mailpit	Mailpit or relay to test addresses	Live relay
Signing key	Test key file	Test key	Test key in a test vault	CODL vault key
Demo clock	Allowed	Allowed	Allowed	Disabled; the app refuses to boot with it on
Data	Seed (sample world)	Seed (sample world)	Seed or masked copy	Real data
Debug tools (Telescope)	On	On	Off	Off

Development and test environments hold no real personal data unless it is masked (NFR-44). Only the stub identity provider and the demo clock are limited to non-production, and the start-up check described in Section 3.6 enforces it (AD-06, AD-18).

10.2 Deployment Architecture

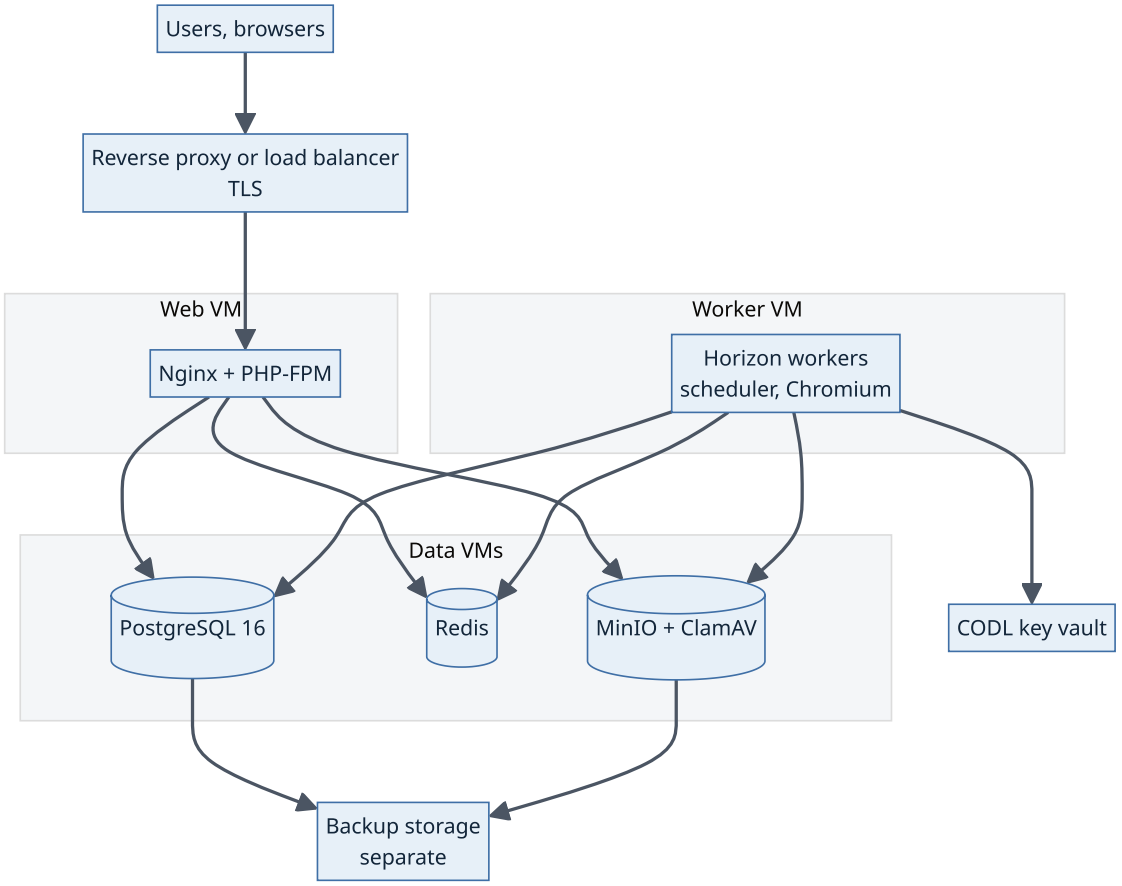


Figure 39: *Production deployment. The web and worker VMs run the same release; a second web VM can be added behind the proxy.*

All traffic uses HTTPS (TLS 1.2 or later) (IR-22, SR-06). The database, Redis and MinIO accept connections only from the web and worker VMs. Only the proxy is reachable from users. The database account used by the application has no UPDATE or DELETE right on the audit table (AD-11); migrations run under a separate account.

10.3 CI/CD Pipeline

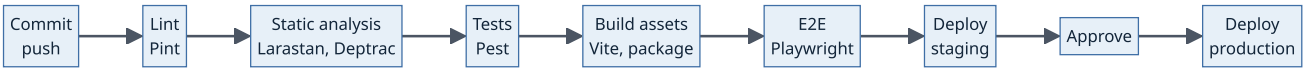


Figure 40: *Pipeline from commit to production. A failing step stops the pipeline.*

Table 109: *Pipeline steps*

Step	What runs	Fails when
Lint	`pint --test`	Code style differs from PSR-12
Static analysis	`phpstan` (Larastan, level 8) and `deptrac`	Type error or forbidden module dependency
Tests	Pest on a PostgreSQL 16 service with migrations and the test seed; includes architecture, contract and language-file tests	Any test fails; a requirement key without a test is reported
Build assets	`npm ci` and `vite build`; contrast check of tokens; a release archive or container image is created once	Build error or token below 4.5:1
End-to-end	Playwright runs the prototype's flows and the hand-off scenario against a built instance, with axe checks	A flow differs from the scenario
Deploy to staging	Same artefact; `php artisan down` only for the migration step; `migrate --force`; cache warm; Horizon restart (`horizon:terminate`); smoke test of `/health`	Migration or smoke test fails
Approve	A named person approves promotion after the staging check	Not approved

Step	What runs	Fails when
Deploy to production	The same artefact as staging; same steps; the previous artefact is kept for rollback	Smoke test fails; rollback to the previous artefact

The artefact is built once and promoted unchanged; only configuration differs between environments (NFR-31, NFR-33). The deployment procedure is a script in `deploy/` and is documented in the operations guide so that it is repeatable. A rollback restores the previous artefact; a migration that cannot be reversed is announced as such in the release notes and a database backup is taken before it runs.

10.4 Configuration and Secrets

- Environment values are held in `.env` files that are created on each server by the deployment, not committed to the repository.
- Secrets (database password, Redis password, MinIO keys, Entra ID client secret, SMTP credentials) are held in the secret store of UTeM ICT or, until one is agreed, in a file readable only by the application account (assumption). Secrets are rotated when a person with access leaves.
- The credential signing private key is never an environment value. It stays in the CODL key vault and the application asks the vault to sign (AD-09, SR-08). Key rotation does not invalidate credentials already issued.
- Logs and error pages never print secrets (NFR-41).

10.5 Database Migrations and Seed Data

- **Migrations.** Versioned migrations are the only way to change the schema (NFR-32). They run in the deployment under the migration account. A migration is small, backward compatible with the previous release where practicable, and tested on a copy of staging.
- **Reference seeder.** `ReferenceSeeder` loads reference data that production needs: roles, status lists, default settings, notification templates, PTj and programme lists as agreed with the offices (FR-24, DR-11).
- **Sample world.** `SampleWorldSeeder` loads the prototype's sample world: users in every role, three offerings of the three categories, one participant followed through every stage, and the stub adapter behaviour of PRD Section 7. It is the seed of the development, test and staging environments and the base of the Pest and Playwright tests. It is never run in production.
- **No legacy credentials.** No credential is created for results before go-live (DR-13).
- **Production data.** Production starts with the reference seed and the first administrators. All other data comes from use.

10.6 Monitoring and Alerting

Table 110: *Monitored items and alert rules*

Item	Alert when	To	Basis
Availability	`/up` or `/health` fails for 2 checks in a row	Operations team	NFR-40
Response time	The 95th percentile page time stays above 2 seconds for 10 minutes	Operations team	NFR-01
Queue length and wait	A queue waits longer than an agreed time (assumption: 5 minutes, 30 for reports)	Operations team	NFR-11
Failed jobs	Any failed job in `credentials` or `integration`; more than a set number elsewhere	Operations team	NFR-40
Scheduler	A timed job misses its run or ends with an error	Operations team	NFR-43
Disk	Database, MinIO or log disk above 80%	Operations team	NFR-40
Backup	A backup is missing or a restore test is overdue	Operations team	NFR-16
Adapter	An adapter is failing or retrying beyond the limit	Administrator	FR-25
Security	Many failed authorisations, mass export of personal data, key use outside issue jobs	UTeM ICT security	SR-19, SR-18
Audit chain	The integrity check finds a break	Operations team and security	SR-10
Certificate	A TLS certificate expires within 30 days	Operations team	SR-06

Planned maintenance is announced 48 hours ahead in the banner (NFR-15). An incident procedure covering detection, containment, notification and recovery is written before the first production release (SR-20), and a security test is carried out first (SR-21).

10.7 Release Flags: Release 1 and Later

Table 111: *Release switches*

Switch	Release 1 value	Effect when turned on	Release
`features.live_ucreds`	off (manual CSV)	Binds the live `LearningPlatformGateway` (FR-117, FR-118)	Later release (assumed 2)
`features.live_abc`	off (manual)	Binds a live `AbcGateway` (FR-192)	Later release (assumed 2)
`features.live_qv`	off (manual)	Binds a live `QvGateway` (FR-176)	Later release (assumed 2)
`features.live_payment`	off (stub)	Binds the live payment service when UTeM provides it (A-03)	Later release (assumed 2)
`features.sis_link`	off	Admission and graduation records from the student information system (FR-236)	Later release (assumed 3)
`features.wallet_share`	off	Credential wallet integration (FR-149)	Later release (assumed 3)
`demo_clock`	on in non-production	Allows a pretend date	Never in production

Release 1 contains all four increments (1.1 to 1.4) of the PRD. A flag that is off hides its screens and routes, and the manual path remains in use. Turning a flag on in production follows the pipeline: change the configuration in staging, test, approve, then change production.

11. Verification, Traceability and Open Items

11.1 Test Strategy

Every SHALL requirement of the SRS is proved by at least one automated test that names its requirement ID (D-01, AD-17). The levels below run in the pipeline of Section 10; a failure at any level stops the deployment.

Table 112: *Test levels*

Level	What it proves	Tool	Run
Unit	Rules inside actions, policies, enums and calculators: SLT totals, 40 hours per credit, 30% UTeM credit, QV validity of at most 3 years, credit expiry of 10 years, status transitions	Pest	Every commit
Feature (requirement)	One test per requirement, named after its key, for example `it_verifies_results_without_issuing_credentials` for FR-123; uses the database and fake adapters	Pest with Laravel testing	Every commit
Policy matrix	Every role against every academic action: CODL never passes; separation of duties per record	Pest data sets	Every commit
Adapter contract	Each adapter (Stub, Manual, Live) against the same interface tests: UCreds CSV, payment, SMTP, SSO claims, signing	Pest	Every commit
End to end	The 26 flows of the approved prototype (UI/UX spec Section 7) and its role hand-off scenario, run against the real application. FL-19 follows the demonstration scenario of PRD Section 7 (D-31); the build follows the PRD increments (D-32)	Playwright	Every merge to main
Accessibility	Automated WCAG 2.1 AA checks on every screen of the UI/UX spec inventory, plus keyboard walks of the main flows	axe-core with Playwright	Every merge to main
Security	Static analysis, dependency audit, OWASP ZAP baseline scan of staging, authorisation fuzzing of routes	PHPStan, composer audit, ZAP	Nightly
Performance	Page, search, verification, import and issuing targets of the NFR at the D-21 sizing	k6	Before each release

11.2 Traceability

Table 113 links each functional area of the SRS to the module that delivers it, the screens that show it and the end-to-end flows that test it. The per-requirement links are kept in the test suite: each feature test carries its requirement key, and a report generated in the pipeline lists any SHALL requirement without a passing test.

Table 113: *SRS sections, modules, screens and test flows*

SRS section	Module	Screens (UI/UX spec v1.2)	Flows
3.1 Identity, roles and access	MD-01, MD-00	CMN sign-in and profile; CODL users, settings, templates, reference data	FL-01, FL-17
3.2 Category and proposal	MD-02	LEC new proposal, proposal workspace, proposals	FL-02
3.3 Review, approval and versioning	MD-03	LEC review, change request; EXT review; DD proposals, Senate decision, decision pack	FL-03, FL-04, FL-20
3.4 Offering, catalogue and readiness	MD-04	LEC create offering, readiness; DD readiness; PUB catalogue and offering	FL-05, FL-22
3.5 Registration, enrolment and cohorts	MD-05	ST register, registrations, enrolment; LEC cohorts	FL-06, FL-06b, FL-21
3.6 Learning platform connection	MD-06	CODL UCreds exchange; import report; enrolment export	FL-07
3.7 Assessment and verification	MD-07	LEC cohort results; DD results, verify	FL-07, FL-08
3.8 Achievement Record	MD-08	ST achievement record; DD completion	FL-08
3.9 Credentials and verification	MD-09	DD completion, credentials; ST credentials; PUB verify; certificate and credential documents	FL-08, FL-09, FL-10, FL-23
3.10 Quality Verification	MD-10	LEC QV evidence; DD quality, QV record; QV package	FL-14
3.11 ACR and ABC deposit; 3.13 single redemption	MD-11	ST ledger, ACR; DD ACR, deposit; ABC export	FL-11, FL-12
3.12 Credit transfer	MD-12	ST transfer; DD transfer; transfer letter	FL-12
3.14 Stackable pathway	MD-13	DD pathways, plans; ST plan; PUB pathways	FL-13
3.15 Monitoring and CQI	MD-14	LEC quality review; DD quality review, periodic review, suspension	FL-15, FL-16

SRS section	Module	Screens (UI/UX spec v1.2)	Flows
3.16 Records, audit and reporting	MD-15	DD traceability, reports; CODL audit, reports, retention; report documents	FL-18, FL-24
3.17 Notifications and integration	MD-16	Notifications, preferences; CODL adapters, test mailbox	FL-07, FL-17

> **Note:** FL-19 (end-to-end journey of SRS use case A.5) crosses every module and is the Release 1 acceptance walk (PRD Section 7).

Table 114 lists requirements that apply across modules and the design element that meets each. Table 115 gives the build order of the PRD (D-32) with the modules and test flows each increment completes.

Table 114: *Cross-cutting requirements and where the design meets them*

Requirement	Where the design meets it
BR-09, BR-12, BR-13	MD-02 UMC structure checks: a UMC proposal must name an accredited parent course and map module outcomes to its CLOs; badges or modules from different UMCs or parents cannot be combined without an equivalence decision (MD-08 completion check); a module is a credit unit only when the separately recognisable flag is Yes, set by a recorded academic approval.
BR-10, BR-15	MD-02 SLT check: under the credit-based model every module must reach the SLT minimum (setting); reaching it gives no credit value or MQF level by itself, which come only from the approval or the QV decision.
BR-11	MD-08 and MD-09: module badges and module completion are recorded as badges and module results only; UMC completion is a separate deputy dean confirmation (EV-31).
BR-14	MD-10 QV: credit value and MQF level of a SAMC are copied only from the recorded QV decision (D-17).
IR-05	Shared Livewire form components (Section 4): validation on entry, error beside the field in plain language, entered data kept after an error.
IR-08	DESIGN.md tokens built by Vite; assumption DA-18.
IR-09	CMN-08 Help and the context-help panel on each screen; PUB-05 explains the separate recognition stages.
IR-12	A4 Blade templates for every printable document (DOC-01 to DOC-16), checked on a standard office printer in the acceptance test.
NFR-23	Task page and step counts: the Playwright flows fail if a common task needs more than five steps from the task page.
NFR-35	Guide compliance: each BR test cites its Guide clause; the Guide copy is Versi 1.0 (D-03).
NFR-39	Credential signing and public verification (AD-09, MD-09).
NFR-51, NFR-52	Process items, not code: role training and quick reference cards are part of go-live handover (Section 10).

Table 115: *Build order: PRD increments, modules and test flows*

Increment (PRD 5.3)	Modules	Flows that must pass at its end
1.1 Foundation	MD-00, MD-01, MD-15 (audit trail), MD-16	FL-00, FL-01, FL-17
1.2 Curriculum	MD-02, MD-03, MD-04	FL-02, FL-03, FL-04, FL-05, FL-20, FL-22
1.3 Delivery and credentials	MD-05, MD-06, MD-07, MD-08, MD-09	FL-06, FL-06b, FL-07, FL-08, FL-09, FL-10, FL-21, FL-23
1.4 Credit and quality	MD-10, MD-11, MD-12, MD-13, MD-14, MD-15 (reports)	FL-11, FL-12, FL-13, FL-14, FL-15, FL-16, FL-18, FL-24, and FL-19 (end to end)

11.3 Design Assumptions

These assumptions fill gaps the owner has not yet decided. Each is safe to change: the affected parts are named.

ID	Assumption	Affects	Closes with
DA-01	UTeM provides Linux virtual machines in its data centre for staging and production, with outbound HTTPS to Microsoft Entra ID.	Section 10, AD-18	CODL infrastructure
DA-02	PostgreSQL 16 is available; if not, MySQL 8 is used with the same migrations, losing only database-level check constraints (enforced in code instead).	AD-03, Section 5	OI-02
DA-03	The UTeM Microsoft Entra tenant allows an MCMS app registration with OpenID Connect, group claims for staff roles, and Conditional Access requiring MFA for staff.	AD-06, Section 8	OI-03
DA-04	CODL owns the credential signing keys and the public verification domain, and can provide a key vault (HashiCorp Vault or Azure Key Vault).	AD-09, Section 8	OI-08
DA-05	Redis, MinIO and a headless Chromium runtime may be installed on the application servers.	AD-05, AD-10, AD-14	CODL infrastructure

ID	Assumption	Affects	Closes with
DA-06	Programme admission, completion and withdrawal are recorded manually in MCMS by the deputy dean of the degree's faculty, from programme records, until a student-system link exists.	MD-11, MD-12, MD-13	OI-10
DA-07	UCreds exchanges data by CSV files in the template versions of Section 6 until its connection method is known, and issues module badges itself.	MD-06, AD-08	OI-04
DA-08	The ABC deposit export uses the ACR fields of Guide Appendix F3 in CSV and PDF until ABC publishes its format.	MD-11, Section 6	OI-11
DA-09	Backups follow the SRS targets: recovery point within 24 hours and restore within 8 working hours, by nightly full database backups and archived write-ahead logs kept 35 days. The monitoring system and the secret store are those UTeM ICT provides; until a secret store exists, secrets sit in a file readable only by the application account.	Sections 9 and 10	UTeM ICT
DA-14	Where the SRS state names and the status words of the UI/UX spec (Section 4.8) differ, the database stores the SRS value and the interface shows the UI/UX word (Section 5, status mapping): QV "Not verified" and "Re-verification required" show as Preparing; "Not supported" and "Rejected" show as Not approved, with the stage kept in `approval_decisions`; the plan decision "Not suitable" shows as Not yet suitable; deposit outcomes Successful, In process and Correction needed map to Deposited, ACR prepared and Correction needed.	MD-03, MD-10, MD-11, MD-13	UI/UX spec v1.2
DA-16	Public verification codes are 10 random Crockford base32 characters, shown in groups for reading; the shorter codes in the prototype's sample data are illustrative only. The host name in examples (`mcms.example.my`) is a placeholder for the domain CODL assigns.	MD-09, Section 6	OI-08
DA-17	The staff or matric number arrives in an Entra ID token attribute whose name is configured and confirmed with CODL. Rate limits for the public verification page and retry counts are configuration values set at deployment.	MD-01, Section 6, Section 8	OI-03
DA-15	A Digital Badge is issued for each module when its results are verified; the certificate and Digital Credential wait for the separate completion confirmation, after which the system issues them.	MD-07, MD-09	Matches the approved prototype and D-29
DA-13	Release flags: live UCreds, ABC, QVC/MQA and payment connections are Release 2; the student information system link and wallet sharing are Release 3. Each is a configuration switch over an existing adapter interface.	Sections 3 and 10	PRD release plan review
DA-10	A place is counted from Registered, so a registration in Pending payment holds its place until the payment window ends; an unpaid registration ends as Cancelled, and only an Enrolled participant can withdraw.	MD-05, Section 7.4	OI-12
DA-11	Revocation is final. A revocation made in error is put right by issuing a replacement credential, never by reversing the revocation.	MD-09, Section 7.7	CODL and Senate practice
DA-12	Notification recipients the SRS leaves open: a submitted proposal notifies the deputy dean of its PTj; an offering that opens is announced only in the catalogue and to its lecturer.	MD-16	CODL
DA-18	The interface follows DESIGN.md (Carbon with uLearn Navy) as the UTeM corporate identity until UTeM supplies its own guidelines (SRS IR-08, D-33, A-10).	Section 4, UI/UX spec 4	UTeM identity guidelines
DA-19	Laravel 13 is the framework line for the build. If a package of Section 4 does not yet support it at build start, the newest Laravel line that it supports and that is in security support is used (AD-02).	Section 4	Start of the build

11.4 Open Items Carried into the Design

Open item	How the design handles it until it closes
OI-02 Database	PostgreSQL assumed; portable queries keep MySQL possible (DA-02).
OI-03 SSO and applicants	Entra OIDC assumed (DA-03); outside applicants are recognised by an account attribute or an imported list held in MD-01.
OI-04 UCreds connection	CSV exchange through the Manual adapter (DA-07); a Live adapter can replace it without module changes.
OI-07 Retention periods	Retention per record category is a setting with no default disposal; ACR are kept indefinitely; disposal needs approval outside CODL.
OI-08 Signing keys	CODL ownership assumed (DA-04); key IDs are recorded on every credential so keys can rotate.
OI-09 Transfer limits	Programme limit is a programme setting; when unset, the transfer screen says no limit is applied and the decision is unaffected.

Open item	How the design handles it until it closes
OI-10 Admission and completion source	Manual recording by an authorised user (DA-06).
OI-11 ABC format	Export of Appendix F3 fields (DA-08); the export template is versioned so the format can change.
OI-12 Fees and refunds	Fee per offering; refunds handled outside MCMS; withdrawal deadline is a setting.
OI-13 Completion window	Setting `umc.completion_window_months` (no value until answered) that an offering may override; no reminder is sent while neither is set.
OI-15 Identity verification for online assessment	Method recorded per assessment component; no system enforcement in Release 1.

11.5 Design Risks

Risk	Mitigation in the design
Entra tenant settings differ from DA-03	Sign-in is one adapter in MD-01; the stub keeps development and tests independent of the tenant.
Signing key compromise or loss	Keys in a vault, never in the database or code; key ID on each credential; rotation and revocation procedure in Section 8.
Deputy deans become a bottleneck (PRD RK-10)	Task queue with age and reminders; bulk verification and batch issuing jobs; delegation hook for Release 2.
Audit table growth	Partitioning by year and archived partitions; the hash chain is verified per partition.
PDF rendering load at peaks	Rendering in queue workers with back-pressure; documents cached by version and regenerated only on change.
No mature PHP library for RDFC-1.0 canonicalisation, which the eddsa-rdfc-2022 proof needs	Spike before the build; if no library is fit, sign the same credential as a VC-JOSE (JWT) proof with the same Ed25519 key, which needs no canonicalisation. Only the signing service in MD-09 changes.
Small team maintaining a large codebase	Modular monolith with module boundaries checked in the pipeline, conventional Laravel structure, and the prototype as an executable reference for behaviour.